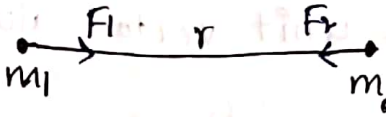
Gravitation

23/12/24/ Monday

Newton's universal law of gravitation:



m_1, m_2 are at rest.

① $F \propto m_1 m_2$

② $F \propto \frac{1}{r^2}$ — inverse square law.

On combining,

$$F \propto \frac{m_1 m_2}{r^2}$$

$$\boxed{F = \frac{G m_1 m_2}{r^2}} \ *$$

* G = universal gravitational const.

$= 6.67 \times 10^{-11} \ N m^2/kg^2$

* G is a scalar quantity.

* It doesnot depends on

1) medium b/w the masses

2) location, time & distance b/w the masses.

* gravitional force is always attractive force and acts only along line joining the masses.

Vector form:

$\bar{F}_{12}$ = force on m_1 due to m_2.

$\bar{F}_{21}$ = force on m_2 due to m_1.

$\bar{r}_{12} = \bar{r}_1 - \bar{r}_2$ = position vector of m_1 w.r.to m_2
(direction is from m_2)

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$\bar{r}_{21} = \bar{r}_2 - \bar{r}_1$ = position vector of m_2 w.r.t to m_1

(direction is from m_1)

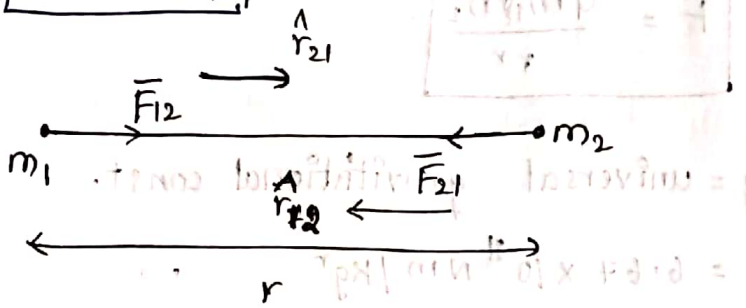
now, $\hat{r}_{12} = \dfrac{\bar{r}_{12}}{|\bar{r}_{12}|}$ = unit vector along $\bar{r}_{12}$

$\hat{r}_{21} = \dfrac{\bar{r}_{21}}{|\bar{r}_{21}|}$ = unit vector along $\bar{r}_{21}$

clearly, $\hat{r}_{12} = -\hat{r}_{21}$

$\Rightarrow |\hat{r}_{12}| = |\hat{r}_{21}|$

$$\boxed{|\bar{r}_{12}| = |\bar{r}_{21}|}$$



here $|\bar{r}_{12}| = |\bar{r}_{21}| = r$

now, $\bar{F}_{12} = \dfrac{Gm_1m_2}{|\bar{r}_{12}|^2}(-\hat{r}_{12})$

& $\bar{F}_{21} = \dfrac{Gm_1m_2}{|\bar{r}_{21}|^2}(-\hat{r}_{21})$

$= -\dfrac{Gm_1m_2}{|\bar{r}_{12}|^2}(\hat{r}_{21})$

$$\boxed{\bar{F}_{21} = -\bar{F}_{12}}$$

action reaction.

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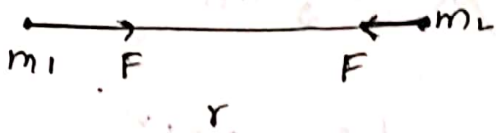
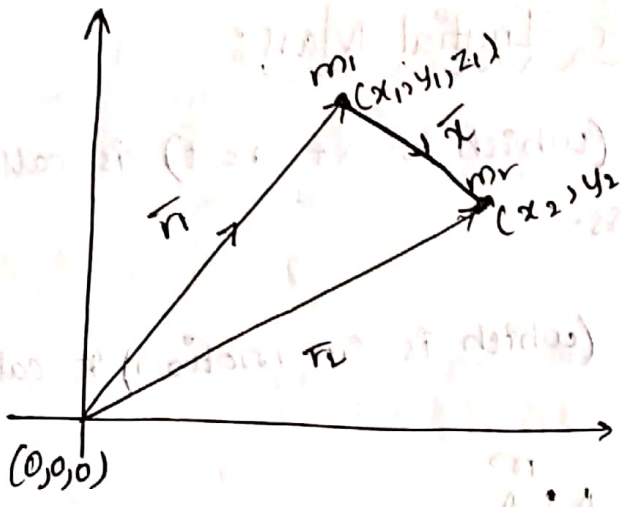
Q $|\bar{F}_{21}| = |\bar{F}_{12}|$



$$F = \frac{Gm_1m_2}{r^2}$$

* gravitational force is conservative force.

Ex:6



$\bar{r}_1 + \bar{x} = \bar{r}_2$

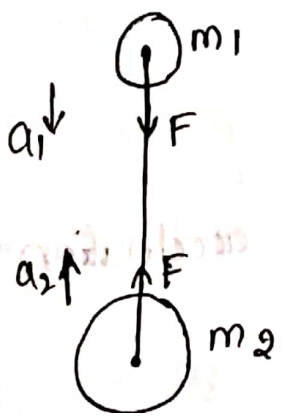
$\bar{x} = \bar{r}_2 - \bar{r}_1$

$\bar{r}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

$\bar{r}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

$$F_{12} = \frac{Gm_1m_2}{|\bar{r}_{21}|^2} \frac{\bar{r}_{21}}{|\bar{r}_{21}|}$$

$$F_{12} = \frac{Gm_1m_2}{|r_{21}|^3} (\bar{r}_{21})$$



$a_1 = \frac{F_{N1}}{m_1}$, $a_2 = \frac{F}{m_2}$

If $m_1 << m_2$

then, $a_2 >> a_1$

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* Time for collision of masses :

$$S_{rel} = u_{rel}^{0} t + \frac{1}{2} a_{rel} t^2$$

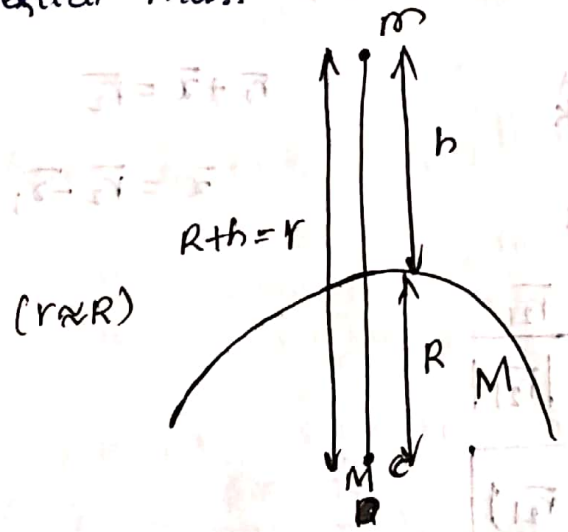
$$S_{rel} = \frac{1}{2}(a_1 + a_2) t^2$$

$$\boxed{t = \sqrt{\frac{2s}{a_1 + a_2}}}$$

* Gravitational Mass & Inertial Mass :

* mass of object (which is at rest) is called as gravitational mass.

* mass of object (which is in motion) is called inertial mass.



$h \ll\ll R$

$R_E = 6400 Km$

when body is at rest,

$$F = \frac{GMmg}{r^2}$$

when body is released, its acceleration is $a = g$

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$$F = m_i a$$

$$m_i a = \frac{G m_g M}{r^2}$$

$$a = \frac{G M m_g}{r^2 (m_i)}$$

clearly, $m_g \propto m_i$ *

If $g = \frac{GM}{r^2}$, then $\boxed{m_g = m_i}$

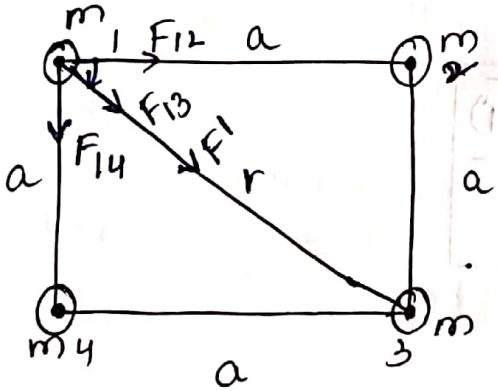
Principle of Superposition:

To find net force acting on any mass due to remaining mass.

$$\boxed{\bar{F}_1 = \bar{F}_{12} + \bar{F}_{13} + \bar{F}_{14} + \cdots}$$

vector sum.

Ex:



$F' = \sqrt{2} F_{12}$

$r = \sqrt{2} a$

$$\bar{F}_{12} = \frac{G m^2}{a^2}$$

$$\bar{F}_{14} = \frac{G m^2}{a^2}$$

$$\bar{F}_{13} = \frac{G m^2}{r^2} = \frac{G m^2}{2a^2}$$

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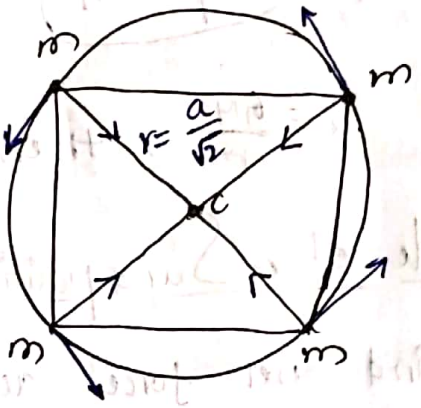
$$\bar{F}_1 = \bar{F}_{13} + \bar{F}_1$$

$$= \frac{Gm^2}{2a^2} + \frac{\sqrt{2}\,Gm^2}{a^2}$$

$$\boxed{\bar{F}_1 = \frac{Gm^2}{2a^2}(2\sqrt{2}+1)}$$

* If system moves on circular path with angular velocity ω.



$F_c = F_g$

↓ centripetal force

$$m r\omega^2 = \frac{Gm^2}{2a^2}(2\sqrt{2}+1)$$

$$\frac{a}{\sqrt{2}}\omega^2 = \frac{Gm}{2a^2}(2\sqrt{2}+1)$$

$$\omega^2 = \frac{Gm(2\sqrt{2}+1)}{\sqrt{2}a^3}$$

$$\boxed{\omega = \sqrt{\frac{Gm(2\sqrt{2}+1)}{\sqrt{2}a^3}}}$$

* $T = \frac{2\pi}{\omega}$

* M.I of system about 'C',

$$I = 4mr^2 = 4m\left(\frac{a}{\sqrt{2}}\right)^2$$

$$\boxed{I = 2ma^2}$$

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* Angular momentum of system about C.M,

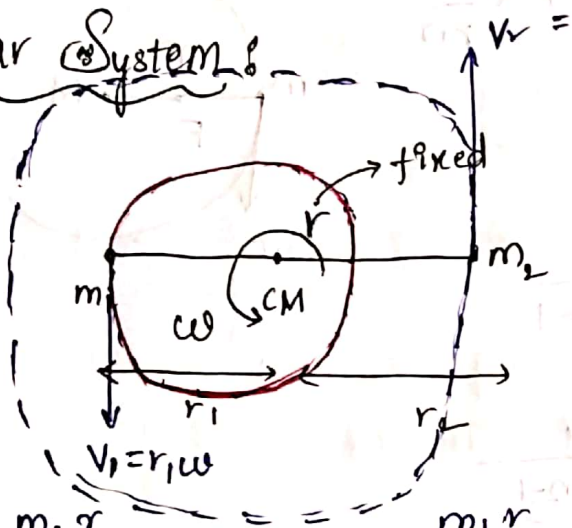
$$\boxed{L = 4I\omega}$$

* K.E of system,

$$K = 4 \times \frac{1}{2} I\omega^2$$

$$\boxed{K = 2I\omega^2}$$

Binary-Star System :-



revolving about C.M with constant ω.

$$r_1 = \frac{m_2 r}{m_1 + m_2}, \quad r_2 = \frac{m_1 r}{m_1 + m_2}$$

when $V_1 = V_2$
$\Rightarrow r_1 = r_2$
$\Rightarrow m_1 = m_2$

$$F_g = \frac{G m_1 m_2}{r^2}$$

$$F_g = F_c$$

$$\frac{G m_1 m_2}{r^2} = m_1 r_1 \omega^2$$

$$\frac{G m_2}{r^2} = \frac{m_2 r}{(m_1 + m_2)} \omega^2$$

$$\boxed{\omega = \sqrt{\frac{G(m_1 + m_2)}{r^3}}} \quad *$$

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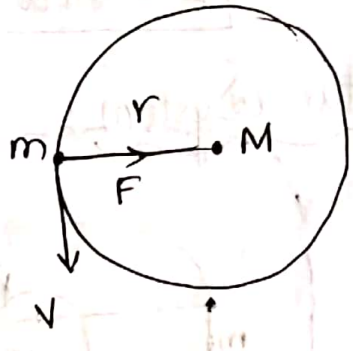
$T = \frac{2\pi}{\omega}$

$$T = 2\pi\sqrt{\frac{r^3}{g(m_1+m_2)}}$$

clearly $T^2 \propto r^3$ *

Ex:

hypothetically, $F \propto \frac{1}{r^n}$



$$\frac{mv^2}{r} \propto \frac{1}{r^n}$$

$$v^2 \propto \frac{1}{r^{n-1}}$$

$$v \propto \frac{1}{r^{\frac{n-1}{2}}}$$

& $\omega = \frac{v}{r} \propto \frac{1}{r^{\frac{n-1}{2}} \cdot r}$

$$\omega \propto \frac{1}{r^{\frac{n+1}{2}}}$$

$$\frac{2\pi}{T} \propto \frac{1}{r^{\frac{n+1}{2}}}$$

$$T \propto r^{\frac{n+1}{2}}$$ **

Ex: mass is split into; parts so that force b/w them is maximum at a fixed seperation b/w them.

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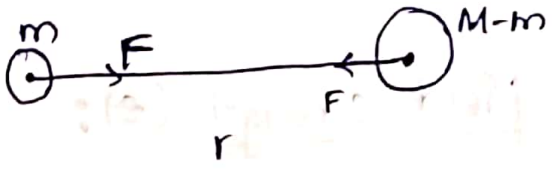


$$F = \frac{Gm(M-m)}{r^2}$$

$$\frac{dF}{dm} = 0 \Rightarrow \frac{d}{dm}(m)(M-m) = 0$$

$$(M-m) + (0-1)(m) = 0$$

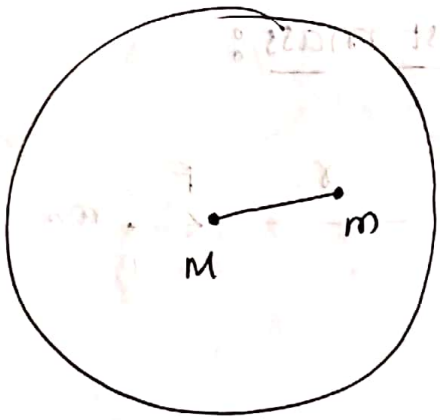
$$M - m - m = 0$$

$$M = 2m$$

$$\boxed{m = \frac{M}{2}}$$

Gravitational field :

* It is the space that surrounds the body upto which it's influence is felt is called gravitational field.



Gravitational Field

gravitational field strength ($\vec{E}$)

gravitational potential (V)

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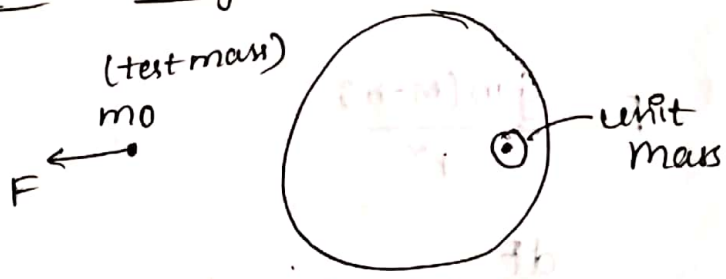
Gravitational field strength (ϵ):



$F \leftarrow m_0$

$\frac{F}{m_0} \leftarrow 1$

$\therefore \boxed{\bar{\epsilon} = \frac{\bar{F}}{m_0}}$ N/Kg

direction of $\bar{\epsilon}$ is along direction of $\bar{F}$.

* The force acting on unit mass when placed at a point in the field is called gravitatic intensity at that pt.

Gravitational intensity at a pt due to pt mass:

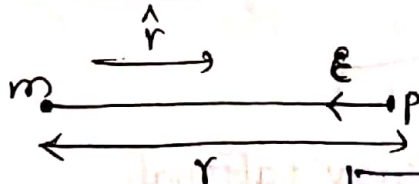
* Force on the test mass:

$F = \frac{Gmm_0}{r^2}$

$\epsilon = \frac{F}{m_0}$

$\boxed{\epsilon = \frac{Gm}{r^2}}$

*



$\therefore \boxed{\bar{\epsilon} = \frac{Gm}{r^2}(-\hat{r}) = \frac{Gm}{r^3}(\bar{r})}$ $\quad \hat{r} = \frac{\bar{r}}{|\bar{r}|}$

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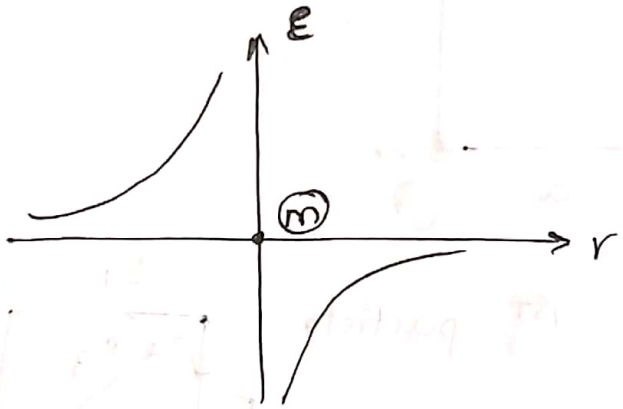
* graphs



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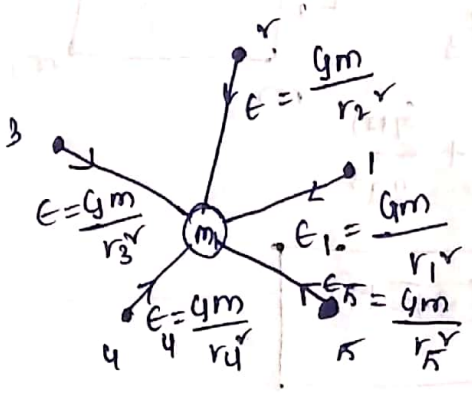
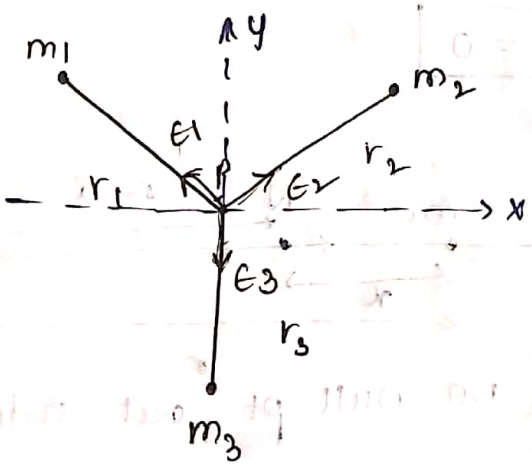


Principle of Superposition:

net gravitational intensity,

$$\bar{\epsilon} = \bar{\epsilon_1} + \bar{\epsilon_2} + \bar{\epsilon_3} + \dots$$



$$\epsilon_1 = \frac{Gm_1}{r_1^2}, \quad \epsilon_2 = \frac{Gm_2}{r_2^2}, \quad \epsilon_3 = \frac{Gm_3}{r_3^2}$$

$$\bar{\epsilon} = \bar{\epsilon_1} + \bar{\epsilon_2} + \bar{\epsilon_3}$$

$$\boxed{\bar{\epsilon} = \bar{\epsilon_x} + \bar{\epsilon_y}}$$

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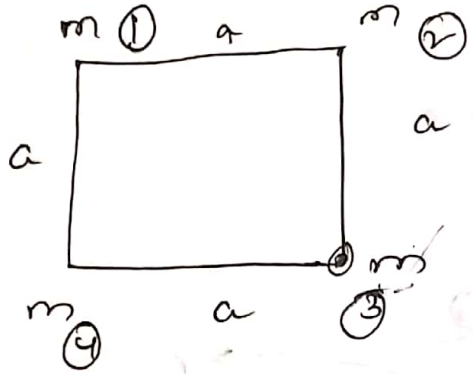
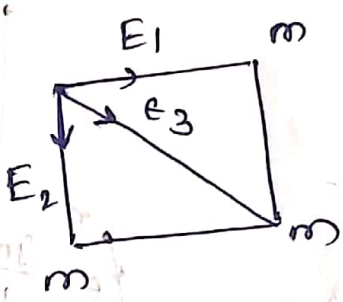
Ex:



ϵ_{net} on 1st particle,



$$\epsilon_{net} = \sqrt{2}\epsilon_1 + \epsilon_3$$

$$= \frac{\sqrt{2}Gm}{a^2} + \frac{Gm}{2a^2}$$

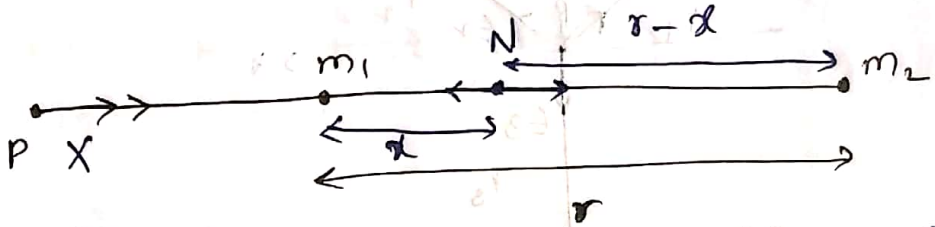
$$\boxed{\epsilon_{net} = \frac{Gm}{2a^2}(2\sqrt{2}+1)}$$

F_{net} on 1st particle,

$$\boxed{F_{net} = m\epsilon_{net}}$$

Null point / Neutral point :

here $\boxed{\epsilon_{net} = 0}$



* There will be no null pt out sides m_1 & m_2

* At null pt,

$$\epsilon_{net} = 0$$

$$\vec{\epsilon_1} + \vec{\epsilon_2} = 0$$

$$\vec{\epsilon_1} = -\vec{\epsilon_2}$$

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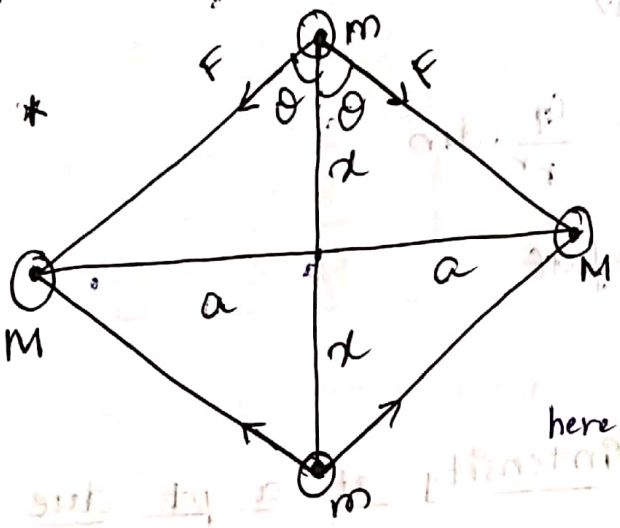
$\because \epsilon_1 = \epsilon_2$

$$\frac{Gm_1}{x^2} = \frac{Gm_2}{(r-x)^2}$$

$$\boxed{\frac{m_1}{x^2} = \frac{m_2}{(r-x)^2}}$$

from this x can be calculated.

Ex:



stable eqm.

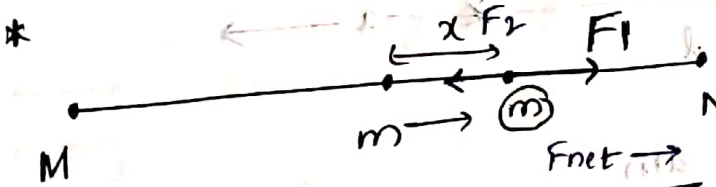
here particle is in SHM.

$$F_{net} = 2F\cos\theta$$

$$\cos\theta = \frac{x}{\sqrt{a^2+x^2}}$$

$x \ll a$

$$ma_0 = 2F\cos\theta$$



unstable eqm.

$\boxed{F_1 > F_2}$

* particle is not coming back to mean position when displaced further. So it is in unstable eqm.

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Gravitational intensity due to extended bodies:

Mass densities:

a) $\lambda = \frac{m}{l} = \frac{dm}{dx}$

b) $\sigma = \frac{m}{A} = \frac{dm}{dA}$

c) $\rho = \frac{m}{V} = \frac{dm}{dV}$

i) $x_{cm} = \frac{\int x \cdot dm}{\int dm}$

ii) $I = \int r^2 \cdot dm$

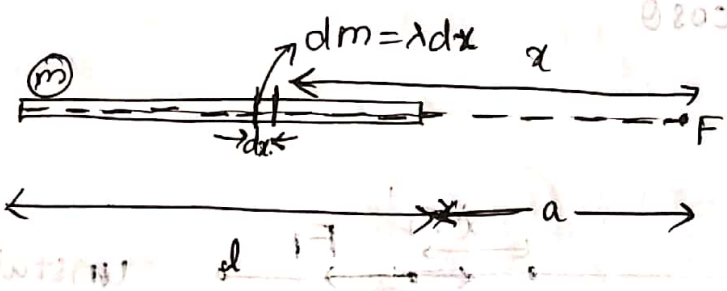
here, 1) $dE = \frac{Gq}{r^2} \cdot dm$

2) $E = \int dE$

Applications:

① Gravitational intensity at a pt due to thin rod:

a) on the axial line:



i) $\lambda = \frac{m}{l} = \frac{dm}{dx}$

Gravitational intensity at a pt due to small element,

$$dE = \frac{G dm}{x^2}$$

$$\Rightarrow dE = \frac{G \lambda dx}{x^2}$$

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∴ Total Gravitational Intensity,

$$E = \int_{x=a}^{x=a+l} dE = G\lambda \int_{a}^{a+l} \frac{1}{x^2}\, dx$$

$$E = G\lambda \left[-\frac{1}{x}\right]_{a}^{a+l}$$

$$\boxed{E = G\lambda\left(\frac{1}{a} - \frac{1}{a+l}\right)} \quad *$$

$$E = \frac{G\lambda(l)}{a(a+l)}$$

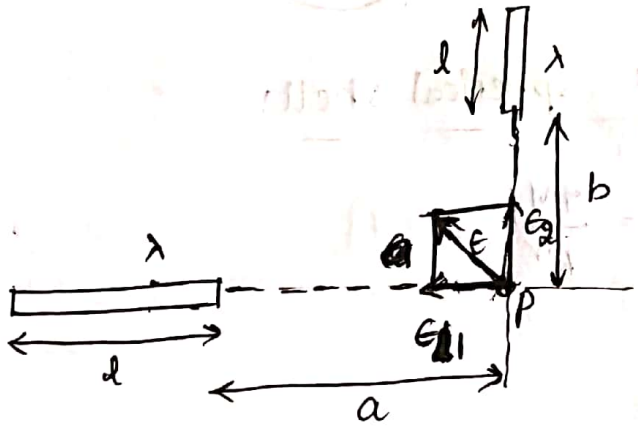
$$\boxed{E = \frac{Gm}{a(a+l)}}$$

* Force on the mass 'm',

$$F = mE$$

$$\Rightarrow \boxed{F = \frac{Gmm_0}{a(a+l)}}$$

Ex:



$$E_1 = \frac{G\lambda l}{a(a+l)}$$

$$E_2 = \frac{G\lambda l}{b(b+l)}$$

$$E = \sqrt{E_1^2 + E_2^2}$$

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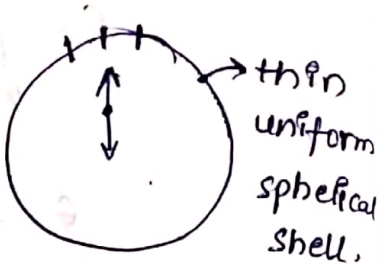
Shell Theorems:

1. At any pt inside the shell,

$$E_{net} = 0$$



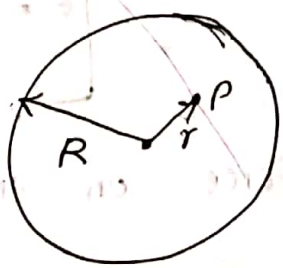
2. For any pt outside the shell / on the surface of the shell, shell can be treated as "pt mass".

Gravitational intensity due to spherical shell:

a) pt lies inside the shell:

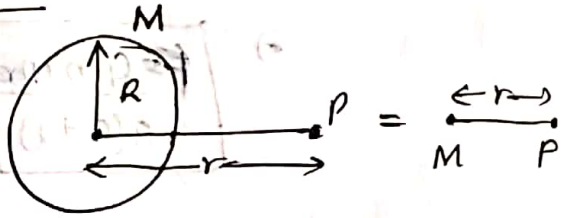
$(r < R)$

$$E = 0$$



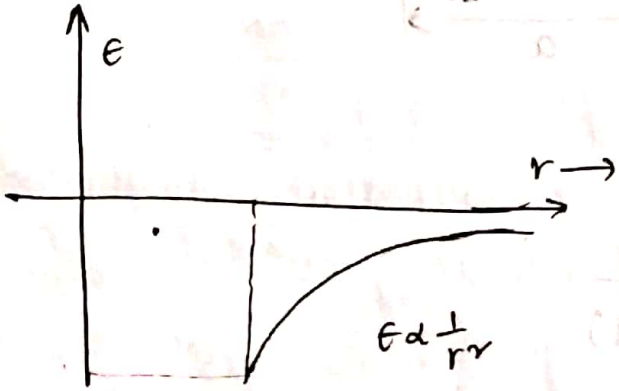
b) pt lies outside the shell:

$$E_0 = \frac{GM}{r^2}$$



c) pt lies on the spherical shell:

$r = R$, $E_0 = \frac{GM}{R^2}$



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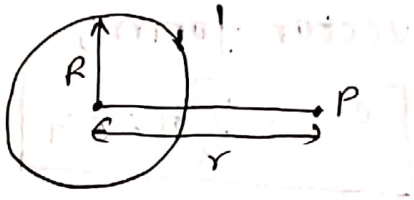
Gravitational Intensity due to solid sphere

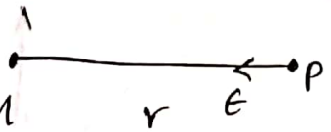
$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \text{uniform}$$

a) pt lies out side :

$$\bar{E} = -\frac{GM}{r^2}\hat{r}$$



cor) $\bar{E} = \frac{GM}{r^2}$



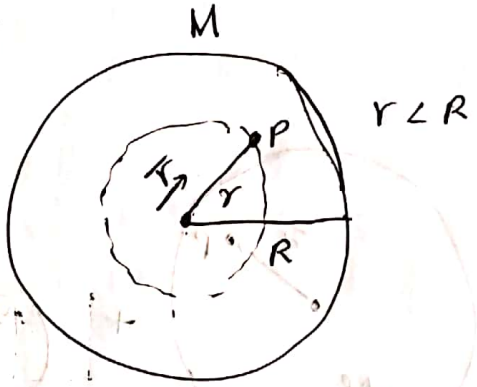
$$\hat{r} = \frac{\bar{r}}{|\bar{r}|}$$

b) on the sphere :

$r = R$

$$E = \frac{GM}{r^2} \Rightarrow \boxed{E = \frac{GM}{R^2}}$$

c) Inside of sphere



$$\mathcal{E} = \frac{GM}{r^2}$$

$$= G\left[\frac{M}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi r^3\right]$$

$$\boxed{\mathcal{E} = \frac{GMr}{R^3}}$$

If ρ is variable,

$dm = \rho dv$

$$m = \int \rho \, dv$$

$$m = \int_{x=0}^{x=r} \rho(4\pi x^2)\,dx$$

$$\mathcal{E} = \frac{G}{R^3} \times \frac{4}{3}\pi R^3 \rho r$$

$$\boxed{\mathcal{E} = \frac{4}{3}\pi \rho r G}$$

$\boxed{\mathcal{E} \propto r}$

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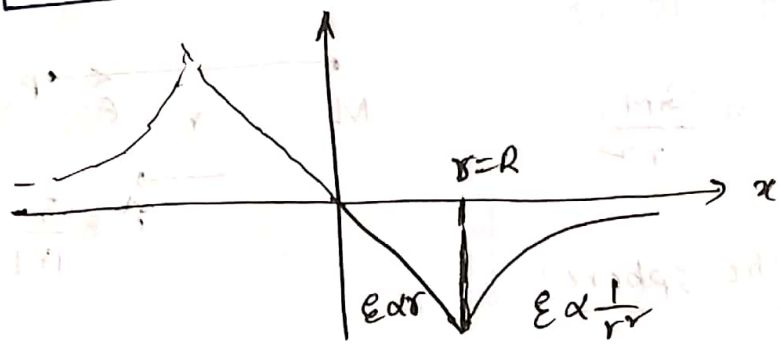
at centre of sphere,

$r = 0$

$\Rightarrow \boxed{E = 0}$

In vector form,

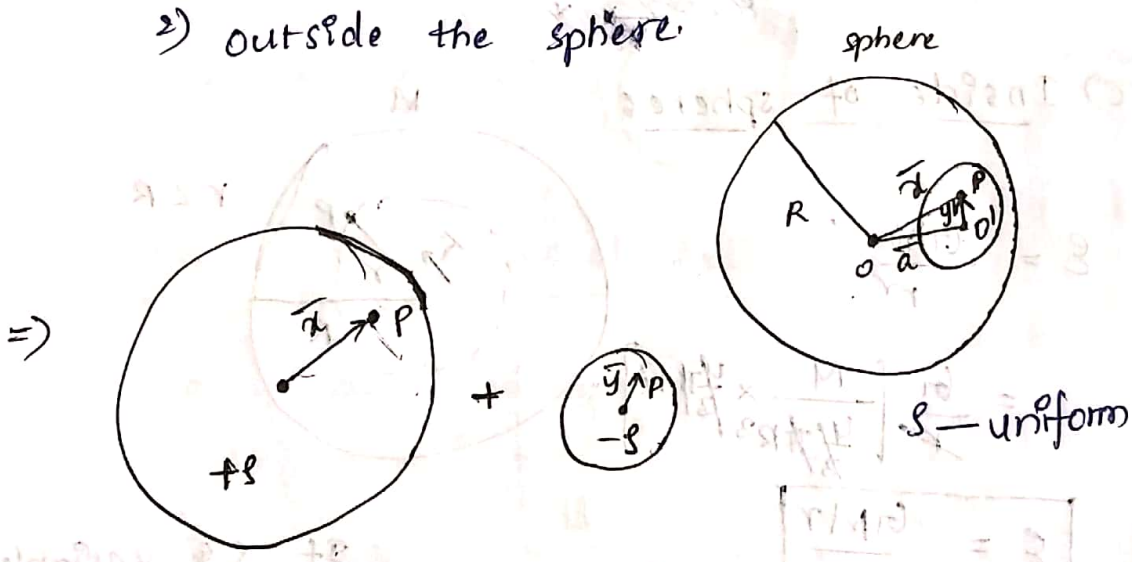
$$\boxed{\mathcal{E} = -\frac{4}{3}\pi \rho \bar{r} G}$$



Gravitational Intensity at a pt:

1) Inside the cavity

2) outside the sphere.



Principle of Super position:

$$\bar{\mathcal{E}} = \bar{\mathcal{E}}_1 + \bar{\mathcal{E}}_2$$

$$= -\frac{4}{3}\pi \rho G \bar{x} + \left[-\frac{4}{3}\pi(-\rho) G \bar{y}\right]$$

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$$\bar{E} = -\frac{4}{3}\pi\rho G\,[\bar{x}-\bar{y}]$$

from vector addition,

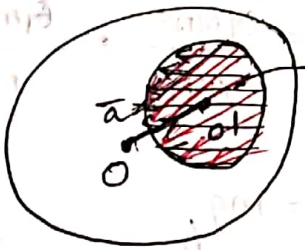
$$\bar{a}+\bar{y}=\bar{x} \Rightarrow \bar{a}=\bar{x}-\bar{y}$$

$$\boxed{\bar{E} = -\frac{4}{3}\pi\rho G(\bar{a})}$$

clearly field is uniform inside the cavity.

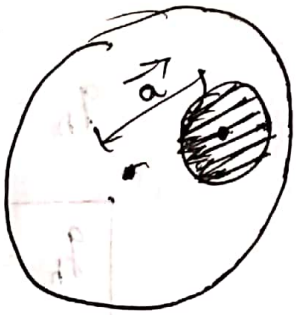
$\bar{a}=0$

$\bar{E}=0$

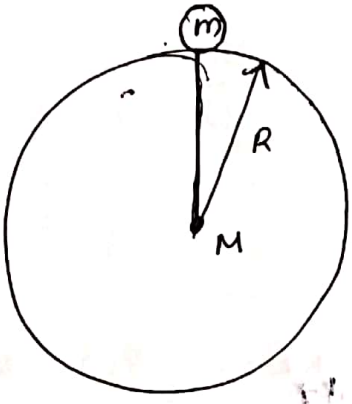


Acceleration due to gravity (g):

* It is the acceleration of freely falling body due to gravitation pull of the planet/earth.



* Relation b/w g & G:



$$F_g = \frac{GMm}{R^2} \quad -①$$

here $F_g = mg$ — ②

$$mg = \frac{GMm}{R^2}$$

$$\boxed{g = \frac{GM}{R^2}}^{*}$$ = gravitational intensity at that place.

(or) $g = G\,\frac{\frac{4}{3}\pi R^3\rho}{R^2}$

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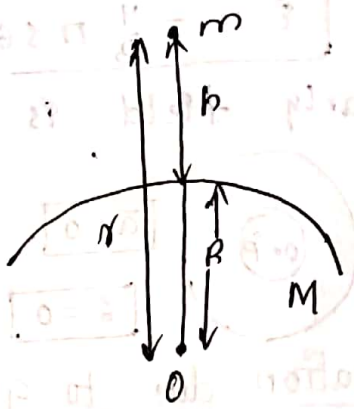
$$g = \frac{4}{3}\pi R \rho G$$

* Variation of g:

① with altitude / height:



$$F_g = \frac{GMm}{(R+h)^2} = \frac{GmM}{r^2}$$

$$F_g = mg_h$$

$$mg_h = \frac{GmM}{r^2}$$

$$g_h = \frac{GM}{r^2}$$

$$\boxed{g_h = \frac{GM}{(R+h)^2}}$$

(or) $$g_h = \frac{gR^2}{(R+h)^2}$$

$$\boxed{g_h = g\left(\frac{R}{R+h}\right)^2} \quad **$$

for small altitudes, $h << R$

$$g_h = g\left(\frac{1}{1+\frac{h}{R}}\right)^2$$

$$g_h = g\left(1-\frac{h}{R}\right)^2$$

$$\boxed{g_h = g\left(1-\frac{2h}{R}\right)}$$

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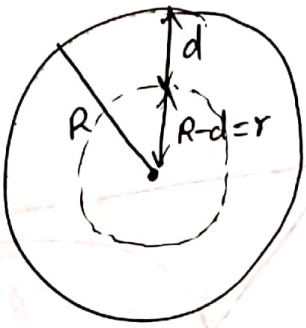
decrease in 'g' with small altitudes,

$$\Delta g_h = g - g_h = \frac{2gh}{R}$$

fractional change in g-value,

$$\frac{\Delta g_h}{}$$

② with depth,



$$g_d = \frac{4}{3}\pi (R-d)\rho G$$

$$\frac{g_d}{g} = \frac{\frac{4}{3}\pi (R-d)\rho G}{\frac{4}{3}\pi R \rho G}$$

$$\frac{g_d}{g} = \frac{R-d}{R} \Rightarrow g_d = \frac{gr}{R}$$

$$g_d = g\left(1 - \frac{d}{R}\right)$$

* decrease in g-value,

$$\Delta g_d = g - g_d$$

$$\Delta g_d = \frac{g_d}{R}$$

* At the centre of earth

$$d = R \Rightarrow g_h = g$$

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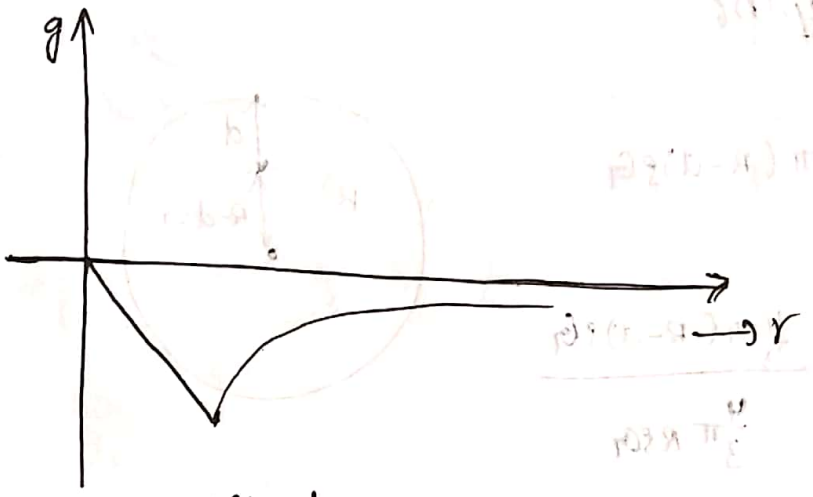
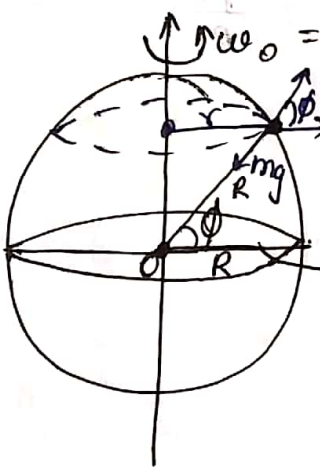
points:

* decrease in 'g' is large for small altitudes than for small depths.
* decrease in 'g' is less for large altitudes than for large depths.



③ with Latitude:



$$\cos\phi = \frac{r}{R}$$

$$r = R\cos\phi$$

* line joining centre and pt on surface makes an angle ϕ with equator where the angle is called latitude.

$$F_{net} = mg - mr\omega_0^2\cos\phi$$

$$mg_\phi = mg - mr\omega_0^2\cos\phi$$

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$$\boxed{g_\phi = g - R\omega_0^2 \cos^2\phi} \quad **$$

* At equator,

$$\boxed{g_\phi = g - R\omega_0^2} \quad (\min)$$

* rotation doesnot effect value of g at poles.

at poles, $\phi = 90°$

$$\boxed{g_\phi = g}$$

* $R\omega_0^2 = 6400 \times \left(\frac{2\pi}{24 \times 3600}\right)^2 \times 10^3 = 0.034 \text{ m/s}^2.$

* As spinning increases ($\omega \uparrow$)

at one particular ω,

$$g_\phi = 0$$

$$\boxed{mg_\phi = 0}$$

$$g - R\omega^2 = 0$$

$$\boxed{\omega = \sqrt{\frac{g}{R}}}$$

(*) every body is weightless at equator.
(for particular ω)

Time period, $T = \frac{2\pi}{\omega}$

$$= 2\pi\sqrt{\frac{R}{g}}$$

$$\boxed{T = 84 \text{ min}, 36 \text{ sec}}$$

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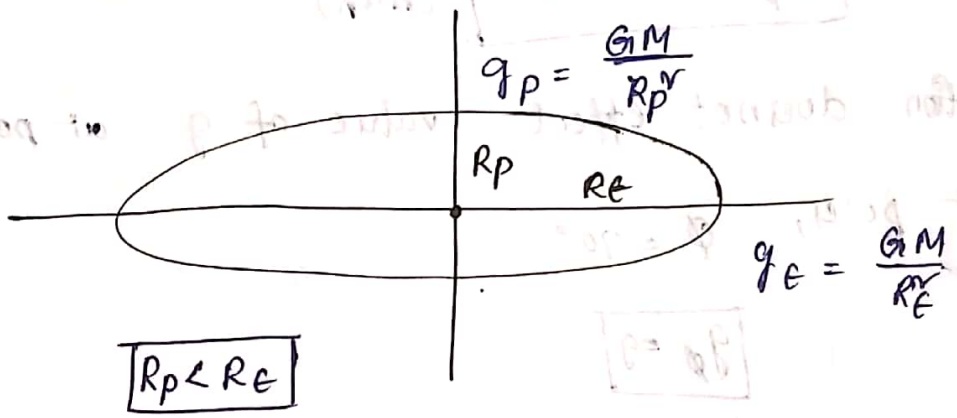
and $\frac{R\omega^2}{R\omega_0^2} = \frac{g'}{0.0034} = 289$

$$\boxed{\omega = 17\omega_0}$$

④ with shape of earth:



$$\boxed{R_p < R_E}$$

clearly $\boxed{g_E < g_p}$

24|12|24|Tuesday

Gravitational potential energy:

It is of two types.

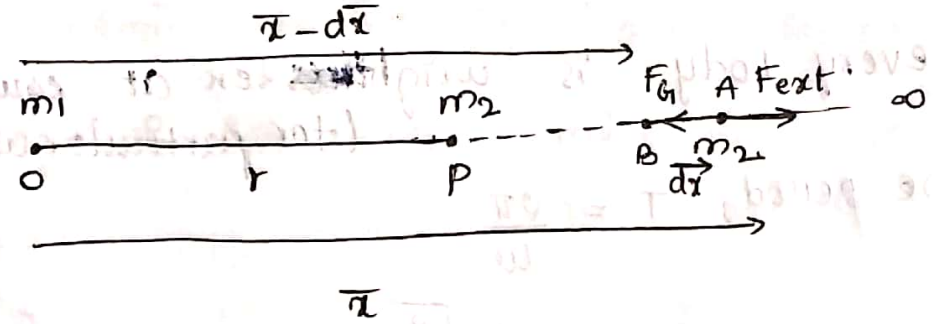
1) Interaction potential energy.

2) Self energy.

Interaction potential energy:

*It is the amount of work done in keeping two particles at certain seperation against gravitation force is called interaction potential energy.

Derivation:



* gravitational force acting on m_2 placed at 'A' is,

$$F_G = \frac{Gm_1m_2}{r^2}, \text{ towards 'O'.}$$

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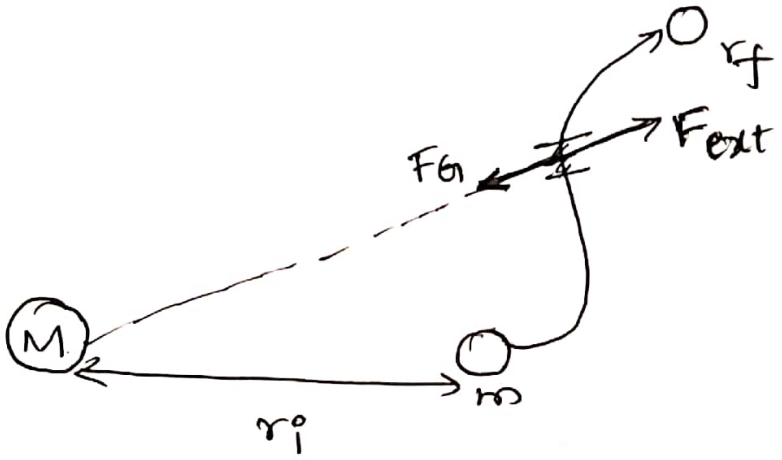
** workdone by an external agent in moving a mass from r_i to r_f.



$$\bar{F}_{Gr} + \bar{F}_{ext} = m\vec{a}$$

Since $\bar{a} = 0$, $\quad \bar{F}_g = -\bar{F}_{ext}$

$$dw_{ext} = \bar{F}_{ext} \cdot d\bar{l}$$

$$= -\bar{F}_g \cdot d\bar{l}$$

$$= -|\bar{F}_g| \cdot |d\bar{l}| \cos(180-\theta)$$

$$dw_{ext} = F_g \cdot dl \cos\theta$$

$$\int_{U=0}^{U} dw_{ext} = \int_{r=\infty}^{r=r} \frac{GMm}{r^2} \cdot dr$$

$$\boxed{U = -\frac{GMm}{r}}$$

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work done by external force in moving the mass m_2 slowly from A to B is

$$dw_{ext} = \bar{F}_{ext} \cdot d\bar{x}$$

$$= F_{ext}\, dx \cos\theta$$

$$= F_{ext} \cdot dx$$

$\bar{F}_{ext} = 0$

$\bar{F}_g + \bar{F}_{ext} = 0$

$$\boxed{\bar{F}_g = -\bar{F}_{ext}} \quad *$$

$$\therefore dw_{ext} = \frac{Gm_1m_2}{x^2} \cdot dx$$

$\Rightarrow$ Total work done,

$$w_{ext} = \int_{x=\infty}^{x=r} \frac{Gm_1m_2}{x^2} \cdot dx$$

$$= Gm_1m_2 \left[-\frac{1}{x}\right]_{\infty}^{r}$$

$$\boxed{w_{ext} = -\frac{Gm_1m_2}{r}}$$

from work-energy theorem,

$$w_g + w_{ext} = \Delta K \quad (\Delta K \to 0)$$

$$-\Delta U + w_{ext} = 0$$

$$\Delta U = w_{ext}$$

$$w_{ext} = U_f - U_i$$

$$-\frac{Gm_1m_2}{r} = U_f - U_i$$

$$\boxed{U = U_i - \frac{Gm_1m_2}{r}} \quad *$$

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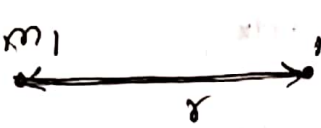
* If $U_i = 0$ then $\boxed{U = -\frac{Gm_1m_2}{r}}$ **

*

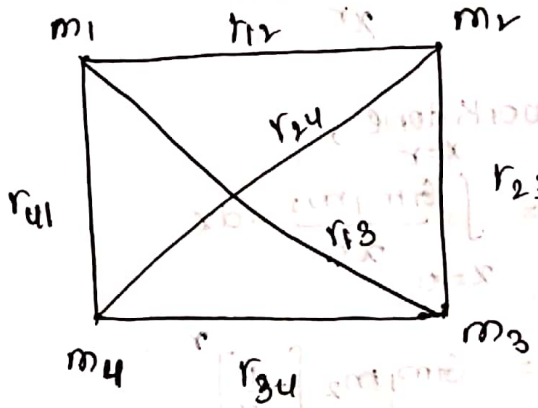


$U = -\frac{Gm_1m_2}{r}$

* for N-particle system,

$$\text{no. of pairs} = \frac{N(N-1)}{2}$$

Ex:

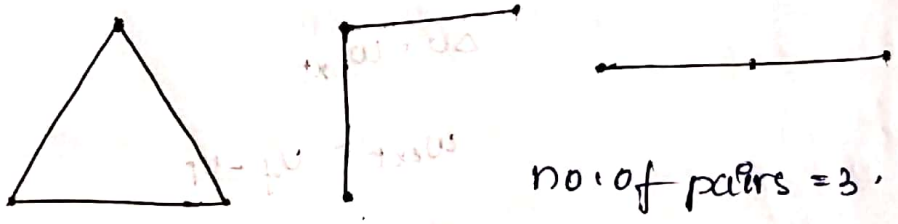


$$\text{no. of pairs} = \frac{4(4-1)}{2} = \frac{4(3)}{2} = 6$$

now, $U_{system} = U_{12} + U_{23} + U_{34} + U_{41} + U_{24} + U_{13}$

$$U_{system} = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_2m_3}{r_{23}} - \frac{Gm_3m_4}{r_{34}} - \frac{Gm_4m_1}{r_{41}} - \frac{Gm_2m_4}{r_{24}} - \frac{Gm_1m_3}{r_{13}}$$

Ex:



no. of pairs = 3.

* In general,

$$U_{system} = -\sum_{i=1}^{n-1} \sum_{j=2}^{n} \frac{Gm_im_j}{r_{ij}}$$

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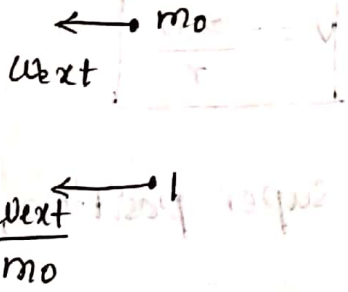
Gravitational Potential : (V) :-

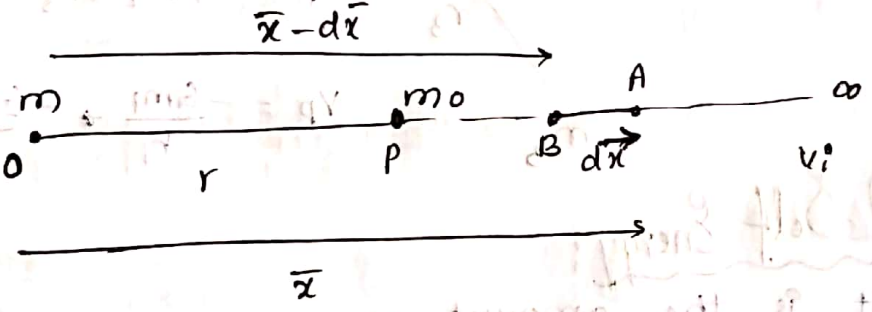
** It is the amount of workdone in bringing a unit pt mass from infinity to a pt in the field & against to it is called gravitational potential at that point.



$$V = \frac{W_{ext}}{m_0}$$ unit: J/kg.

* It is a scalar — has magnitude but no direction.

Potential at a point due to point mass :



$$F_G = \frac{Gmm_0}{x^2}, \text{ towards 'O'}.$$

$$dW_{ext} = \frac{Gmm_0}{x^2} \cdot dx$$

$$W_{ext} = \int_{x=\infty}^{x=r} \frac{Gmm_0}{x^2} \cdot dx$$

$$W_{ext} = -\frac{Gmm_0}{r}$$

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$$V = \frac{W_{ext}}{m_0}$$

$$\boxed{V = \frac{-Gm}{r}}$$

Work done – (+ve)
Potential – ↓

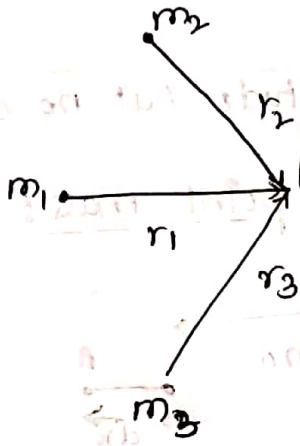
* m —— r

$$\boxed{V = -\frac{Gm}{r}}$$

* principle of super position,

$$V = V_1 + V_2 + V_3 + \cdots$$

Ex:



$$V_P = V_1 + V_2 + V_3.$$

$$V_P = -\frac{Gm_1}{r_1} - \frac{Gm_2}{r_2} - \frac{Gm_3}{r_3}$$

Self Energy:

* It is the amount of work done against the gravitational force in creating a body.

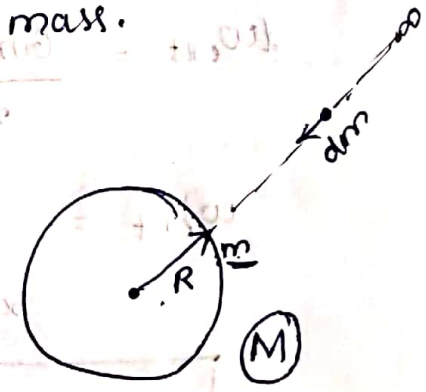
* Self energy is zero for pt mass.

① Self energy of hollow sphere:

In intermediate state,

potential at the surface of sphere,

$$V = -\frac{Gm}{R}$$



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work done in adding a further mass dm on to the shell is,

$$dw_{ext} = V(dm)$$

$$dw_{ext} = -\frac{Gm}{R}(dm)$$

∴ Total work done,

$$w_{ext} = -\frac{G}{R}\int_0^M m\,dm$$

$$= -\frac{G}{R}\times\frac{M^2}{2}$$

$$\boxed{w_{ext} = -\frac{GM^2}{2R}}$$

$$\boxed{U_{self} = -\frac{GM^2}{2R}}$$

② Self energy of Solid Sphere:

ρ = uniform,

* potential at the surface,

$$V = -\frac{Gm}{r}$$

work done in adding a further mass dm.

$$dw_{ext} = (V)\,dm$$

here,

$$m = \frac{4}{3}\pi r^3\rho$$

$$dm = 4\pi r^2 dr\rho$$

$$dw_{ext} = -\frac{Gm}{r}.dm$$

$$= -\frac{Gm}{r}(4\pi r^2\rho\,dr)$$

$$= -G\left(\frac{4}{3}\pi r^3\rho\right)(4\pi r\rho\,dr)$$

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$$dw_{ext} = -G\left(\frac{4}{3}\pi\right)(u\pi)\, s^r r^4 . dr$$

Total work done,

$$w_{ext} = \frac{-G(u\pi)^r}{3}(s^r)\int_{r=0}^{R} r^4 . dr$$

$$= \frac{-G(u\pi)^r}{3}(s^r) \times \left[\frac{r^5}{5}\right]_0^R$$

$$= \frac{-G(u\pi)^r}{3} \times \left(\frac{M}{\frac{4}{3}\pi R^3}\right)^r \times \frac{R^5}{5}$$

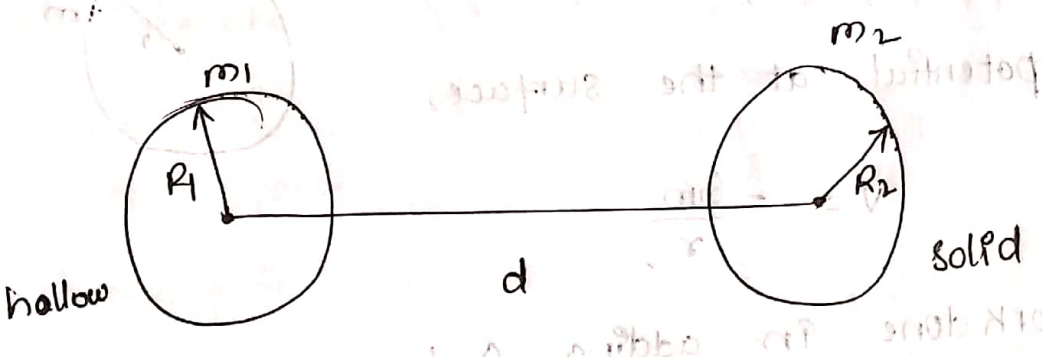
$$\boxed{w_{ext} = \frac{-3GM^r}{5R}}$$

$$\boxed{U_{self} = -\frac{3GM^r}{5R}}$$

* Finally,

$$U_{system} = U_{interaction} + U_{self}$$

Ex:



$$U_{system} = \frac{-Gm_1^r}{2R_1} - \frac{3Gm_2^r}{5R_2} - \frac{Gm_1m_2}{d}$$

Relation b/w $\vec{E}$ & V :

*we have, $\boxed{F_q = \frac{-dU}{dr}}$ ---- @

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* conservative force is negative of gradient of P.E.

$$\bar{F} = \bar{\nabla} U$$

$$F = -\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) U$$

$$\hat{i} F_x + \hat{j} F_y + \hat{k} F_z = -\hat{i}\frac{\partial U}{\partial x} - \hat{j}\frac{\partial U}{\partial y} - \hat{k}\frac{\partial U}{\partial z}$$

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}$$

Ex! $U = 2xy + yz$. Find force,

$$F_x = -\frac{\partial U}{\partial x} = -2y$$

$$F_y = -\frac{\partial U}{\partial y} = -2x - z$$

$$F_z = -\frac{\partial U}{\partial z} = -y$$

$$\therefore \bar{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

* $\boxed{dU = -\bar{F} \cdot d\bar{r}}$ (from eqn a) —(b)

$$\int dU = -\int \bar{F} \cdot d\bar{r}$$

** $\boxed{U_f - U_i = -\int \bar{F} \cdot d\bar{r}}$

* $dU = -m\bar{E} \cdot d\bar{r}$ (from eqn b)

$$\frac{dU}{m} = -\bar{E} \cdot d\bar{r}$$

$$dV = -\bar{E} \cdot d\bar{r}$$

$V = -\frac{Gm_1}{r}$

$U = -\frac{Gm_1m_2}{r} \Rightarrow \boxed{U = Vm_2}$

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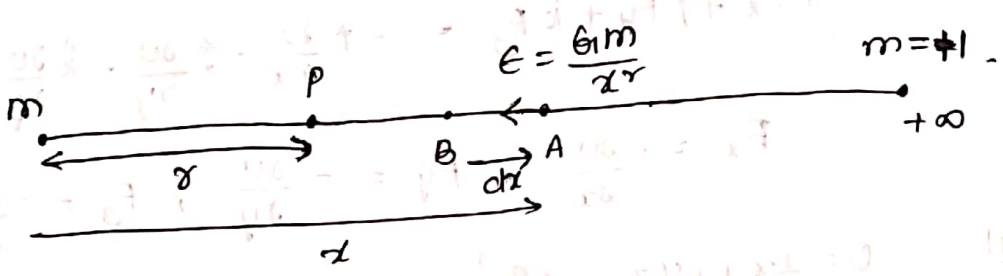
$$\int_{q}^{f} dV = -\int_{i}^{f} \bar{E} \cdot d\bar{r} \quad *$$

$$\Rightarrow \boxed{\bar{E} = \frac{-dV}{dr}}$$

Ex:



$$\int_{V_\infty}^{V_p} dV = -\int_{x=\infty}^{x=r} \frac{Gm}{x^2} dx \cos 180^\circ$$

$$\boxed{V_p - V_\infty = -\frac{Gm}{r}} \quad * \quad \text{(potential difference)}$$

$$\boxed{V_p = V_\infty - \frac{Gm}{r}} \quad *$$

Gravitational potential due to extended bodies:

here we use,

① $dV = \frac{-Gdm}{r}$

$V = \int dV$

(or)

② $\int_{i}^{f} dV = -\int_{q}^{f} \bar{E} \cdot d\bar{r}$

Applications :-

① gravitational potential due to hollow sphere:

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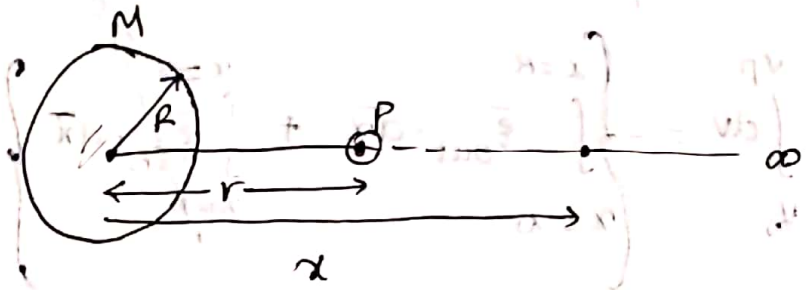
a) pt lies outside the sphere:



$$\int_{V_i}^{V_f} dV = -\int_{i}^{f} \bar{E}\cdot d\bar{r}$$

$$\int_{V_\infty}^{V_p} dV = -\int_{\infty}^{r} \frac{GM}{x^2}\,dx \cos 180^\circ$$

$$V_p - V_\infty = \frac{-GM}{r}$$

$$V_p = -\frac{GM}{r} + V_\infty$$

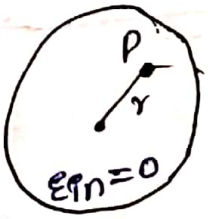
If $V_\infty = 0 \Rightarrow$ $\boxed{V_p = -\frac{GM}{r}}$

b) pt lies on the sphere:

$r = R$, $\boxed{V_s = -\frac{GM}{R}}$

field lines are perpendicular to the equipotential surface.

c) pt lies inside the sphere:



$$E_{out} = \frac{GM}{r^2}$$

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$$\int_i^f dV = -\int_i^f \vec{E} \cdot d\vec{r}$$

$$\int_{V_\infty}^{V_p} dV = -\left\{ \int_{x=\infty}^{x=R} \vec{E}_{out} \cdot d\vec{x} + \int_{x=R}^{x=0} \vec{E}_{in} \cdot d\vec{x} \right\}$$

$$\int_{V_\infty}^{V_p} dV = -\int_{x=\infty}^{x=R} \vec{E}_{out} \cdot d\vec{x}$$

$$V_p - V_\infty = -\frac{GM}{R}$$

$$\boxed{V_p = V_\infty - \frac{GM}{R}}$$

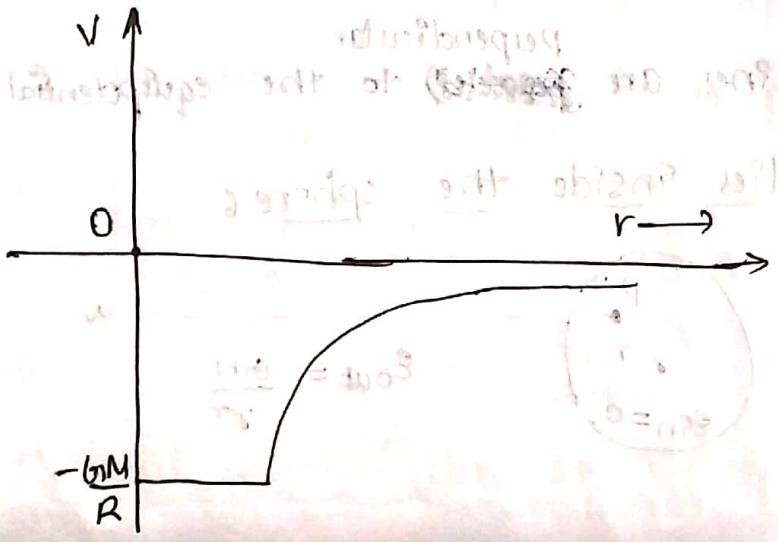
* If $V_\infty = 0 \Rightarrow \boxed{V_p = -\frac{GM}{R}}$

* V is same inside & on the surface.

* on surface - equipotential surface.

  inside - equipotential volume.

Graph:



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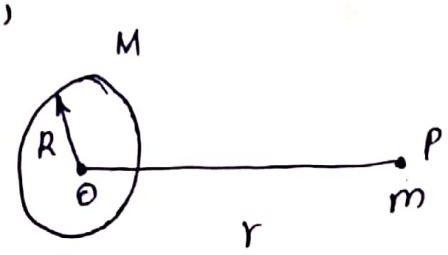
* finally,

1)

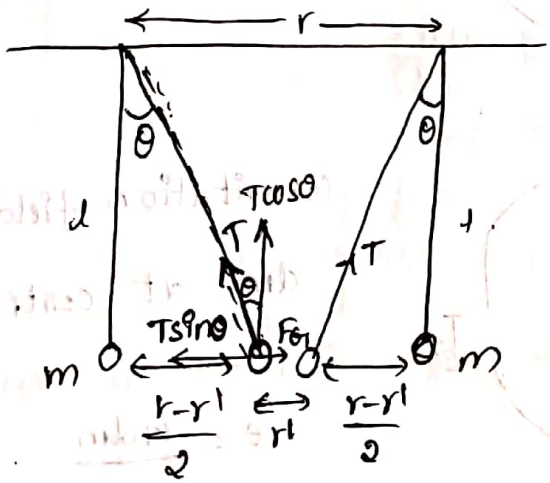


$$U_{system} = -\frac{GM^2}{2R} - \frac{GMm}{r}$$

2)

$$U_{system} = -\frac{GM^2}{2R} - \frac{GMm}{R}$$

Ex: Two balls of mass m each are hung side by side by two long threads of equal length 'l' & the distance blw upper ends is 'R' & distance blw 2 centres after eqm is r'. Find $\frac{q(r')^2(r-r')}{Gml} = ____$.



since, m is in eqm,

$$\Sigma F_x = 0, \quad \Sigma F_y = 0$$

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* $F_G = \frac{GMm}{(r')^2}$

$T\cos\theta = mg$

$T\sin\theta = F_G$

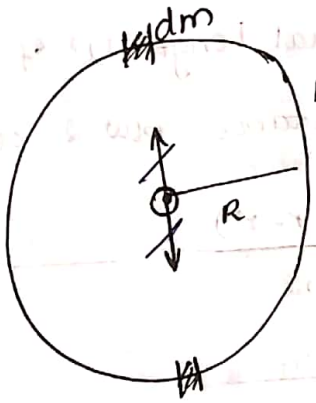
$\tan\theta = \frac{F_G}{mg}$

$\frac{(r-r')}{2l} = \frac{\frac{GMm}{(r')^2}}{mg}$

$$\frac{g(r')^2(r-r')}{GMl} = 2$$

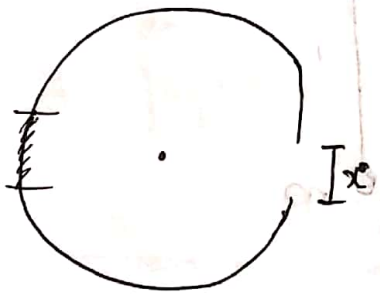
Gravitational Field strength due to ring:

*



$\mathcal{E}_{centre} = 0$

*



Gravitation field due to dm at centre.

$e = \frac{GMdm}{R^2}$

$2\pi R \longrightarrow M$

$1 \longrightarrow \frac{M}{2\pi R}$

$x \longrightarrow \frac{M}{2\pi R}x$

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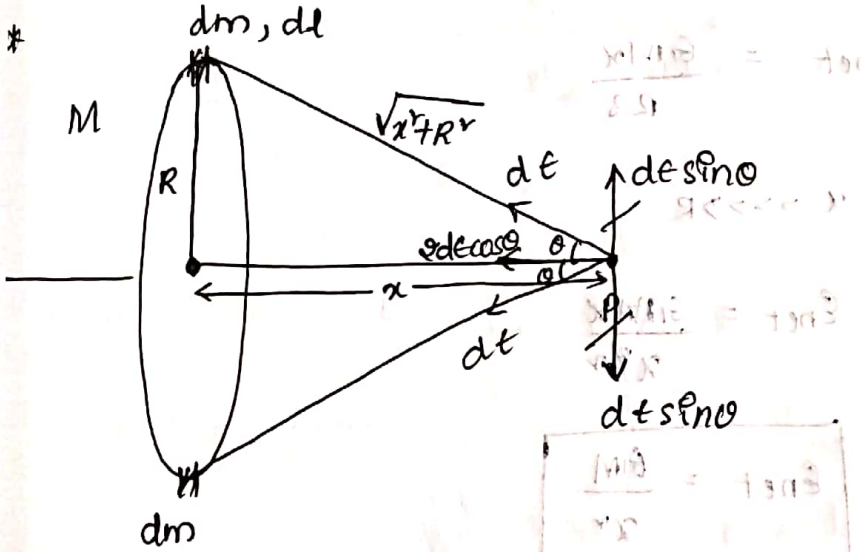
$$E = \frac{Gdm}{R^2}$$

$$E = \frac{G}{R^2}\left(\frac{M\,dl}{2\pi R}\right)$$

* Here $E \neq 0$ because the field due to the elemental part dm which is opposite to the removed part is not cancelled

∴ the field is only due to that elemental part.

On the axis of the ring :



Gravitational field strength at P due to dm element,

$$dE = \frac{Gdm}{(\sqrt{x^2+R^2})^2} = \frac{Gdm}{x^2+R^2}$$

$2\pi R \rightarrow M$

$1 \rightarrow \frac{M}{2\pi R} \Rightarrow dl \rightarrow \frac{M}{2\pi R}dl$

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$$dE = \frac{GM}{2\pi R(x^2+R^2)} \cdot dl$$

$$E_{net} = \int dE \cos\theta$$

$$= \int \frac{GM\, dl}{2\pi R(x^2+R^2)} \cdot \frac{x}{(x^2+R^2)^{1/2}}$$

$$E_{net} = \frac{GMx}{2\pi R(R^2+x^2)^{3/2}} \int dl$$

$$= \frac{GMx(2\pi R)}{(2\pi R)(R^2+x^2)^{3/2}}$$

$$\boxed{E_{net} = \frac{GMx}{(R^2+x^2)^{3/2}}}$$

If $R >>> x$

$$E_{net} = \frac{GMx}{R^3}$$

* If $x >>> R$

$$E_{net} = \frac{GMx}{x^{3/2}}$$

$$\boxed{E_{net} = \frac{GM}{x^2}}$$

Graphs

$x \longrightarrow \infty$, $x \longrightarrow -\infty$, $x = 0$

$E_{net} = 0$ $E_{net} = 0$ $E_{net} = 0$

$$E = \frac{GMx}{(x^2+R^2)^{3/2}} \Rightarrow \frac{dE}{dx} = 0 \quad \frac{d}{dx}\left(\frac{x}{(x^2+R^2)^{3/2}}\right) = 0$$

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$$x \times \frac{3}{2} \times 2(x^2+R^2)^{1/2}(x) - (x^2+R^2)^{3/2} = 0$$

$$3x^2(x^2+R^2)^{1/2} - (x^2+R^2)^{3/2} = 0$$

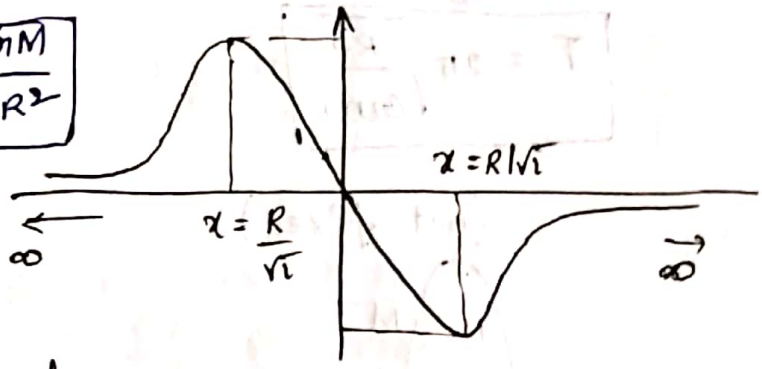
$$3x^2 - x^2 - R^2 = 0$$

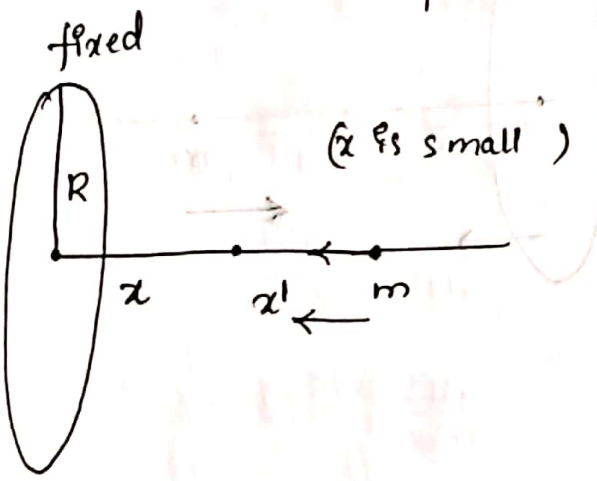
$$2x^2 = R^2$$

$$\boxed{x = \pm R/\sqrt{2}}$$

$$E_{max} = \frac{GM(R/\sqrt{2})}{\left(\frac{3R^2}{2}\right)^{3/2}} = \frac{GMR}{\sqrt{2}\times\left(\frac{3}{2}\right)^{3/2}R^3}$$

$$\boxed{E_{max} = \frac{2}{3^{3/2}}\frac{GM}{R^2}}$$





Find Time period ??

Restoring force on m = $-m \times E_{\text{due to ring at } x}$.

$$R.F = -m\frac{GMx}{(R^2+x^2)^{3/2}}$$

$$ma = -m\frac{GMx}{R^3}$$

$$a = -\frac{GMx}{R^3}$$

$$\boxed{a \propto -x}$$

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So, mass m is doing SHM.

$$\omega^2 = \frac{Gm}{R^3}$$

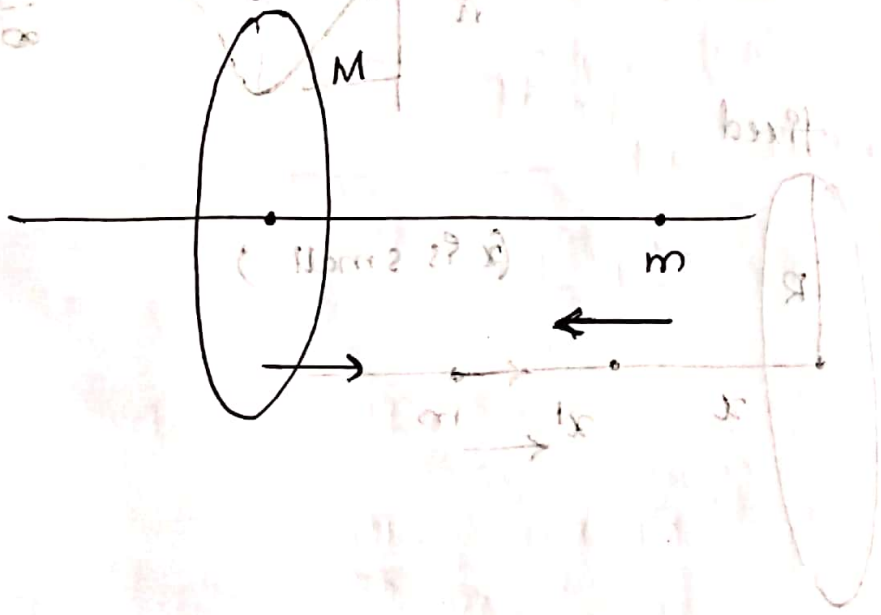
$$\omega = \sqrt{\frac{Gm}{R^3}}$$

$$T = \frac{2\pi}{\omega}$$

$$\boxed{T = 2\pi\sqrt{\frac{R^3}{Gm}}}$$

(not fixed)

* ?



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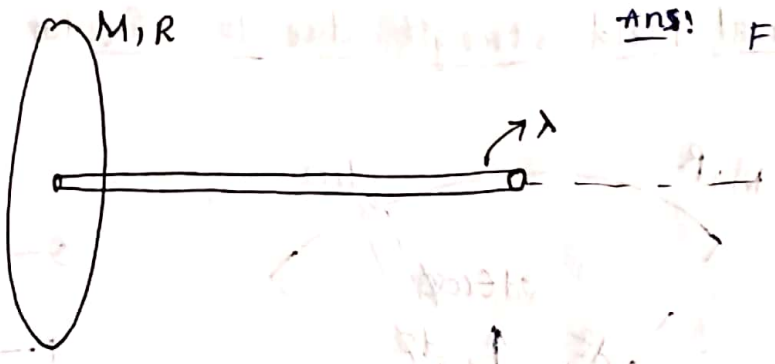
Ex 6



Ans: $F = \frac{GM\lambda}{R}$

Find Force exerted by ring on rod?

Sol^n Force on dm element due to ring,

$$dF = \frac{dm \times GMx}{(R^2+x^2)^{3/2}}$$

$$F_{net} = \int dF = \int_0^\infty \frac{GM\lambda\, x\, dx}{(R^2+x^2)^{3/2}}$$

$$= \frac{GM\lambda}{1} \int_0^\infty \frac{x}{(R^2+x^2)^{3/2}} \cdot dx$$

$$= -\frac{GM\lambda}{R} [\cos\theta]$$

$$= -\frac{GM\lambda}{R} \left[\frac{R}{\sqrt{x^2+R^2}}\right]_0^\infty$$

$$= -\frac{GM\lambda}{R}[0-1]$$

$$= -\frac{GM\lambda}{R}(-1)$$

$x = R\tan\theta$

$dx = R\sec^2\theta \cdot d\theta$

$\int \frac{R\tan\theta \cdot R\sec^2\theta \, d\theta}{R\sec\theta}$

$-\cos\theta$

$$\int dF = \frac{GM\lambda}{R}$$

$$\boxed{F_{net} = \frac{GM\lambda}{R}}$$

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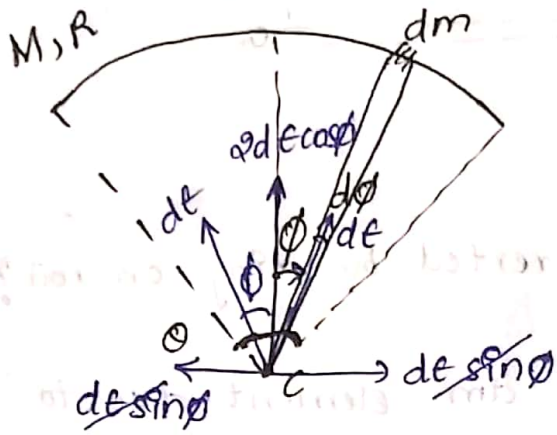
Gravitational Field strength due to circular arc:



$\theta \rightarrow M$

$1 \rightarrow \frac{M}{\theta}$

$d\phi \rightarrow \frac{M}{\theta} \cdot d\phi = dm$

Gravitational field strength due to 'dm' at 'c'

$$d\epsilon = \frac{G\,dm}{R^2}$$

$$d\epsilon = \frac{G\left[\frac{M}{\theta}\right] d\phi}{R^2}$$

$$d\epsilon = \frac{GM}{R^2\theta}\, d\phi$$

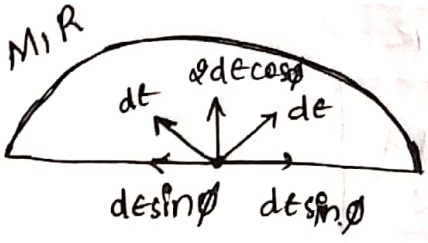
$$\varepsilon_{net} = \int d\epsilon \cos\phi$$

$$= \frac{GM}{R^2\theta} \int_{-\theta/2}^{\theta/2} \cos\phi \cdot d\phi$$

$$\boxed{\varepsilon_{net} = \frac{2GM \sin\theta/2}{R^2\theta}} \quad (\uparrow)$$

* θ should be in radians.

*



semicircular arc.

$\theta = \pi$

$$\boxed{\varepsilon_{net} = \frac{2GM}{\pi R^2}}$$

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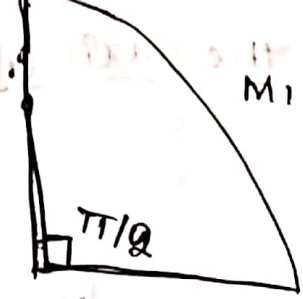


M-I

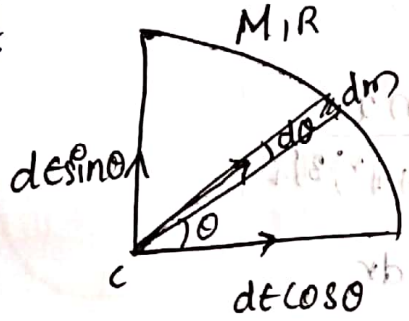
$$\mathcal{E}_x = \mathcal{E}_{net} \cos\pi/4$$

$$\mathcal{E}_y = \mathcal{E}_{net} \sin\pi/4$$

$$\mathcal{E} = \sqrt{\mathcal{E}_x^2 + \mathcal{E}_y^2} = \sqrt{2}\,\mathcal{E}_{net}$$

$$\boxed{\mathcal{E} = \frac{2\sqrt{2}GM}{R^2\pi}}$$

M-II



$$\mathcal{E}_x = \int dE\cos\theta$$

$$\mathcal{E}_y = \int dE \sin\theta$$

$$\mathcal{E}_{net} = \frac{2\sqrt{2}GM}{R^2\pi}$$

$$\mathcal{E}_{net} = \frac{2GM(3M)\sin(30^\circ)}{R^2(\pi/3)}$$

$$\boxed{\mathcal{E}_{net} = \frac{9GMm}{R^2\pi}}$$

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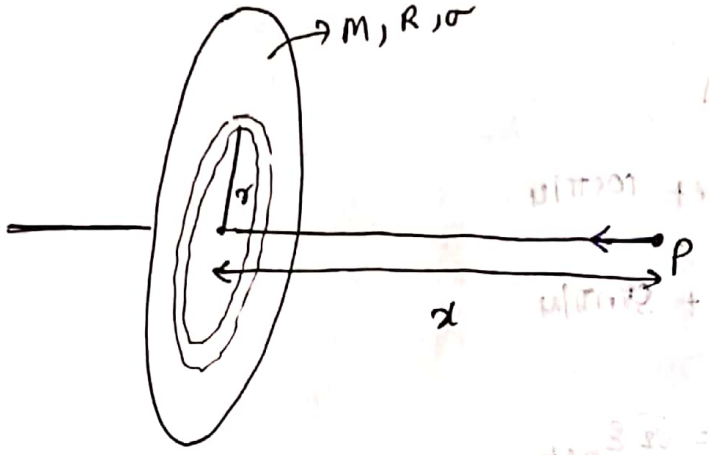
Gravitational field strength on the axis of disc



$dm = \sigma \times dA$

$= \sigma (2\pi r) \cdot dr$

Gravitational field strength due to ring element at point 'P',

$$dE = \frac{G\,dm\,x}{(r^2+x^2)^{3/2}}$$

$$E_{net} = \int dE = \int \frac{G\,dm\,x}{(r^2+x^2)^{3/2}}$$

$$= \int \frac{Gx \cdot \sigma \cdot 2\pi r \cdot dr}{(r^2+x^2)^{3/2}}$$

$$= 2\pi G x \sigma \int_0^R \frac{r \cdot dr}{(r^2+x^2)^{3/2}}$$

$$E_{net} = -2\pi G x \sigma \left[\frac{x}{\sqrt{r^2+x^2}}\right]_0^R$$

$$= -\frac{2\pi G x \sigma}{x}\left[\frac{x}{\sqrt{R^2+x^2}} - \frac{x}{\sqrt{x^2}}\right]$$

$$= -2\pi G x \sigma \left[\frac{x}{\sqrt{R^2+x^2}} - 1\right]$$

$$\boxed{E_{net} = 2\pi G x \sigma \left[1 - \frac{x}{\sqrt{R^2+x^2}}\right]}$$

$$= 2\pi G \frac{M}{\pi R^2}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right] \Rightarrow \boxed{E_{net} = \frac{2GM}{R^2}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]}$$

let $r = x\tan\theta$

$dr = x\sec^2\theta \cdot d\theta$

$$\frac{\sin\theta}{x^2\tan\theta \cdot x\sec^2\theta \cdot d\theta}$$

$$\frac{1}{x}\int \sin\theta \cdot d\theta$$

$$= -\frac{\cos\theta}{x}$$

$$= -\frac{x}{\sqrt{r^2+x^2}} \times \frac{1}{x}$$

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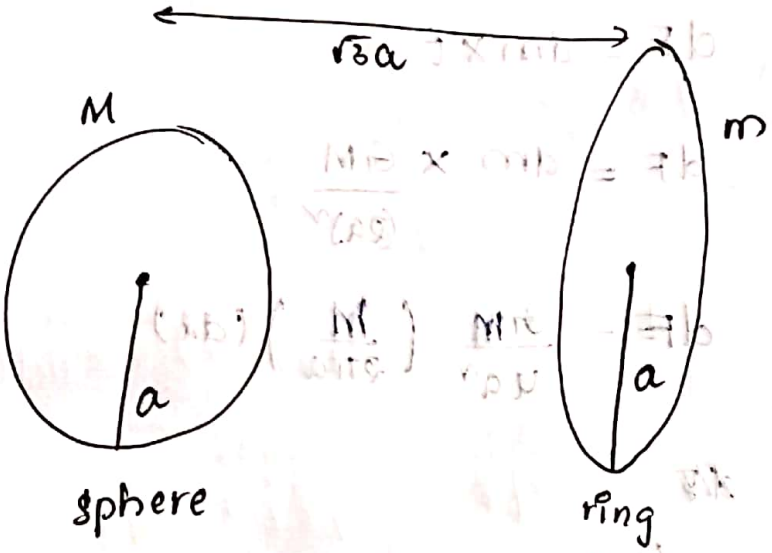
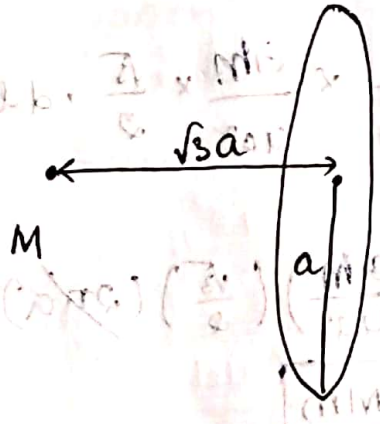
placed on mass M as shown in figure.

Ex! A uniform ring of mass m & radius a is placed on a sphere of mass M and equal radius. The centre of ring is at a distance of $\sqrt{3}a$ from the centre of sphere. Find the gravitational force exerted by sphere on ring.



Sol:



Force on ring due to M = Force on M due to ring.

$$F = M \times f$$

$$F = M \times \frac{Gm(\sqrt{3}a)}{(a^2 + 3a^2)^{3/2}}$$

$$= \frac{\sqrt{3}GMm}{8a^3} = \frac{\sqrt{3}}{8}\frac{GMm}{a^2}$$

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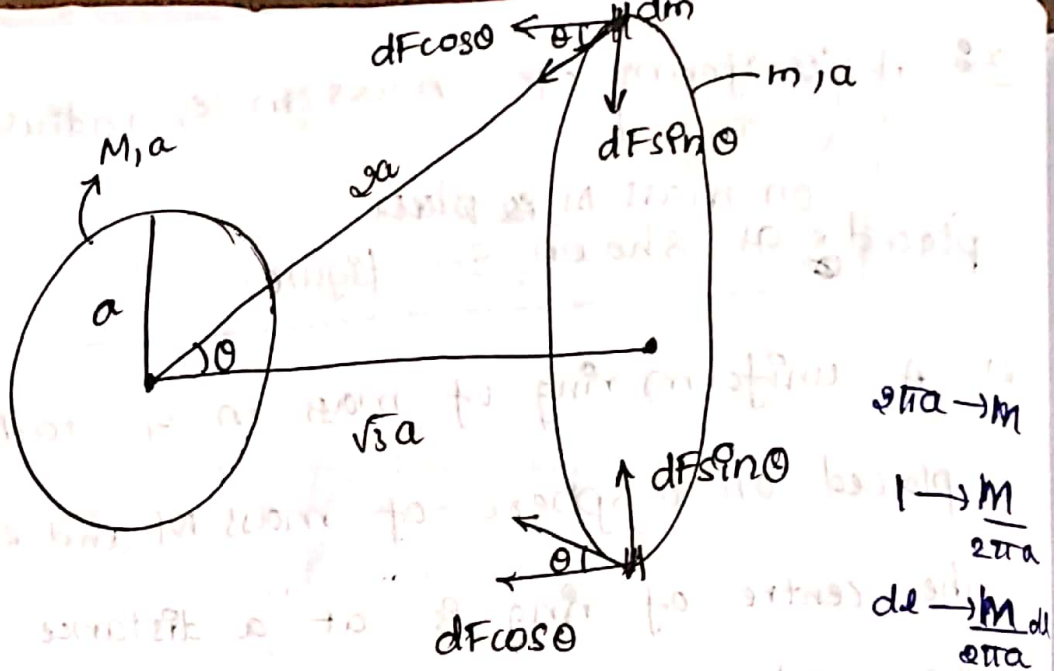
M-II 9.



Force on dm element due to sphere,

$$dF = dm \times f$$

$$dF = dm \times \frac{GM}{(2a)^2}$$

$$dF = \frac{GM}{4a^2}\left(\frac{M}{2\pi a}\right)(dl)$$

$$F_{net} = \int dF\cos\theta$$

$$= \int \frac{M}{2\pi a} \times \frac{GMm}{4a^2} \times \frac{\sqrt{3}}{2} \cdot dl$$

$$= \frac{M}{2\pi a}\left(\frac{GMm}{4a^2}\right)\left(\frac{\sqrt{3}}{2}\right)(2\pi a)$$

$$\boxed{F_{net} = \frac{\sqrt{3}\,GMm}{8a^2}}$$

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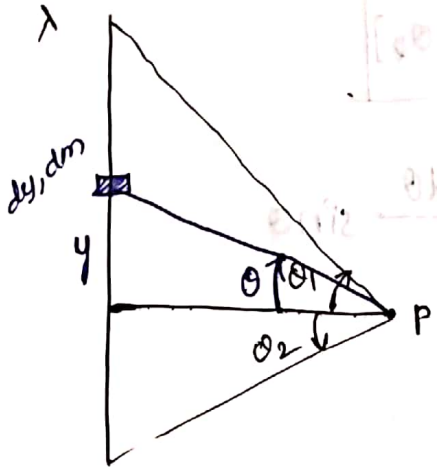
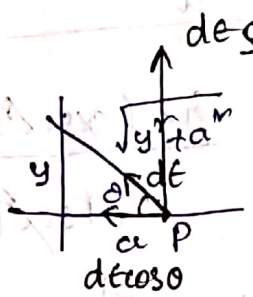
26/12/24 | Thursday

Gravitational Field Intensity at a pt due to finite rod :





$$\cos\theta = \frac{a}{\sqrt{y^2+a^2}}$$

$$\sin\theta = \frac{y}{\sqrt{y^2+a^2}}$$

* Gravitational field intensity at P due to dm.

$$dE = \frac{G\,dm}{(\sqrt{y^2+a^2})^2} = \frac{G\,dm}{y^2+a^2}$$

$$dE = \frac{G\lambda\,dy}{y^2+a^2}$$

* The component which is ⊥ler to rod,

$$E_{\perp ler} = \int dE\cos\theta$$

* The component which is ∥el to rod,

$$E_{\parallel el} = \int dE\sec\theta$$

$$\tan\theta = y/a$$

$$y = a\tan\theta$$

$$dy = a\sec^2\theta\cdot d\theta$$

$$\Rightarrow E_{\perp ler} = \int \frac{G\lambda\, a\sec^2\theta\cdot d\theta}{a^2\sec^2\theta}\cdot\cos\theta$$

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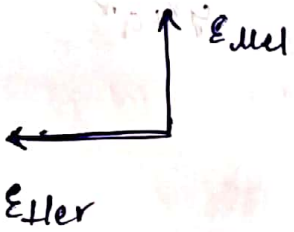
$$= \frac{G\lambda}{a} \int_{-\theta_2}^{\theta_1} \cos\theta \cdot d\theta$$

$$\boxed{E_{\perp ler} = \frac{G\lambda}{a}[\sin\theta_1 + \sin\theta_2]}$$

$$E_{mel} = \int \frac{G\lambda (a\sec^2\theta)\cdot d\theta}{a\sec\theta} \sin\theta$$

$$= \frac{G\lambda}{a} \int_{-\theta_2}^{\theta_1} \sin\theta \cdot d\theta$$

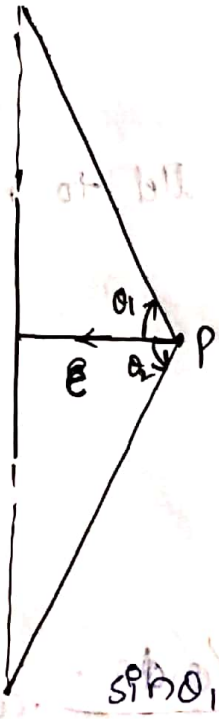
$$\boxed{E_{mel} = \frac{G\lambda}{a}[\cos\theta_2 - \cos\theta_1]}$$

* 

$$\vec{E} = E_{\perp ler}(-\uparrow) + E_{mel}(\uparrow)$$

$$\boxed{|E| = \sqrt{E_{\perp}^2 + E_{mel}^2}}$$

for infinite rod | wire | thread :

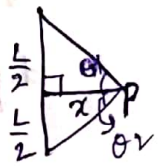


$\theta_1 = 90°$, $\theta_2 = 90°$.

$$\boxed{E_{mel} = 0}$$

$$\boxed{E_{\perp ler} = \frac{2G\lambda}{a}}$$

* at equator pt, $\theta_1 = \theta_2$



$$\sin\theta_1 = \frac{L/2}{\frac{1}{2}\sqrt{4x^2+L^2}}$$

$$E_{\perp ler} = \frac{2G\lambda\sin\theta_1}{x}$$

$$E_{mel} = 0$$

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for semi infinite wire :

$\theta_1 = 90°, \theta_2 = 0°$

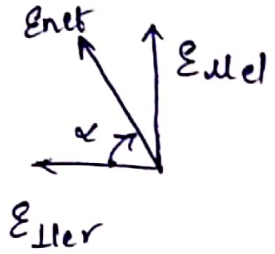
$E_{\parallel el} = \frac{G\lambda}{a}$

$E_{\perp ler} = \frac{G\lambda}{a}$

$\bar{E} = E_{\perp}(-\uparrow) + E_{\parallel el}(\uparrow)$

$$|\bar{E}| = \frac{\sqrt{2}G\lambda}{a}$$

*

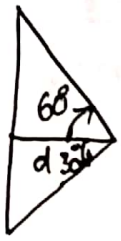


$\tan\alpha = \frac{E_{\parallel el}}{E_{\perp ler}} = 1$

$\alpha = 45°$

***

1)

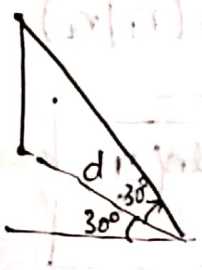


$a = d$

$\theta_1 = 60°$

$\theta_2 = 30°$.

2)

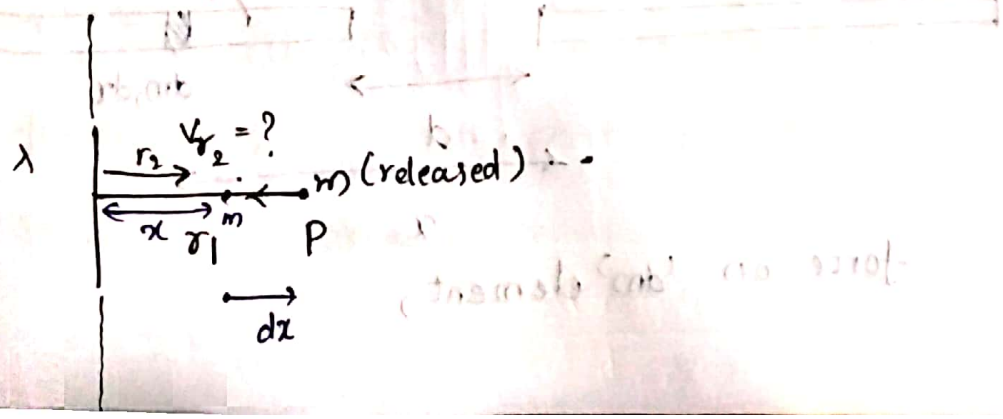


$a = d\cos 30°$

$\theta_1 = 60°$

$\theta_2 = -30°$ (or) $330°$.

Ex:



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Sol:

$$E = \frac{2G\lambda}{x}$$

$$F = \frac{2G\lambda m}{x}$$

$$\frac{F}{m} = \frac{2G\lambda}{x}$$

$$a = \frac{2G\lambda}{x}$$

$$-v\cdot\frac{dv}{dx} = \frac{2G\lambda}{x}$$

$$-v\cdot dv = 2G\lambda\,\frac{dx}{x}$$

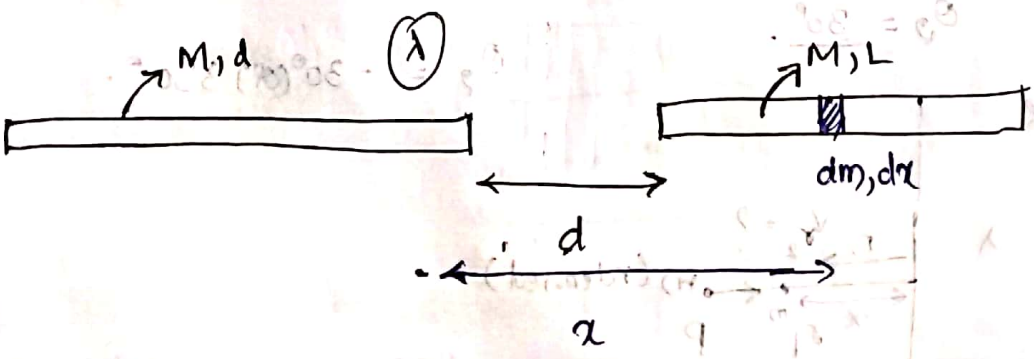
$$\int v\cdot dv = 2G\lambda \int -\frac{1}{x}\cdot dx$$

$$\int_{v}\cdot dv = 2G\lambda \int_{r_1}^{r_2} -x^{-1}\cdot dx$$

$$\frac{v^2}{2} = 2G\lambda \ln(r_1/r_2)$$

$$\boxed{v = 2\sqrt{G\lambda \ln\left(\frac{r_1}{r_2}\right)}}$$

Ex: Find gravitational force of attraction b/w rods.



Sol:

force on 'dm' element,

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$$dF = dm \times \frac{GM}{x(x+l)}$$

$$dF = \lambda dx \times \frac{GM}{x(x+l)}$$

$$F = GM\lambda \int_{d}^{d+l} \frac{dx}{x(x+l)}$$

$$F = \frac{GM^2}{L} \int_{d}^{d+l} \frac{dx}{x(x+l)}$$

$$= \frac{GM^2}{L^2} \int_{d}^{d+l} \left[\frac{1}{x} - \frac{1}{x+l}\right] dx$$

$$= \frac{GM^2}{L^2} \left[\ln\left(\frac{d+l}{d}\right) - \ln\left(\frac{d+2l}{d+l}\right)\right]$$

$$= \frac{GM^2}{L^2} \left[\ln\left(\frac{d+l}{d} \times \frac{d+l}{d+2l}\right)\right]$$

$$\boxed{F = \frac{GM^2}{d^2} \ln\left(\frac{(d+l)^2}{d(d+2l)}\right)}$$

Ex1 A spherically symmetric gravitational system of particles have mass density $\rho = \rho_0$ for $r \le R$

$\rho = 0$ for $r > R$

where $\rho_0 = \text{const}$ at test mass m_0 can go circular motion under the influence of gravitation -al force of particles. Calculate its speed v as a function of r, from centre of sphere & draw graph of speed vs r.

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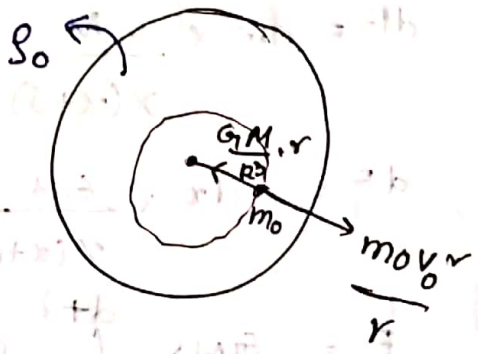
Sol:



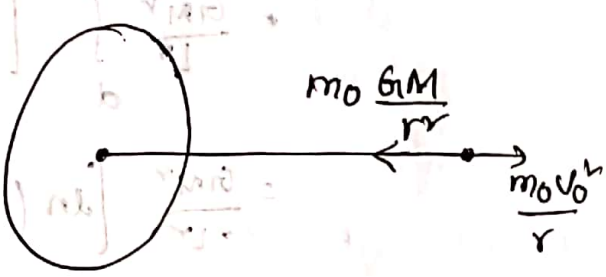
for $r \leq R$!

$$m_0 \times \frac{GM}{R^3} \times r = \frac{m_0 v_0^2}{r}$$

$$\frac{GMr^2}{R^3} = v_0^2$$

$$\boxed{V_0 = K r}$$

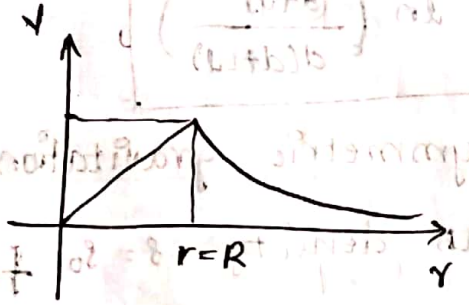
for $r \geq R$!



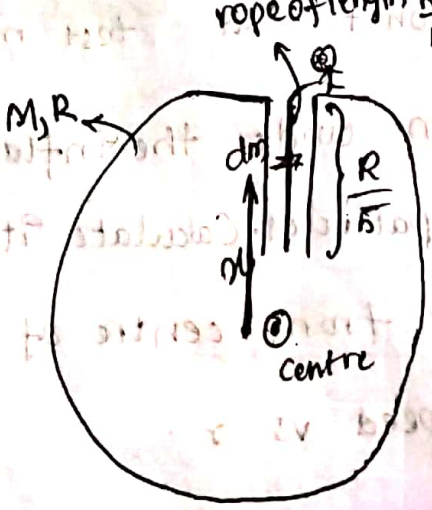
$$m_0 \frac{GM}{r^2} = \frac{m_0 v_0^2}{r}$$

$$\boxed{V_0 = K \frac{1}{\sqrt{r}}}$$

Graph:



Ex: Find force required to hold the rope at the top.



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2016 M-I:

Force on dm element,

$$dF = dm \times \frac{GM}{R^3} \times$$

$$dF = \left(\frac{M}{R/5}\right) \frac{GM}{R^3} \cdot x \cdot dx$$

$$F_{net} = \int dF = \frac{5M}{R^4} \times GM \int_{\frac{4R}{5}}^{R} x\,dx$$

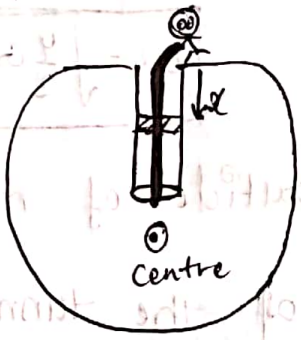
$$F_{net} = \frac{5GMm}{R^4} \times \frac{1}{2}\left[R^2 - \left(\frac{4R}{5}\right)^2\right]$$

$$= \frac{5GMm}{2R^4} \times \frac{9R^2}{25}$$

$$\boxed{F_{net} = \frac{9GMm}{10R^2}}$$

M-II:

force on dm element,



$$mgl = dF$$

$$= dm \times g\left(1 - \frac{x}{R}\right)$$

$$dF = \left(\frac{5m}{R}\right) g\left(1 - \frac{x}{R}\right) \cdot dx$$

$$F_{net} = \int dF = \frac{GM}{R^2}\left(\frac{5m}{R}\right) \int_0^{R/5} \left(1 - \frac{x}{R}\right) \cdot dx$$

$$F_{net} = \frac{5GMm}{R^3}\left[\frac{R}{5} - \frac{R^2/25}{2R}\right]$$

$$= \frac{5GMm}{R^2}\left(\frac{9R}{50}\right) \Rightarrow \boxed{F_{net} = \frac{9GMm}{10R^2}}$$

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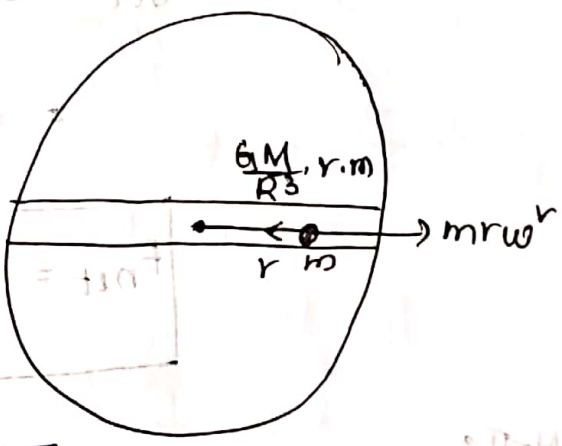
Ex: A straight smooth tunnel is dug through a spherical planet whose mass density ρ_0 is constant. The tunnel passes through centre of planet & other to planet's axis of rotation. The planet rotates with angular velocity ω so that object in tunnel have no acceleration. Find ω.

Ans: $\omega = \sqrt{\frac{4}{3}G\pi\rho_0}$.

Sol:



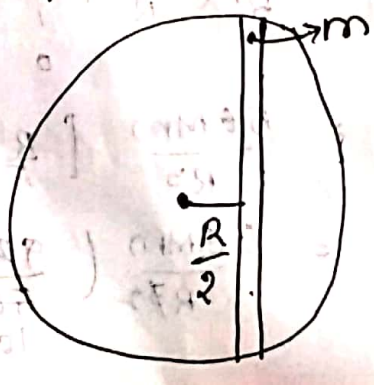
$$\frac{GM}{R^3} \cdot r \cdot m = m r \omega^2$$

$$\frac{G \times \frac{4}{3}\pi R^3 \rho_0}{R^3} = \omega^2$$

$$\boxed{\omega = \sqrt{\frac{4}{3}G\pi\rho_0}}$$

Ex: A particle of mass m released at one end of the tunnel. The pressing force by the particle on the wall & the acceleration of the particle varies with x where x is the distance of particle from centre of earth.

Sol:



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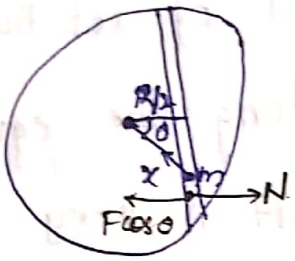
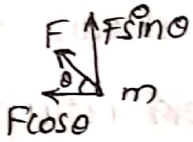
Force on m

$$= \left(\frac{GM}{R^3}x\right)m$$



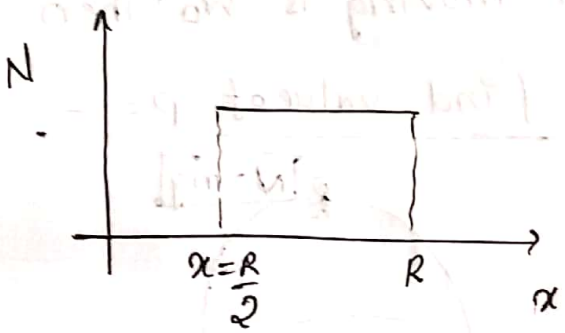
* pressing force,



$$N = F\cos\theta$$

$$= \frac{GMm}{R^3} \times x \times \frac{R}{2x}$$

$$\boxed{N = \frac{GMm}{2R^2}}$$

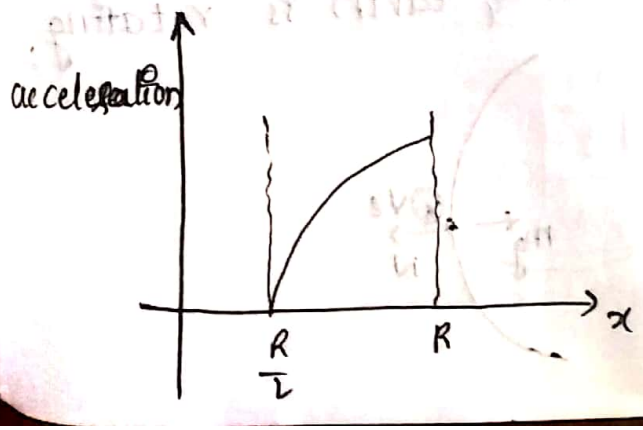


* acceleration,

f_{net} along tunnel $= F\sin\theta$

$$ma = \frac{GMm}{R^3} \times \sqrt{x^2 - \frac{R^2}{4}}$$

$$\boxed{a = \frac{GM}{R^3}\sqrt{x^2 - \frac{R^2}{4}}}$$



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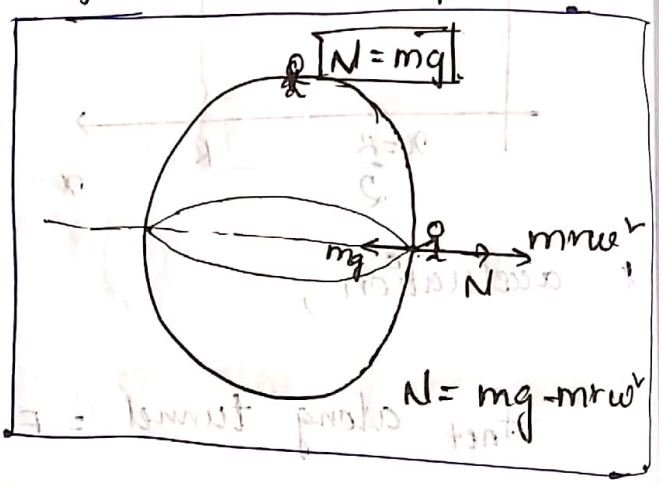
Ex: Angular speed of rotation is ω_0, the train is running along the equator with speed v from west to east. A very sensitive balance inside the train shows the wt of object as W_1 during the return journey, the train running with same speed from east to west, the balance now shows the reading W_2. wt of object shown by the balance when earth is not rotating & train is not moving is W_0. Then $|W_2 - W_1| = \frac{W_0 \omega_0 v}{Rg}$, then find value of $p = __$.

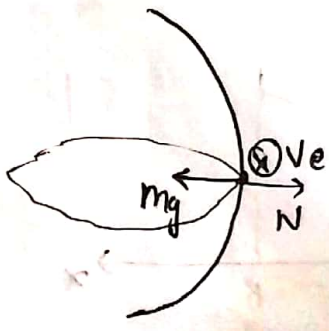
Sol:



* when train is not moving & earth is not rotating,

$$W_0 = mg \Rightarrow m = \frac{W_0}{g}$$

* when train is at rest & earth is rotating,



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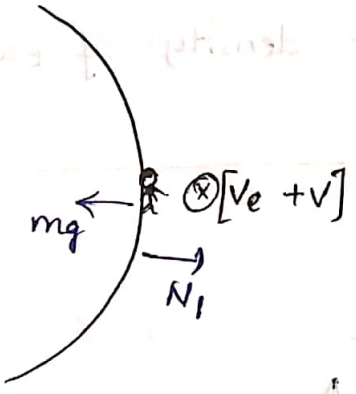
$$W' = N = mg - \frac{mv_e^2}{r} \quad \text{---} ①$$

* when train is moving from west to east.



$$W_1 = N_1 = mg - \frac{m[v_e+V]^2}{r}$$

* when train is moving from east to west.

$$W_2 = mg - \frac{m(v_e-V)^2}{r}$$

$$W_2 - W_1 = -\frac{m}{r}\left((v_e-V)^2 - (v_e+V)^2\right)$$

$$= +\frac{m}{r}(+4v_eV)$$

$$= \frac{4mv_eV}{r}$$

$$= \frac{4mr\omega_0 V}{r}$$

$$W_2 - W_1 = 4m\omega_0 V$$

$$W_2 - W_1 = \frac{4W_0}{g}\omega_0 V$$

$$= \frac{W_0\omega_0 V}{0.25g}$$

$$\boxed{P = 0.25}$$

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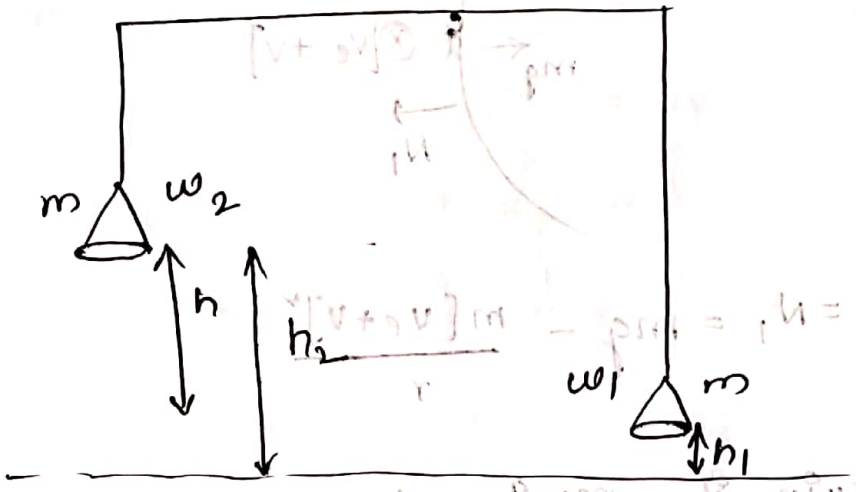
Ex! Two wts of equal masses m & m are hung from a balance whose scale pans differ in vertical height by h. Calculate error in weighing in terms of density of earth ρ.

Sol:



$$\text{error} = w_1 - w_2$$

$$= mg\left(1 - \frac{2h_1}{R}\right) - mg\left(1 - \frac{2h_2}{R}\right)$$

$$= -\frac{2mgh_1}{R} + \frac{2mgh_2}{R}$$

$$= \frac{2mg}{R}(h_2 - h_1)$$

$$= \frac{2mgh}{R}$$

$$= \frac{2mGMh}{R^3}$$

$$= \frac{2mG \times \rho \times \frac{4}{3}\pi R^3 h}{R^3}$$

$$\boxed{\text{error} = \frac{8}{3}\pi mGh\rho}$$

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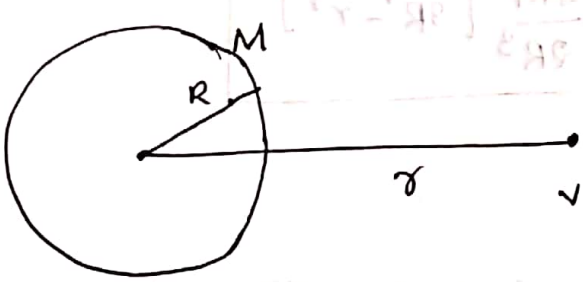
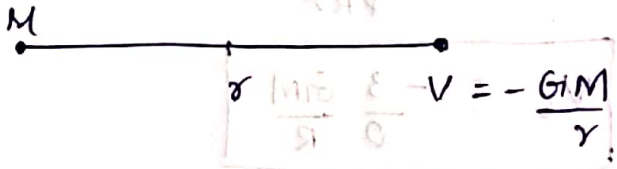
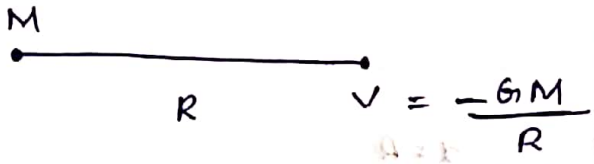
* Potential due to solid Sphere :

(1) Out side point :

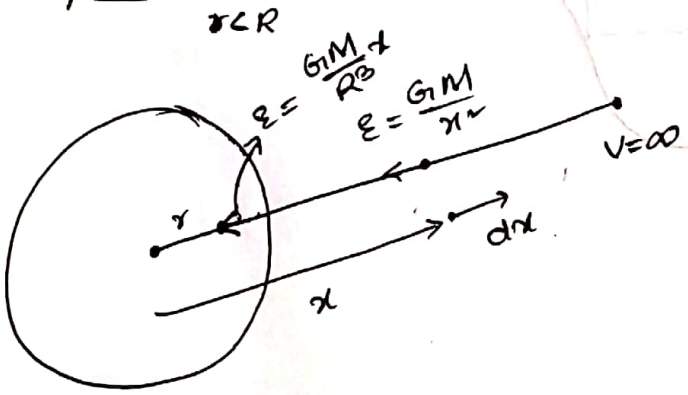




(2) On the surface :



(3) Inside point :



$$\int dV = -\int \vec{E} \cdot d\vec{x}$$

$$V_p - V_\infty = \int_\infty^R \frac{GM}{x^2} \cdot dx + \int_R^r \frac{GM}{R^3} x \cdot dx$$

$$= GM \left[\frac{-1}{x}\right]_\infty^R + \frac{GM}{2R^3} [x^2]_R^r$$

$$= -GM \left[\frac{1}{R}\right] + \frac{GM}{2R^3} [r^2 - R^2]$$

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$$V_p - \cancel{V_\infty}^{0} = \frac{-3GMR^2}{2R^3} + \frac{GM}{2R^3}[r^2 - R^2]$$

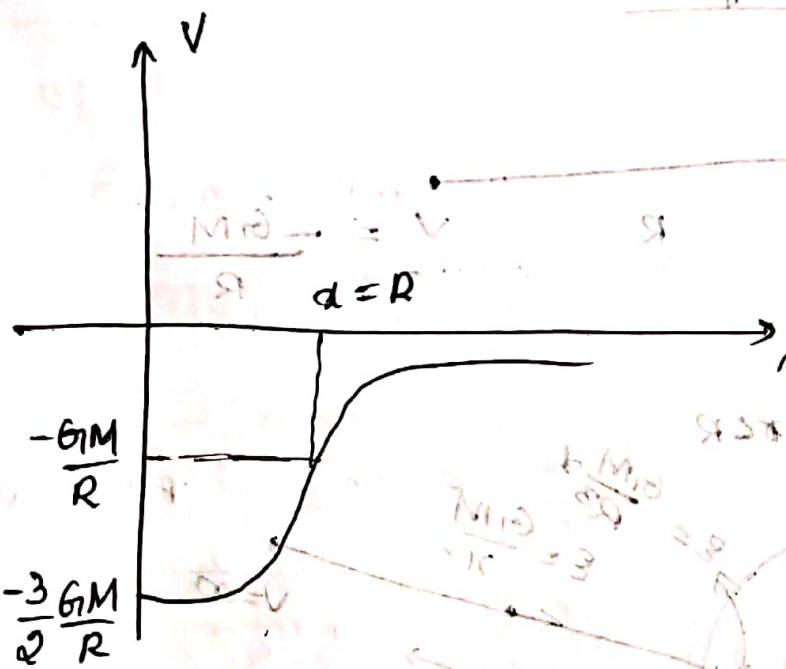
$$\boxed{V_p = -\frac{GM}{2R^3}[3R^2 - r^2]}$$

* at centre!

$$V_c = -\frac{GM}{2R^3}(3R^2)$$

$$\boxed{V_c = -\frac{3}{2}\frac{GM}{R}}$$

Graph:



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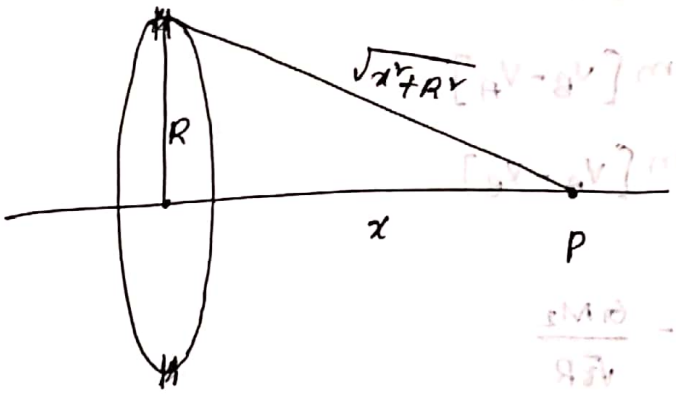
Potential due to ring on the axis :



Potential at 'p' due to 'dm'.

$$dV = -\frac{G\,dm}{\sqrt{x^2+R^2}}$$

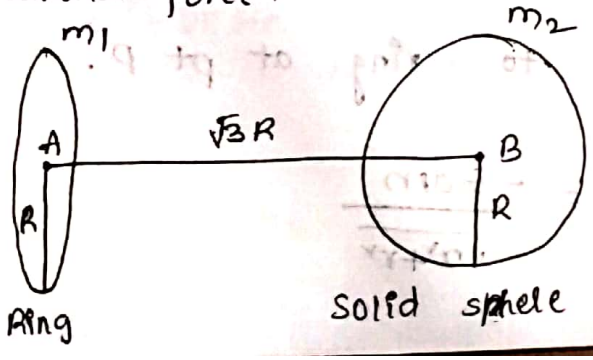
$$\int dV = \frac{-G}{\sqrt{x^2+R^2}} \int dm$$

$$\boxed{V = -\frac{GM}{\sqrt{x^2+R^2}}}$$

at centre, $x = 0$

$$\boxed{V = -\frac{GM}{R}}$$

Ex: Figure shows a ring of mass m_1 & sphere of mass m_2 sapeated by $\sqrt{3}R$. A small object of mass m_1 is displaced from A to B. Find the work done by gravitational force.



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$$W_g = -\Delta U$$

$$= -m[V_F - V_I]$$

$$= -m[V_B - V_A]$$

$$W_g = m[V_A - V_B]$$

$$V_A = -\frac{GM_1}{R} - \frac{GM_2}{\sqrt{3}R}$$

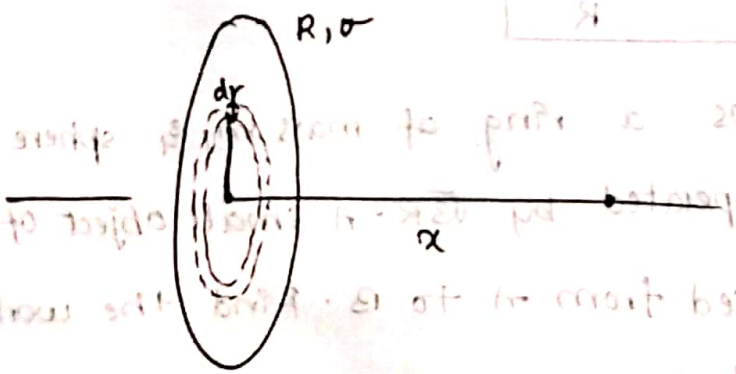
$$V_B = -\frac{GM_1}{2R} - \frac{3}{2}\frac{GM_2}{R}$$

$$W_g = -m\left[\frac{GM_1}{R} + \frac{GM_2}{\sqrt{3}R} - \frac{GM_1}{2R} - \frac{3}{2}\frac{GM_2}{R}\right]$$

$$W_g = -\frac{mG}{R}\left[\frac{M_1}{2} + M_2\left(\frac{1}{\sqrt{3}} - \frac{3}{2}\right)\right]$$

$$\boxed{W_g = -\frac{mGM_1}{2R} - \frac{GM_2 m(2-3\sqrt{3})}{2\sqrt{3}}}$$

Potential due to disc on the axis



G.P due to ring at pt P.

$$dV = -\frac{G\,dm}{\sqrt{x^2+r^2}}$$

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$$dm = (2\pi r)\sigma \cdot dr$$

$$dV = \frac{-G\sigma(2\pi r)\cdot dr}{\sqrt{x^2+r^2}}$$

$$V_p = \int dV = -2\pi G\sigma \int_0^R \frac{r}{\sqrt{x^2+r^2}} \cdot dr$$

~~$r = x\tan\theta$~~

~~$dr = x\sec^2\theta\, d\theta$~~ ~~$V_p = -2\pi G\sigma \int_0^R \frac{x\tan\theta}{x\sec\theta} \cdot (x\sec^2\theta \cdot d\theta)$~~

~~$= -2\pi G\sigma x \int_0^R \sin\theta(\sec^2\theta)\cdot d\theta$~~

$$V_p = -2\pi G\sigma \int_0^R \frac{t\cdot dt}{t}$$

$x^2+r^2 = t^2$

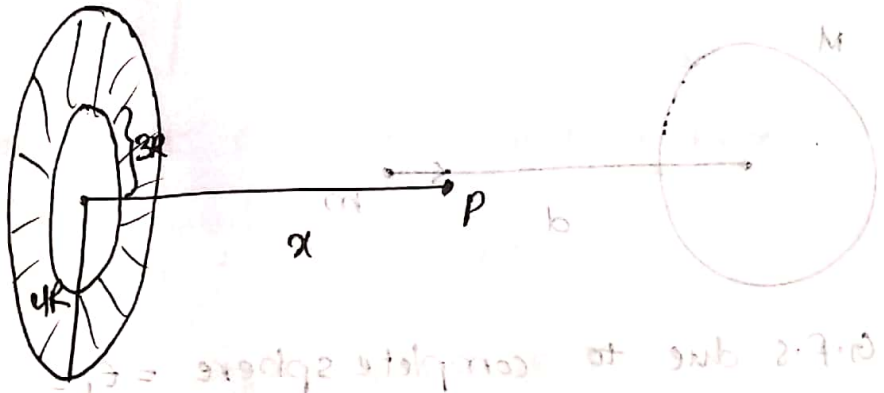
$2r\cdot dr = 2t\cdot dt$

$$= -2\pi G\sigma \left[\sqrt{x^2+r^2}\right]_0^R$$

$$= -2\pi G\sigma \left[\sqrt{x^2+R^2} - \sqrt{x^2}\right]$$

$$\boxed{V_p = -2\pi G\sigma \left[\sqrt{x^2+R^2} - x\right]}$$

Ex6



$$V_p = -2\pi G\sigma \int \frac{r}{x^2+r^2}$$

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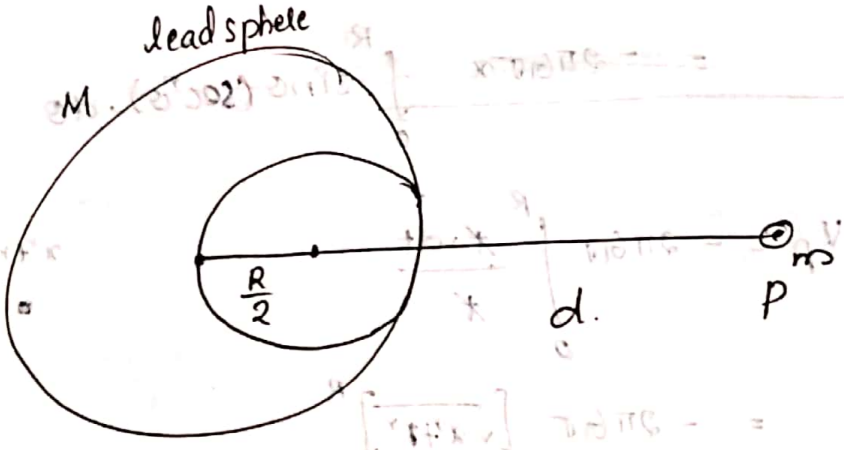
Cavity Problems:

Ex: A spherical hollow is made in a lead sphere of radius R such that it's surface touches the outer surface & passes through centre. The mass of the sphere before hollowing is M. With what gravitational force the hollowed lead sphere attracts a small mass m which lies at a distance d from centre of lead sphere.

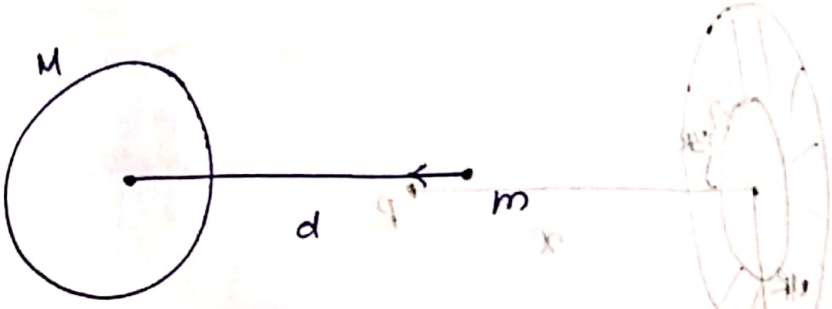
Sol:



* At pt P,

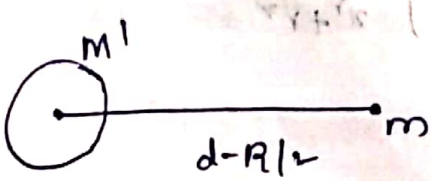
Gravitational field strength of remaining pt = Total − cavity

*



G.F.S due to complete sphere $= e_1 = \frac{GM}{d^2}$

*



$$e_2 = \frac{GM'}{(d-R/2)^2}$$

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at pt P,

$\epsilon_1 \longleftarrow \bullet \longrightarrow \epsilon_2$

$M' = S \times \frac{4}{3}\pi\left(\frac{R}{2}\right)^3$

$= S \times \frac{4}{3}\pi R^3 \times \frac{1}{8}$

$\boxed{M' = M/8}$

$\mathcal{E}_{net} = \mathcal{E}_1 - \mathcal{E}_2$

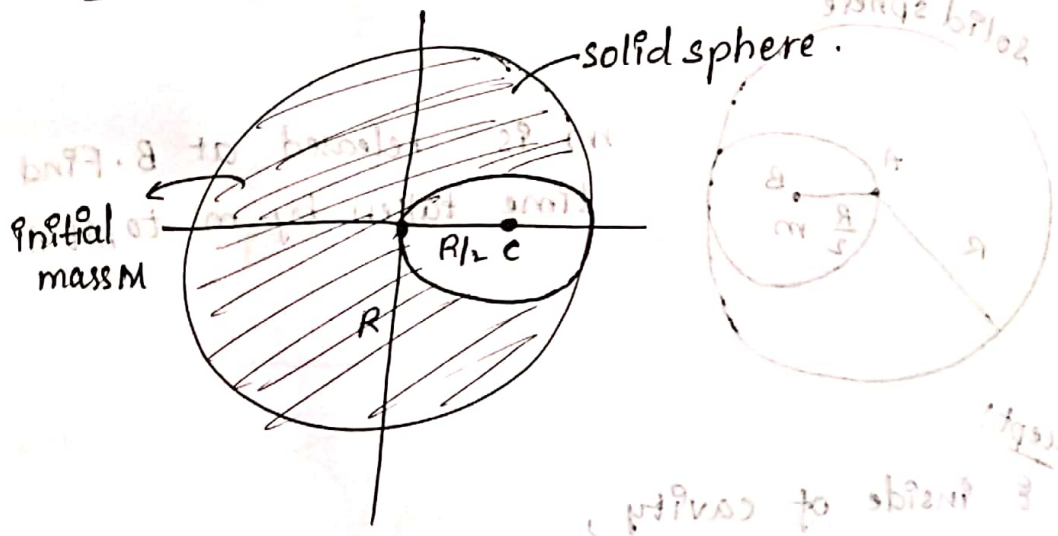
$= \frac{GM}{d^2} - \frac{G(M/8)}{(d-R/2)^2}$

$$\boxed{\mathcal{E}_{net} = GM\left(\frac{1}{d^2} - \frac{1}{8(d-R/2)^2}\right)}$$

force on m $= m \times \mathcal{E}_{net}$

$= GMm\left(\frac{1}{d^2} - \frac{1}{8(d-R/2)^2}\right)$

Eg: Find the potential at the centre of cavity?



potential at C for total sphere,

$$V_T = -\frac{GM}{2R^3}\left[3R^2 - \frac{R^2}{4}\right]$$

$$= -\frac{GM}{2R^3}\left[\frac{11R^2}{4}\right]$$

$$V_T = \frac{-11GM}{8R}$$

potential at C for cavity,

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$$V_C = -\frac{3}{2}\frac{GM/8}{R/2}$$

$$= -\frac{3GM}{16R/8}$$

$$= -\frac{3GM}{8R}$$

$$V_{remaining} = -\frac{11GM}{8R} + \frac{3GM}{8R}$$

$$= \frac{-8GM}{8R}$$

$$\boxed{V_{rem} = -\frac{GM}{R}}$$

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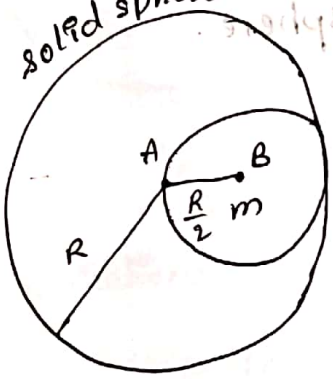

m is released at B. Find time taken by m to hit A.

Sol: Concept:

E inside of cavity,

$$E = \frac{GM}{R^3}(x)$$

$$E = \frac{G}{R^3} \times \frac{4}{3}\pi R^3 \rho x$$

$$E = \frac{4G}{3}\pi\rho x$$

$$\boxed{E = \frac{4}{3}\pi\rho G x}$$

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Force on m!

$$F = mE$$

$$F = m \times \frac{4}{3}\pi \rho G \times \frac{R}{2}$$

$$F = \frac{2\pi \rho R G M}{3}$$

$$ma = \frac{2\pi \rho G R}{3}$$

$$\boxed{a = \frac{2\pi}{3} R \rho G}$$

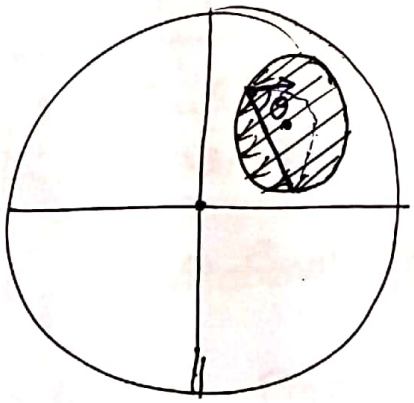
$$s = \cancel{u t}^{0} + \frac{1}{2} a t^2$$

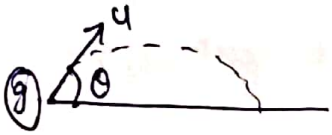
$$s = \frac{1}{2} \times \frac{2\pi}{3} \rho G R t^2$$

$$\frac{R}{2} = \frac{\pi}{3} \rho G R t^2$$

$$\boxed{t = \sqrt{\frac{3}{2\pi \rho G}}}$$

***





$$H = \frac{u y^2}{2g}$$

$$\boxed{a = \frac{4}{3}\pi \rho G (x)}$$

$$H = \frac{u y^2}{2a}$$

$$= \frac{u^2 \sin^2\theta}{2\left[\frac{4}{3}\pi \rho G x\right]}$$

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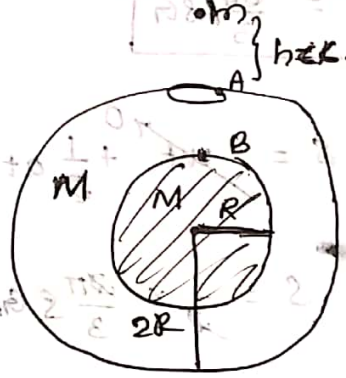
Ex: A solid sphere of mass M & radius R is surrounded by thin spherical sphere of same mass & radius 2R as shown. A small particle of mass m is released from rest from height h ($h \ll R$) above the shell. there is a hole in the shell as shown. neglect the effect of any other body including the earth.

a) In what time will it enter the hole at A.



Force on 'm' due to both the spheres,

$$F = m \times [g_1 + g_2]$$

$$F = m \times \left[\frac{GM}{(2R)^2} + \frac{GM}{(2R)^2} \right]$$

$$\boxed{F = \frac{GMm}{2R^2}}$$

$$ma = \frac{GMm}{2R^2}$$

$$\boxed{a = \frac{GM}{2R^2}}$$

$$s = ut^{0} + \frac{1}{2}at^2$$

$$h = \frac{1}{2} \times \frac{GM}{2R^2} t^2$$

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$$\boxed{t = 2R\sqrt{\frac{h}{GM}}}$$

ff) The time it takes to move from A to B is t

A) $\frac{R^2}{\sqrt{GMh}}$ B) $> \frac{R^2}{\sqrt{GMh}}$ C) $< \frac{R^2}{\sqrt{GMh}}$ D) $\frac{2R^2}{\sqrt{GMh}}$

$$V_A = \sqrt{2ah}$$

$$= \sqrt{2 \times \frac{GM}{2R^2} h}$$

$$\boxed{V_A = \frac{\sqrt{GMh}}{R}}$$

$$t < \frac{R}{V_A}$$

$$t < \frac{R}{\frac{\sqrt{GMh}}{R}}$$

$$\boxed{t < \frac{R^2}{\sqrt{GMh}}}$$

Ex: The gravitational field in a region is given by $\vec{E} = 3\hat{i} - 4\hat{j}$. Find the work done by gravity in displacing a particle by 1m along the line $4y = 3x + 9$.

Sol:

$$W_g = -\Delta U$$

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$= -m(\Delta V)$

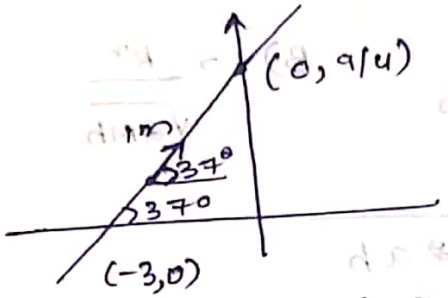
$\Delta V = -\vec{E} \cdot d\vec{r}$

$W_g = m(\vec{E} \cdot d\vec{r})$

$d\vec{r} = dx\uparrow + dy\uparrow$

$= 1\cos 37^\circ \uparrow + 1\sin 37 \uparrow$

$= \frac{4}{5}\uparrow + \frac{3}{5}\uparrow$



$\tan\theta = \frac{9/4}{3} = 3/4$

$\theta = 37^\circ$

$W_g = m\left[3\left(\frac{4}{5}\right) - 4\left(\frac{3}{5}\right)\right]$

$= m(0)$

$\boxed{W_g = 0}$

Slope of line $\Rightarrow \frac{3}{4}$

Slope of $\vec{E} \Rightarrow -\frac{4}{3}$

$m_1 \times m_2 = -1$

$\vec{E}, d\vec{r} \rightarrow \perp$ ler

$W_g = m(0) = 0$

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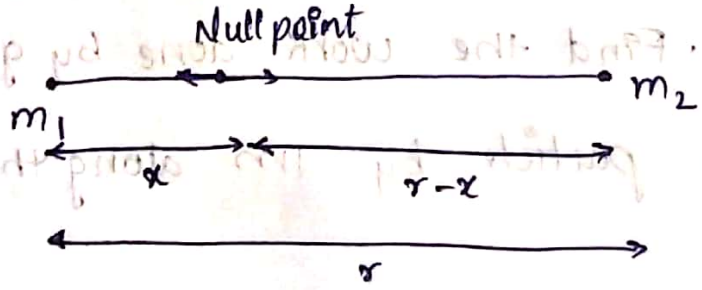
Important Points:

1. For pt masses:



$E_1 = E_2$

$\frac{Gm_1}{x^2} = \frac{Gm_2}{(r-x)^2}$

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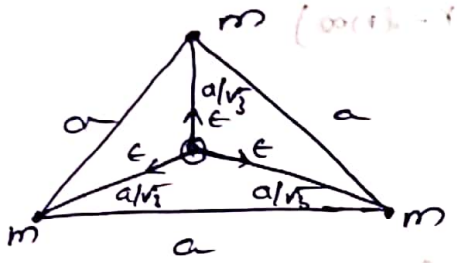
* at null pt, ($\mathcal{E}_{net} = 0$)

$V = V_1 + V_2$

$$V = -\frac{GM_1}{x} - \frac{GM_2}{r-x}$$

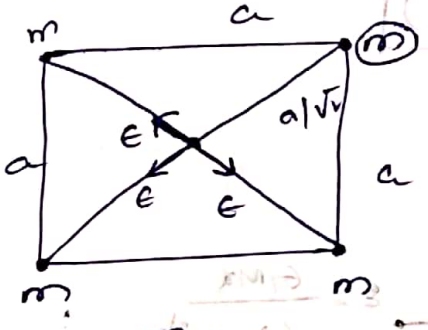
*



Null pt is at centroid of the Δle.

$$V = -\frac{3GM}{a/\sqrt{2}} = -\frac{3\sqrt{2}GM}{a}$$

*

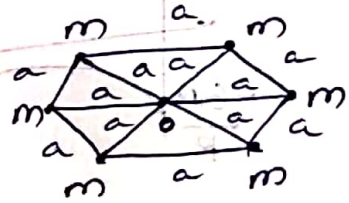


$$\mathcal{E} = \frac{GM}{(a/\sqrt{2})^2} = \frac{2GM}{a^2}$$

$$V = -\frac{GM}{a/\sqrt{2}} \times 3$$

$$V = -\frac{3\sqrt{2}GM}{a}$$

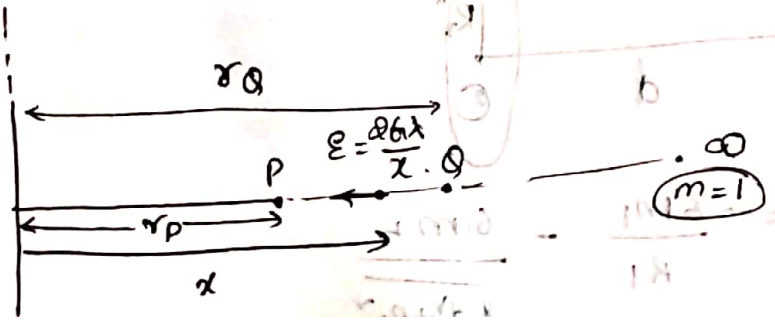
*



* $F = m\mathcal{E}$ — force acting on mass 'm' placed in gravitational field $\mathcal{E}$.

* $U = mV$ — P.E stored in a mass 'm' placed at a point of net potential V.

*



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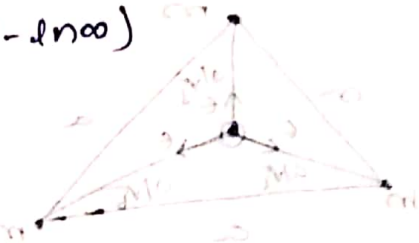
* $\int_{V_0}^{V} dV = -\int_{i}^{f} \vec{E} \cdot d\vec{x}$

$$V_p - V_\infty = -\int_{\infty}^{r} \frac{2G\lambda}{x} \cdot dx \cos 180^\circ$$

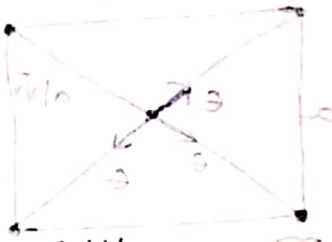
$$V_p - V_\infty = 2G\lambda (\ln r - \ln \infty)$$

$\therefore V_p$ doesnot exist



* $\int_{V_Q}^{V_P} dV = -\int_{r_Q}^{r_P} \frac{2G\lambda}{x} \cdot dx \cos 180^\circ$

$$\boxed{V_p - V_Q = 2G\lambda \ln (r_p / r_Q)}$$



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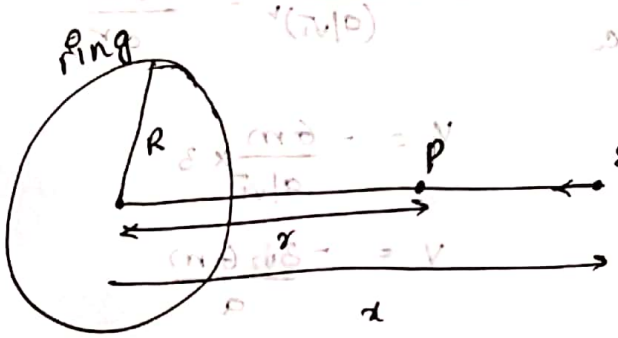
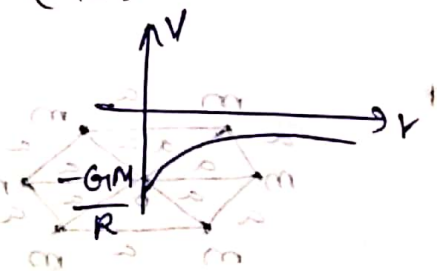


$$E = \frac{GMx}{(x^2+R^2)^{3/2}}$$

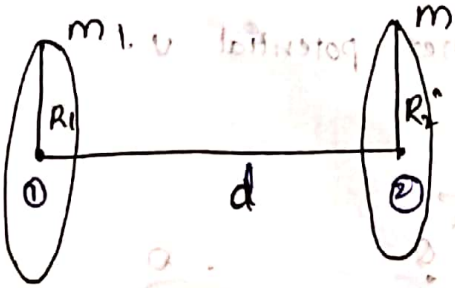


1) $\int dV = -\int \frac{GMx}{(x^2+R^2)^{3/2}} \cdot dx$

2) $V = \int dV = \int \frac{-G\,dm}{\sqrt{x^2+R^2}} = \frac{-Gm}{\sqrt{x^2+R^2}}$

Principle of Superpositons

*



$$V_1 = \frac{-Gm_1}{R_1} - \frac{Gm_2}{\sqrt{d^2+R_2^2}}$$

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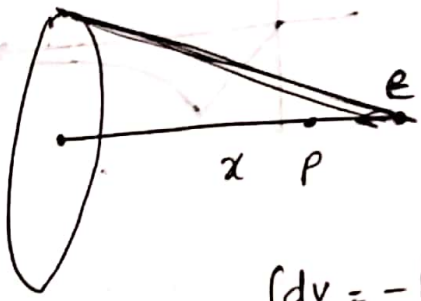
$$V_Q = -\frac{Gm_2}{R_2} - \frac{Gm_1}{\sqrt{d^2+R_1^2}}$$



$$E = \frac{2GM}{R^2}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$$

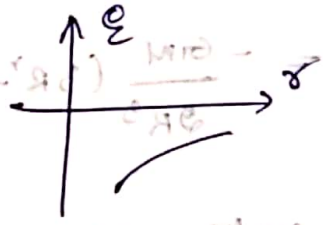
$$\int dV = -\int \vec{E}.d\vec{r}$$

* Hollow spheres

① $E = \frac{GM}{r^2}$ $(r > R)$

② $E = \frac{GM}{R^2}$ $(r = R)$

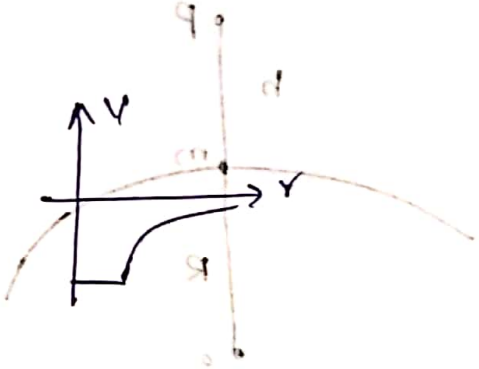
③ $E = 0$ $(r < R)$



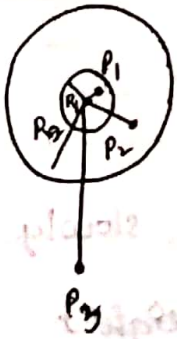
a) $V = -\frac{GM}{r}$ $(r > R)$

b) $V = -\frac{GM}{R}$ $(r = R)$

c) $V = \frac{-GM}{R}$ $(r < R)$



Eg:



for P_3: $U = -\frac{Gm_1^2}{2R_1} - \frac{Gm_2^2}{2R_2} - \frac{Gm_1m_2}{R_2}$

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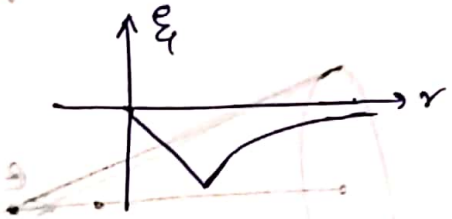
* for solid sphere:

① $E = \frac{GM}{r^2}$ $r > R$

② $E = \frac{GM}{R^2}$ $r = R$

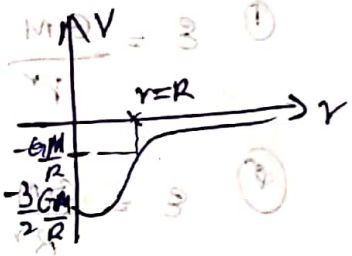
③ $E = \frac{GM}{R^3} r$ $r < R$



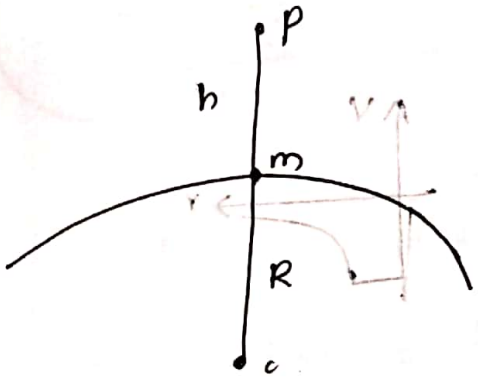
a) $V = -\frac{GM}{r}$ $r > R$

b) $V = -\frac{GM}{R}$ $r = R$

c) $V = \frac{-GM}{2R^3}(3R^2 - r^2)$ $r < R$



at centre, $V = -\frac{3}{2}\frac{GM}{R}$



on surface, $U_S = \frac{-GMm}{R}$

$U_h = -\frac{GMm}{R+h}$

* work done in bringing the mass slowly from surface of earth to certain height:

$W = \Delta U = U_f - U_i$

$= U_h - U_S$

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$$W = -GMm\left(\frac{1}{R+h} - \frac{1}{R}\right)$$

$$= -GMm\left(\frac{-h}{R(R+h)}\right)$$

$$= \frac{GMmh}{R(R+h)} \qquad g = \frac{GM}{R^2}$$

$$= \frac{gRmh}{R+h} \qquad gR = \frac{GM}{R}$$

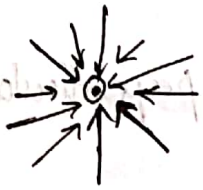
$$\boxed{W = \frac{mgh}{1+h/R}}$$

* $h \ll\ll R \Rightarrow W = mgh = U_f - U_i$

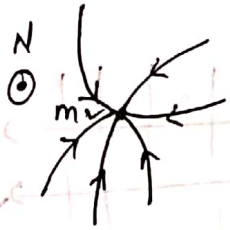
Gravitational field lines:

* It is the line (or) curve along which unit mass would travell in the field.

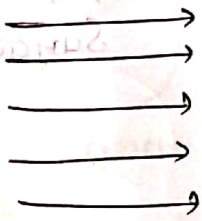
① for pt mass:



②



③



uniform gravitational field

* tangent drawn at any pt to the field line gives the direction of gravitational field at the pt.

* No. of lines passing through unit area gives the magnitude of gravitational field at that pt.

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* No two field lines will intersect with each other bcoz field has 2 diff directions at the intersecting pt is not possible.

* Equipotential Surfaces:

* The surfaces on which potential is same at all pts.

i.e, $V = 0$

$$dV = 0$$

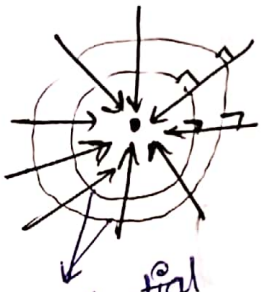
$$dV = \frac{\omega}{m} = 0$$

$$\int \bar{F} \cdot d\bar{r} = 0$$

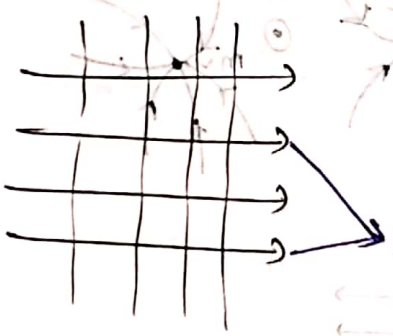
$$\int \bar{\varepsilon} \cdot d\bar{r} = 0$$

$$\boxed{\bar{\varepsilon} \perp d\bar{r}}$$

i.e, gravitational field lines are perpendicular to position vector.



Equipotential Surfaces



Equipotential Surface

* we have,

$$\varepsilon = \frac{-dV}{dr}$$

-ve of slope 'V' vs 'r' gives gravitational intensity at that pt.

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& $dV = -E\,dr$

$$V_f - V_i = -(\text{area of } 'E' \text{ vs } 'r' \text{ graph})$$

* $E = -\frac{dV}{dr}$

-ve sign indicates $\vec{E}$ is always along the direction of decreasing potential.

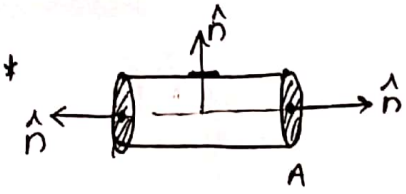
Gravitational Flux:

$$\phi_g = \oint_S \vec{E} \cdot d\vec{s}$$

Note:

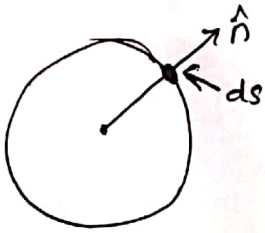
* Area of body is scalar.
* It direction is always normal to the area.

*



$\vec{A} = A \cdot \hat{n}$

*



sphere

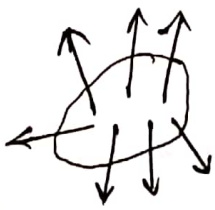
$d\vec{s} = (ds)\hat{n}$

* Gravitational flux is a scalar.
* It can be +ve / -ve / zero.

*

$\phi_g = -ve$

*



$\phi_g = +ve$

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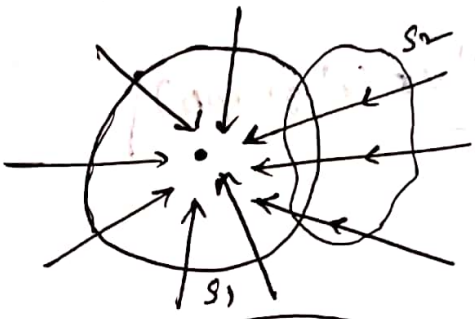
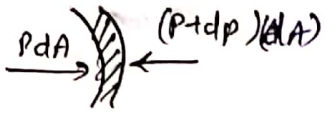
* for ex,

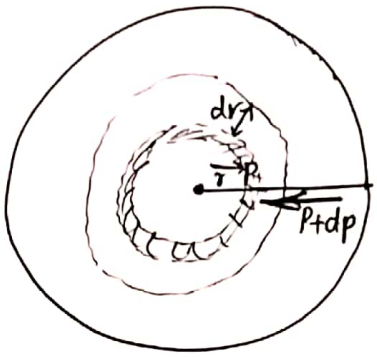


* $\phi_{S_1} = -ve$

Gravitational Pressure!

* Pressure is max at centre.





Since shell is in eqm,

Net upward force = Net downward force

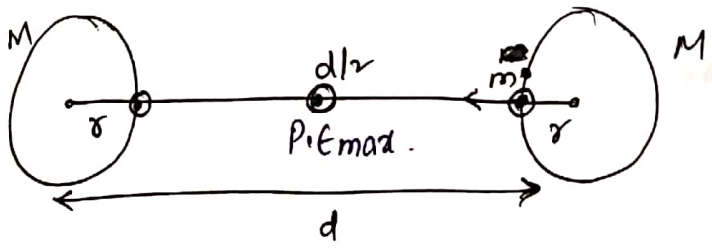
$$P(dA) - (P+dp)(dA) = \frac{GM'}{r^2} \cdot dm$$

$$-dp \cdot dA = \frac{G \times \rho \times \frac{4}{3}\pi r^3}{r^2} \times \rho\, dA \times dr$$

$$\int_P^0 -dp = G\rho^2 \times \frac{4}{3}\pi \int_{r=r}^{r=R} r \cdot dr$$

$$\boxed{P = \frac{2G\rho^2\pi}{3}[R^2 - r^2]}$$

Ex:



$$V_A = \frac{-Gm}{d} - \frac{GM}{r}$$

$$V_B = -\frac{Gm}{r} - \frac{Gm}{d}$$

P.E ⇒ first increases & then decreases.

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Ex: The potential at the surface of a planet of mass M & radius R is assumed to be zero. Choose the correct options.

1) $V_\infty = +\frac{GM}{R}$

2) $V_C = -\frac{GM}{2R}$

3) both A & B are correct.

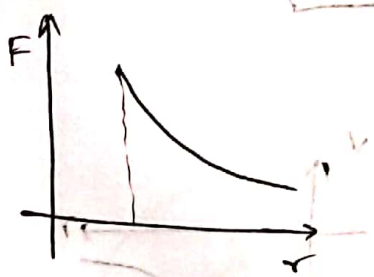
*

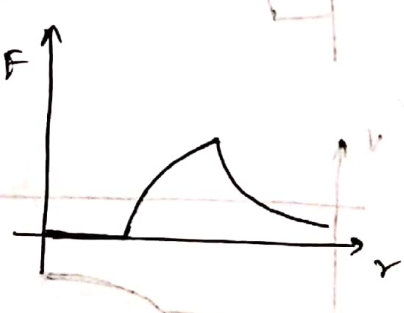
$$V_\infty = 0 + \frac{GM}{R}$$

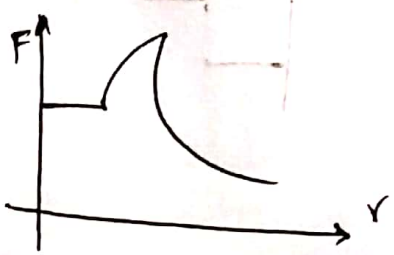
$$V_S = -\frac{GM}{R} + \frac{GM}{R} = 0$$

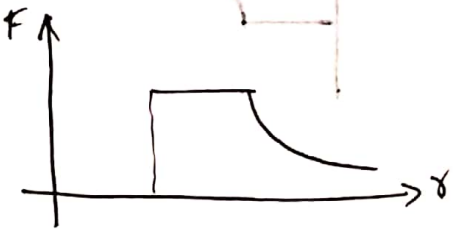
$$V_C = -\frac{3GM}{2R} + \frac{GM}{R} = -\frac{GM}{2R}.$$

Ex: A sphere of mass M & radius R_2 has a concentric cavity of radius R_1 as shown in Figure. The force F exerted by the sphere on a particle of mass m located at a distance r from the centre of sphere varies as

1) 

2) 

3) 

4) 

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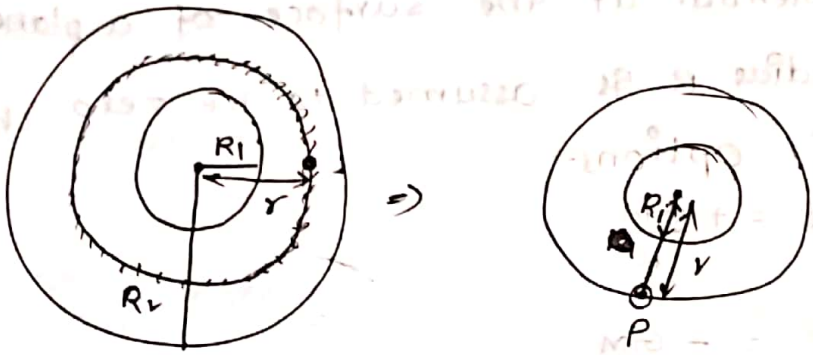
Sol 8



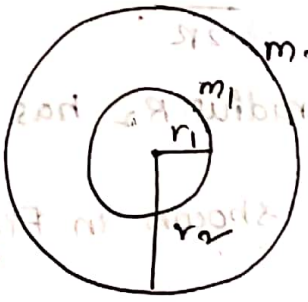
$$\varepsilon_p = \frac{GM}{r^2} - \frac{GM'}{r^2}$$

$$= \frac{G}{r^2}\left(\rho \times \frac{4}{3}\pi r^3 - \rho \times \frac{4}{3}\pi R_1^3\right)$$

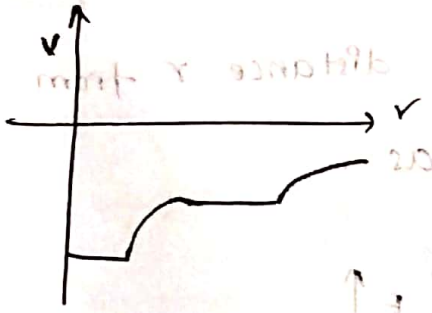
$$\boxed{\varepsilon_p = \frac{4}{3}\pi G\rho\,\frac{(r^3 - R_1^3)}{r^2}}$$

$$\varepsilon_p = \frac{4}{3}\pi G\rho r - \frac{4}{3}\pi\rho G R_1^3 \frac{1}{r^2}$$

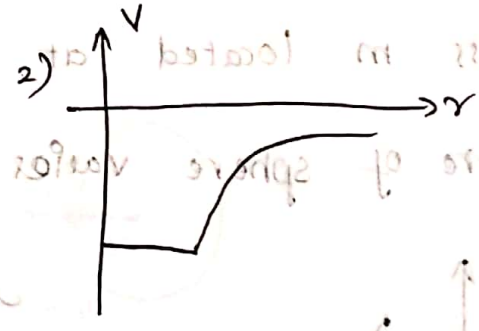
Sol 8



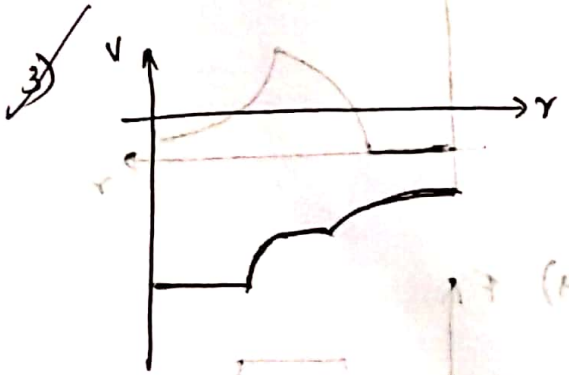
1)



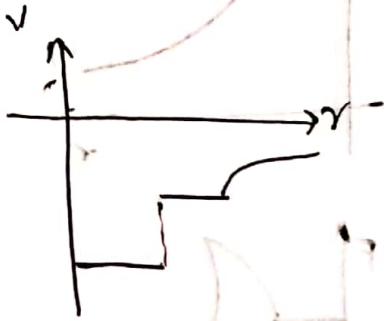
2)



3)



4)



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Sol!

$0 \longrightarrow r_1$

$$V_1 = -\frac{Gm_1}{r_1} - \frac{Gm_2}{r_2}$$

$r_1 \le r \le r_2$

$$V_2 = -\frac{Gm_1}{r} - \frac{Gm_2}{r_2}$$

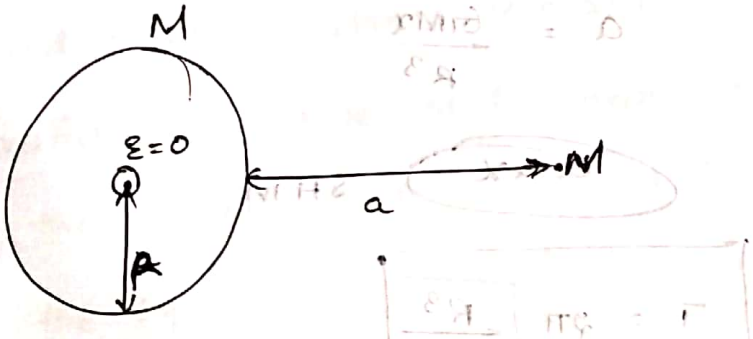
$r > r_2$

$$V_3 = -\frac{Gm_1}{r} - \frac{Gm_2}{r}$$

$$\frac{dV_2}{dr} = \frac{Gm_1}{r^2} \qquad \frac{dV_3}{dr} = \frac{G(m_1+m_2)}{r^2}$$

$$\frac{dV_3}{dr} > \frac{dV_2}{dr}$$

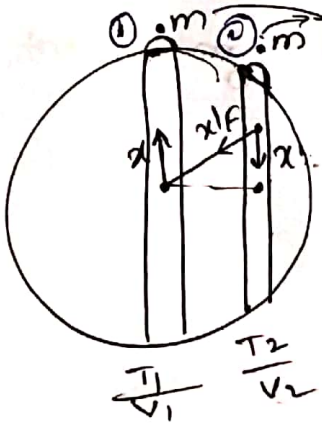
Ex:



ε due to particle $M = \frac{GM}{(R+a)^2}$

$$V = -\frac{GM}{R} - \frac{GM}{R+a}$$

Ex:



compare T & V.

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① → $F_{net} = \frac{GM}{R^3} x m$

$ma = \frac{GMm}{R^3} x$

$a = \frac{GMx}{R^3}$

$a \propto x$ SHM

$T = 2\pi\sqrt{\frac{R^3}{GM}}$

② → $F\sin\theta = F_{net}$

$\frac{GM}{R^3} x \times m \times \frac{x}{x} = ma$

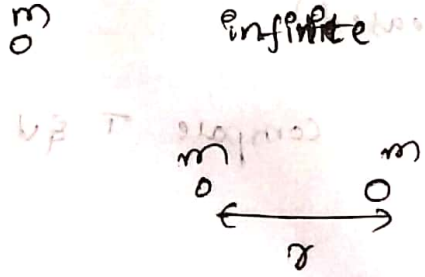
$a = \frac{GMx}{R^3}$

$a \propto x$ SHM

$T = 2\pi\sqrt{\frac{R^3}{GM}}$

$T_1 = T_2$ & $V_1 \neq V_2$

Ex:



1) $T - E = 0$

2) $K.E = \frac{4GM^2}{r}$

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$P.E = 0$

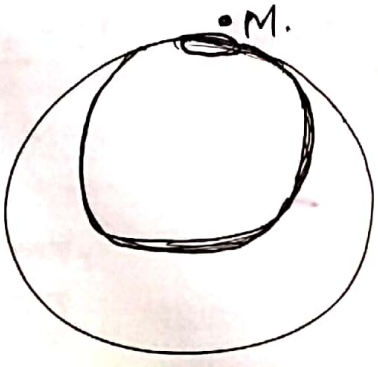
at infinite.

$K.E = 0$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}(um)v_2^2 - \frac{Gm(um)}{r} = 0$$

$$\boxed{K.E = \frac{4Gm^2}{r}}$$

Ex: From a uniform solid spherical planet of radius R, density ρ, a solide spherical material of radius $\frac{2R}{3}$ is scooped out to leave an eccentric hole as shown. Acceleration due to gravity on the surface of the planet before scooping is g and mass of planet left after scooping is M. A particle of same M is kept at the end of the eccentric hole as shown in the figure and the system is released from rest. Assuming head-on elastic collision b/w particle & planet and Newton's law of gravitation to be valid, after what time will the particle come back to the initial position for the first time?



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30|12|24| Monday

Law of Conservation of Energy:

Here gravitational force is a conservative force.

$\therefore$ T.E = const

$\Rightarrow U_1 + K_1 = U_2 + K_2$

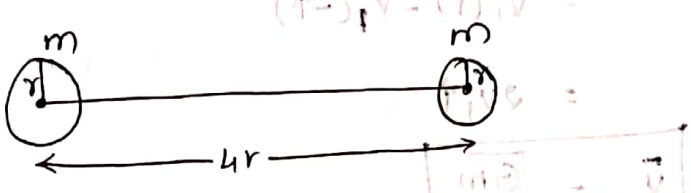
Law of conservation of Linear momentum:

If $\bar{F}_{ext} = 0$ then $\boxed{\bar{P}_{system} = const}$

$\boxed{\bar{P}_i = \bar{P}_f}$

Examples:

→



Two identical spheres initially at rest are seperated by 4r due to gravitational force they approach each other and collide. At the time of collision.

Find

1) Velocity of each sphere.

2) relative velocity of both spheres.

3) time for collision.

Sol: ① using L.C.E,

$U_1 + K_1 = U_2 + K_2$

$\Rightarrow -\frac{Gm^2}{4r} + 0 = -\frac{Gm^2}{2r} + \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2$

from L.C.L.M,

$0 = mv_1 - mv_2$

$\boxed{v_1 = v_2}$

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$$-\frac{Gm^2}{4r} + \frac{Gm^2}{2r} = \frac{1}{2}mv^2 + \frac{1}{2}mv^2$$

$$\frac{Gm^2}{4r} = \frac{1}{2} m v^2 \times 2$$

$$\boxed{v = \sqrt{\frac{Gm}{4r}}}$$

2) relative velocity,

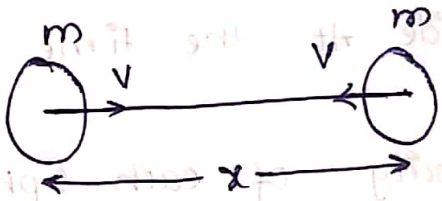
$$\bar{v}_{rel} = \bar{v}_1 - \bar{v}_2$$

$$= v_1(\uparrow) - v_2(-\uparrow)$$

$$= 2v_1 \uparrow$$

$$\boxed{\bar{v}_{rel} = \sqrt{\frac{Gm}{r}}}$$

3)



L.C.E,

at any position,

$$U_1 + K_1 = U_2 + K_2$$

$$-\frac{Gm^2}{4r} + 0 = -\frac{Gm^2}{x} + 2 \times \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{Gm}{x} - \frac{Gm}{4r}}$$

$$\boxed{v = \sqrt{Gm\left(\frac{4r - x}{4rx}\right)}}$$

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we have,

$$v_{rel} = 2v$$

$$-\frac{dx}{dt} = 2v$$

$$v = -\frac{1}{2}\frac{dx}{dt}$$

$$-\frac{1}{2}\frac{dx}{dt} = \sqrt{\frac{Gm}{4r}} \times \sqrt{\frac{4r-x}{x}}$$

$$-\sqrt{\frac{x}{4r-x}}\cdot dx = 2\sqrt{\frac{Gm}{4r}}\cdot dt$$

$$\Rightarrow \int_{x=2r}^{x=4r} \frac{\sqrt{x}}{\sqrt{4r-x}}\cdot dx = 2\sqrt{\frac{Gm}{4r}}\int_0^t dt$$

$$\int_{x=2r}^{x=4r} \sqrt{\frac{x}{4r-x}}\cdot dx = \sqrt{\frac{Gm}{r}}\,(t)$$

let $x = 4r\cos^2\theta$

$$dx = -8r\sin\theta\cos\theta\cdot d\theta$$

$$\int_{x=2r}^{x=4r} \sqrt{\frac{4r\cos^2\theta}{4r(\sin^2\theta)}}\cdot(-8r\sin\theta\cos\theta)\,d\theta = \sqrt{\frac{Gm}{r}}\cdot t$$

$$\int_{x=2r}^{x=4r} \frac{\cos\theta}{\sin\theta}(-8r\sin\theta\cos\theta)\cdot d\theta = \sqrt{\frac{Gm}{r}}\,t$$

$$-8r\int_{x=2r}^{x=4r}\cos^2\theta\cdot d\theta = \sqrt{\frac{Gm}{r}}\cdot t$$

$$-4r\int_{x=2r}^{x=4r}(\cos 2\theta+1)\cdot d\theta = \sqrt{\frac{Gm}{r}}\cdot t$$

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$$-4r \int_{x=2r}^{x=4r} \left[\frac{\sin 2\theta}{2} + \theta\right]_{x=2r}^{x=4r} = \sqrt{\frac{Gm}{r}} \cdot t$$

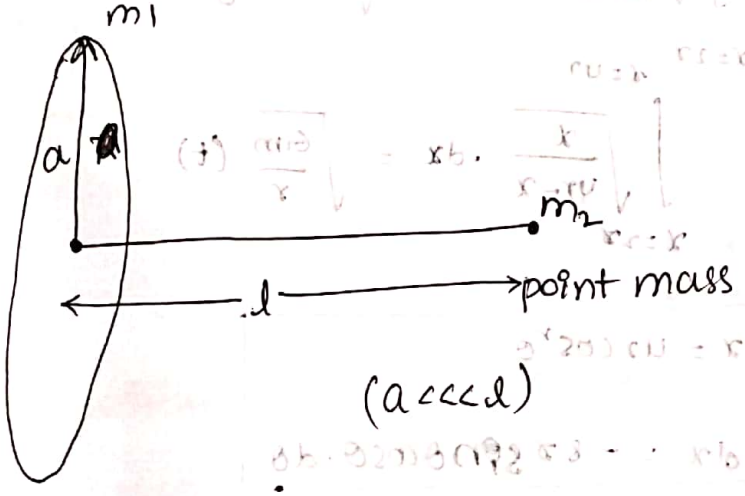
$$-4r\left[\frac{\sin 2\theta}{2} + \theta\right]_{\theta=\pi/4}^{\theta=0} = \sqrt{\frac{Gm}{r^3}}\, t$$

$$-4\left[0 - \left(\frac{1}{2} + \frac{\pi}{4}\right)\right] = \sqrt{\frac{Gm}{r^3}} \cdot t$$

$$(\pi+2)\sqrt{\frac{\theta r^3}{Gm}} = t$$

$$\therefore \boxed{t = (\pi+2)\sqrt{\frac{\theta r^3}{Gm}}}$$

Ex:



uniform disc

Initially both are at rest in free space & approached towards each other due to gravitation -al force. Find relative velocity of approach at the time of collision.

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Sol: L.C.E,

$$U_1 + K_1 = U_2 + K_2$$

we have,

$$U_1 = \frac{-2Gm_1m_2}{a^2}\left(\sqrt{a^2+d^2} - d\right)$$

$$= -\frac{2Gm_1m_2}{a^2} d\left(\left(1+\frac{a^2}{d^2}\right)^{1/2} - 1\right)$$

$$= -\frac{2Gm_1m_2}{a^2} d\left(1 + \frac{1}{2}\frac{a^2}{d^2} - 1\right)$$

$$= -\frac{2Gm_1m_2}{a^2} \times \frac{1}{2}\frac{a^2}{d}$$

$$\boxed{U_1 = -\frac{Gm_1m_2}{d}}$$

& $U_2 = -\frac{2Gm_1m_2}{a^2}\left(\sqrt{a^2}\right) \quad (d=0)$

$$\boxed{U_2 = -\frac{2Gm_1m_2}{a}}$$

now,

$$-\frac{Gm_1m_2}{d} + 0 = -\frac{2Gm_1m_2}{a} + \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = Gm_1m_2\left(\frac{2}{a} - \frac{1}{d}\right) \quad -(1)$$

Again, L.C.L.M,

$$m_1v_1 = m_2v_2 \quad -(2)$$

from eqns (1) & (2),

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from ② ⇒ $v_2 = \frac{m_1}{m_2}(v_1)$ —— ③

① ⇒ $\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2 \times \frac{m_1^2}{m_2^2}(v_1^2) = Gm_1m_2\left(\frac{2}{a} - \frac{1}{d}\right)$

$\frac{1}{2}v_1^2\left(\frac{m_1m_2 + m_1^2}{m_2}\right) = Gm_1m_2\left(\frac{2}{a} - \frac{1}{d}\right)$

$\frac{v_1^2}{2} = \frac{Gm_2^2}{m_1+m_2}\left(\frac{2}{a} - \frac{1}{d}\right)$

$$v_1 = \sqrt{\frac{2Gm_2^2}{m_1+m_2}\left(\frac{2}{a} - \frac{1}{d}\right)}$$

put value of v_1 in ③

$$v_2 = \sqrt{\frac{m_1^2}{m_2^2} \times \frac{2Gm_2^2}{(m_1+m_2)}\left(\frac{2}{a} - \frac{1}{d}\right)}$$

$$v_2 = \sqrt{\frac{2Gm_1^2}{m_1+m_2}\left(\frac{2}{a} - \frac{1}{d}\right)}$$

$v_{rel} = v_1 + v_2$

$= \sqrt{\frac{2G}{m_1+m_2}\left(\frac{2}{a} - \frac{1}{d}\right)}\left(\sqrt{m_1^2} + \sqrt{m_2^2}\right)$

$$v_{rel} = \sqrt{2G(m_1+m_2)\left(\frac{2}{a} - \frac{1}{d}\right)}$$

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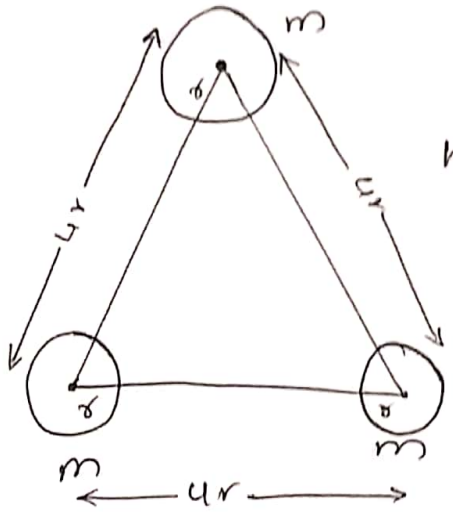
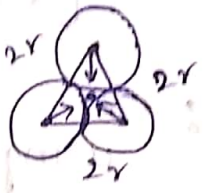
Ex:



Identical spheres kept on smooth horizontal table.

Find the speed of each ball at the time of collision.

Sol:



By symmetry, velocities are equal.

L.C.E)

$$U_1 + K_1 = U_2 + K_2$$

$$-\frac{Gm^2}{4r} \times \beta + 0 = -\frac{Gm^2}{2r} \times \beta + \frac{1}{2} mv^2 \times \beta$$

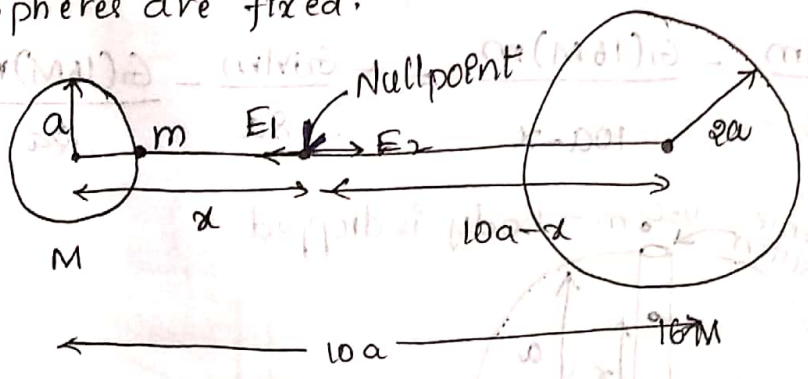
$$\frac{Gm^2}{4r} = \frac{1}{2} mv^2$$

$$v = \sqrt{\frac{Gm}{2r}}$$

***

Ex:

Spheres are fixed.



Find min velocity of projection required for a body to reach on to the surface of other ~~sphere~~ sphere.

Sol:

Step-1:

To find null point due to spheres -

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i.e, $E_1 = E_2$

$$\frac{GMm}{x^2} = \frac{G(16M)}{(10-x)^2}$$

$$10a - x = 4x$$

$$5x = 10a$$

$$\boxed{x = 2a}$$

step-2: using L.C.E,

$$U_1 + K_1 = U_2 + K_2$$

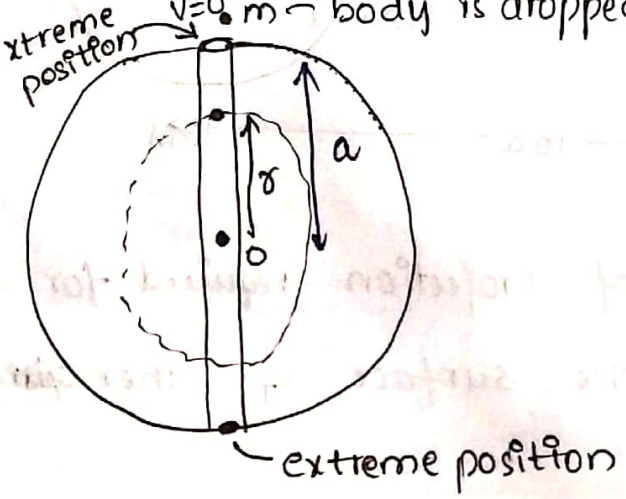
$$-\frac{GMm}{a} - \frac{G(16M)M}{9a} + \frac{1}{2}mv_{min}^2 = -\frac{GMm}{x} - \frac{G(16M)M}{10a-x} + 0$$

solve for V_{min}.

To find speed of body while reaching the surface of other sphere:

$$-\frac{GMm}{x} - \frac{G(16M)m}{10a-x} = -\frac{GMm}{8a} - \frac{G(16M)m}{9a} + \frac{1}{2}mv^2$$

Eg:



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$F = m\varepsilon$

$ma = m\left(-\frac{GM}{R^3}\right) \cdot r$

$a = -\frac{GM}{R^3} \cdot r$

$$\boxed{a = -\omega^2 \cdot r}$$

$a \propto r \Rightarrow$ body is in S.H.M.

$$\omega = \sqrt{\frac{GM}{R^3}} = \sqrt{\frac{gR^2}{R^3}} = \sqrt{\frac{g}{R}}$$

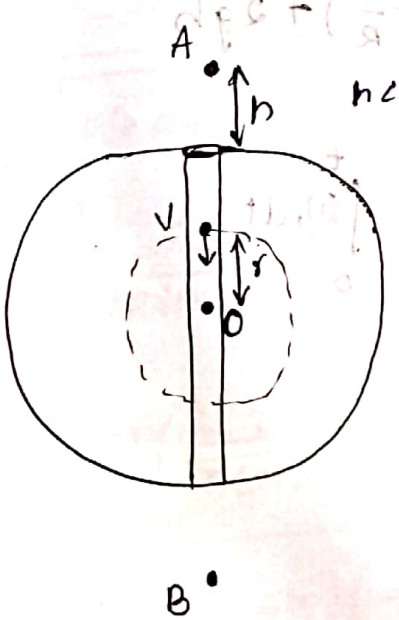
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R}{g}}$$

$$\boxed{T = 84.3 \text{ min}}$$

* velocity at any position,

$$\boxed{v = \omega\sqrt{a^2 - r^2}}$$

Ex:



— Find the time period of oscillation

Sol: velocity of body at A,

$$v_A = \sqrt{2gh} \quad \text{—①}$$

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& time taken $= \sqrt{\frac{2h}{g}} = t_1$

now, from L.C.E,

$U_1 + K_1 = U_2 + K_2$

wrong.

$$-\frac{GMm}{R} + \frac{1}{2}mv_A^2 = -\frac{GMm}{r} + \frac{1}{2}mv^2$$

$$-\frac{GM}{R} + \frac{GM}{r} + \frac{1}{2}v_A^2 = \frac{1}{2}v^2$$

$$GM\left(\frac{1}{r} - \frac{1}{R}\right) + \frac{1}{2} \times 2gh = \frac{1}{2}v^2$$

$$\frac{v^2}{2} = GM\left(\frac{1}{r} - \frac{1}{R}\right) + gh$$

$$v = \sqrt{2GM\left(\frac{1}{r} - \frac{1}{R}\right) + 2gh}$$

wrong

$$-\frac{dr}{dt} = \sqrt{2GM\left(\frac{1}{r} - \frac{1}{R}\right) + 2gh}$$

$$\int_{r=R}^{r=0} \frac{+dr}{\sqrt{2GM\left(\frac{1}{r} - \frac{1}{R}\right) + 2gh}} = \int_0^t (1).dt$$

from L.C.E,

$U_1 + K_1 = U_2 + K_2$

$$-\frac{GMm}{R} + \frac{1}{2}mv_A^2 = -\frac{GMm}{2R^3}(3R^2 - r^2) + \frac{1}{2}mv^2$$

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$$-\frac{GMm}{R} + \frac{GMm}{2R^3}(3R^2 - r^2) + \frac{1}{2}(m)(gh) = \frac{1}{2}mv^2$$

$$-\frac{GM}{R} + \frac{GM}{2R^3}(3R^2 - r^2) + gh = \frac{v^2}{2}$$

$$-\frac{\omega^2 R^2}{R} + \frac{\omega^2 R^3}{2R^3}(3R^2 - r^2) + \omega^2 Rh = \frac{v^2}{2}$$

$$-\omega^2 R^2 + \frac{3}{2}\omega^2 R^2 - \frac{\omega^2 r^2}{2} + \omega^2 Rh = \frac{v^2}{2}$$

$$\frac{\omega^2 R^2}{2} - \frac{\omega^2 r^2}{2} + \omega^2 Rh = \frac{v^2}{2}$$

$$\omega^2 R^2 - \omega^2 r^2 + 2\omega^2 Rh = v^2$$

$$v = \omega\sqrt{R(R+2h) - r^2}$$

$$\frac{-dr}{dt} = \omega\sqrt{R(R+2h) - r^2}$$

$$\int_{r=0}^{r=R} \frac{-dr}{\sqrt{R(R+2h) - r^2}} = \omega\int_0^t dt$$

$$\int_R^0 \frac{dr}{\sqrt{R(R+2h) - r^2}} = \omega t$$

$$\sin^{-1}\left(\frac{r}{\sqrt{R(R+2h) - r^2}}\right)_0^R = \omega t$$

$$\boxed{t = \frac{1}{\omega}\sin^{-1}\left(\frac{R}{\sqrt{(R+2h)R}}\right)}$$

(t time taken from A to B)

Time period of oscillation,

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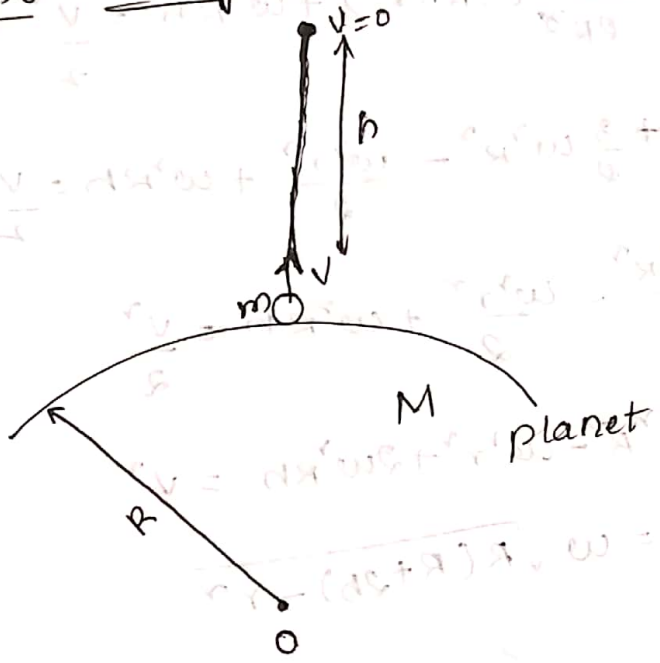
$$T = 4t + 4t_1$$

$$\boxed{T = 4\left[\frac{1}{\omega}\sin^{-1}\left(\frac{R}{\sqrt{R(R+2h)}}\right) + \sqrt{\frac{2h}{g}}\right]}$$

Escape Velocity (V_e) :-

*



To find max height :-

use, C.C.E,

$$U_1 + K_1 = U_2 + K_2$$

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h} + 0$$

$$GM\left(\frac{1}{R+h} - \frac{1}{R}\right) + \frac{1}{2}v^2 = 0$$

$$\boxed{\frac{v^2}{2} = GM\left(\frac{h}{R(R+h)}\right)}$$

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$$v = \sqrt{\frac{2GMh}{R(R+h)}}$$

$$v = \sqrt{\frac{2ghR}{R+h}}$$

$$\boxed{v = \sqrt{\frac{2gh}{1+h/R}}}$$

Cor) $$\boxed{v^2 = \frac{2gh}{1+h/R}}$$

$$v^2 + \frac{v^2 h}{R} = 2gh$$

$$v^2 = 2gh - h\left(\frac{v^2}{R}\right)$$

$$\boxed{h = \frac{v^2}{2g - v^2/R}}$$

As $v \uparrow \Rightarrow h(\uparrow)$

If $v << R$

then $\boxed{h \to \frac{v^2}{2g}}$

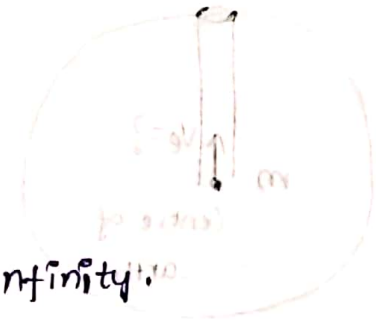
* As $v \uparrow \Rightarrow h(\uparrow)$

at a particular velocity.

$v = v_e$; $h \to \infty$.

i.e, body just reaches to infinity.

i.e, $\boxed{v_\infty = 0}$



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* L.C.E)

$$U_1 + K_1 = U_2 + K_2$$

$$-\frac{GMm}{R} + \frac{1}{2}mv_e^2 = -\frac{GMm}{R+h} + \frac{1}{2}mv_\infty^2$$

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R}$$

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\boxed{v_e = \sqrt{\frac{8}{3}\pi\rho G R^2}} \quad \text{or} \quad \boxed{v_e = \sqrt{2gR}}$$

$$g = \frac{GM}{R^2} \quad \& \quad M = \frac{4}{3}\pi R^3 \rho$$

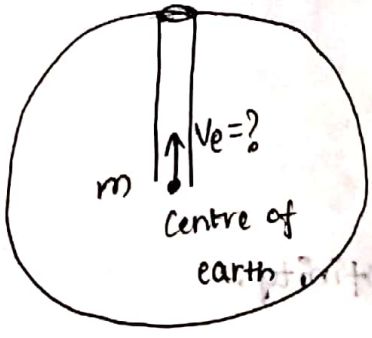
* points:

① v_e depends on

1) mass of planet
2) Radius of planet
3) density of planet
4) place at which body is projected.

② v_e doesnot depend on

1) mass of projected body.
2) Angle of projection.



Just to escape,

$$T.E = 0$$

$$U + K = 0$$

$$-\frac{3}{2}\frac{GMm}{R} + \frac{1}{2}mv_e^2 = 0$$

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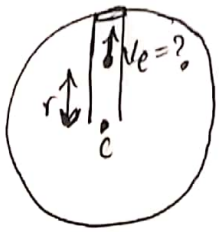
$$V_e = \sqrt{\frac{2GM}{R}}$$

* 

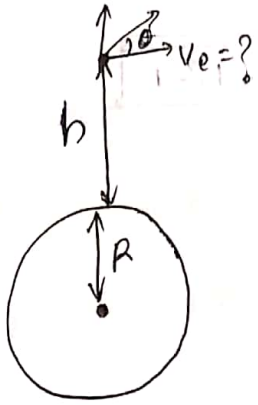
Just to escape,

$T \cdot E = 0$

$U + K = 0$

$$-\frac{GMm}{R^3}(3R^2 - r^2) + \frac{1}{2}mV_e^2 = 0$$

$$V_e = \sqrt{\frac{GM}{R^3}(3R^2 - r^2)}$$

* 

Just to escape,

$T \cdot E = 0$

$U + K = 0$

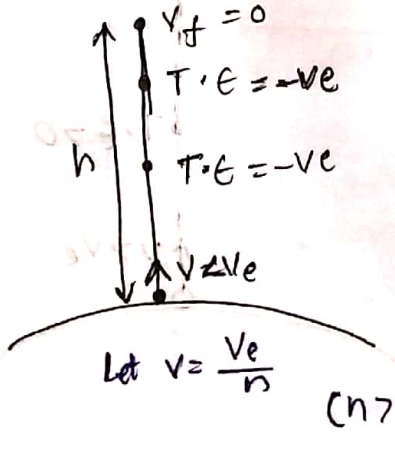
$$-\frac{GMm}{(R+h)} + \frac{1}{2}mV_e^2 = 0$$

$$V_e = \sqrt{\frac{2GM}{R+h}}$$

(or) $$V_e = \sqrt{\frac{2gR^2}{R+h}}$$

$$g_h = \frac{gR^2}{(R+h)^2}$$

$$V_e = \sqrt{2g_h(R+h)}$$

*** 

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** **Binding Energy = $-T.E$**

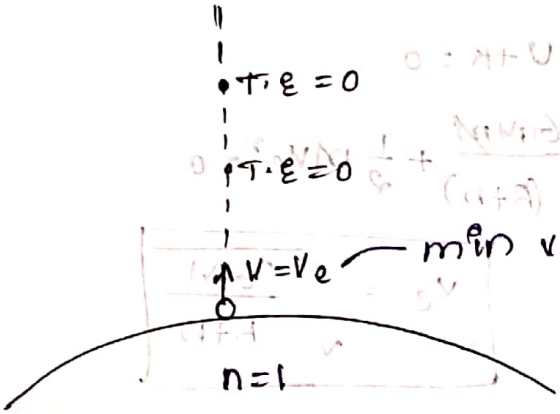
* L.C.E,

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h} + \frac{1}{2}m(0)^2$$

$$h = \frac{v^2}{2g - v^2/R} = \frac{\frac{v_e^2}{n^2}}{2g - \frac{v_e^2}{n^2 R}} = \frac{\frac{2gR}{n^2}}{2g - \frac{1}{n^2}(2g)}$$

** $$\boxed{h = \frac{R}{n^2 - 1}} \Rightarrow \boxed{v_1 = v_e\sqrt{n^2-1}}$$

*



min velocity of body just to escape the gravitational pull

$\Rightarrow h = \infty$

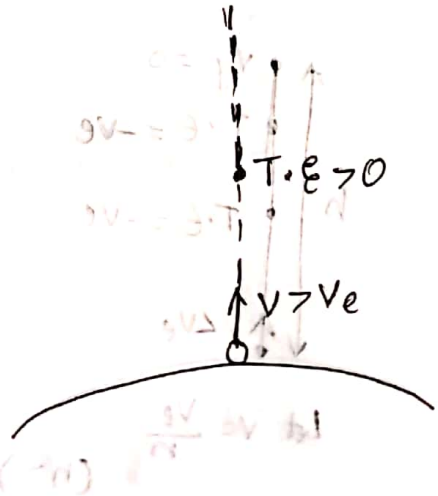
at infinity,

$T.E = 0$

$$-\frac{GMm}{R+h} + \frac{1}{2}mv_\infty^2 = 0$$

$$\boxed{v_\infty = 0}$$

*



At infinity,

$T.E > 0$

$$-\frac{GMm}{R+h} + \frac{1}{2}mv_\infty^2 > 0$$

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$V_\infty = 0$

from L.C.E,

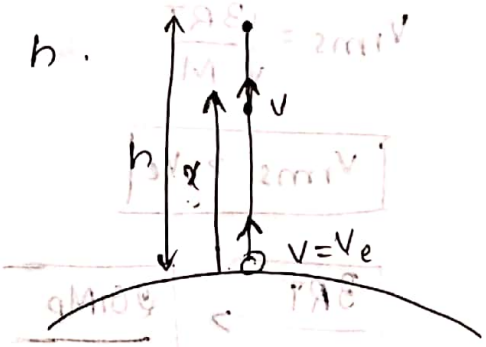
$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h} + \frac{1}{2}mv_\infty^2$$

$$-\frac{v_e^2}{2} + \frac{v^2}{2} = \frac{v_\infty^2}{2}$$

$$v_\infty^2 = v^2 - v_e^2$$

$$\boxed{v_\infty = \sqrt{v^2 - v_e^2}}$$

Ex: A body is projected from the surface of earth just to reach infinity. find the time taken by the body to reach the height h.



Sol: From L.C.E,

$$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{2GM}{R}\right)$$

$$= -\frac{GMm}{R+x} + \frac{1}{2}mv^2$$

$$\boxed{v_{esc} = \sqrt{\frac{2GM}{R+x}}}$$

$$\frac{dx}{dt} = \sqrt{\frac{2GM}{R+x}}$$

$$\int_0^h \sqrt{R+x}\,dx = \sqrt{2GM}\int_0^t dt$$

$$\left[\frac{(R+x)^{3/2}}{3/2}\right]_0^h = \sqrt{2GM}\,(t)$$

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$$\frac{2}{3}\left[(\sqrt{R+h})^3-(\sqrt{R})^3\right] = \sqrt{2GM}\ (t)$$

$$t = \sqrt{\frac{4}{9(2GM)}}\left[(\sqrt{R+h})^3-(\sqrt{R})^3\right]$$

$$\boxed{t = \sqrt{\frac{2}{9GM}}\left((\sqrt{R+h})^3-(\sqrt{R})^3\right)}$$

$$\boxed{t = \sqrt{\frac{2}{9GM}}\left[(R+h)^{3/2} - R^{3/2}\right]}$$

* moon has no atmosphere,

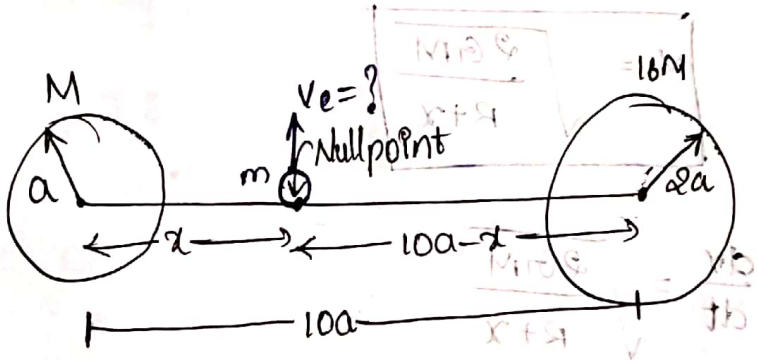
$$\boxed{V_e = 2.4\ km/s}$$

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\boxed{V_{rms} > V_e}$$

$$\sqrt{\frac{3RT}{M_g}} > \sqrt{\frac{2GM_p}{\text{Radius of planet}}}$$

Ex:



Sol:

$$\boxed{x = 2a}$$

here $T.E = 0$

$$U+K=0$$

$$-\frac{GMm}{x} + -\frac{G\times 16M\,m}{10a-x} + \frac{1}{2}mV_e^2 = 0$$

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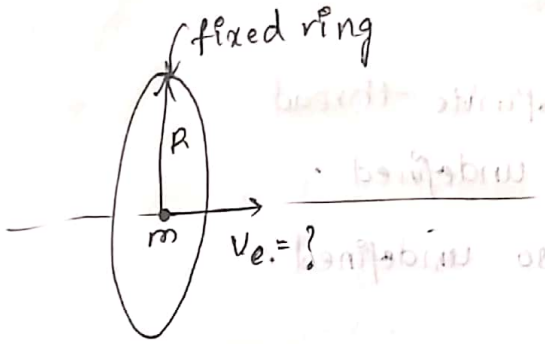
$$\frac{GM}{a} + \frac{16GM}{8a} = \frac{1}{2}v_e^2 \Rightarrow \boxed{v_e = \sqrt{\frac{5GM}{a}}}$$

ex:



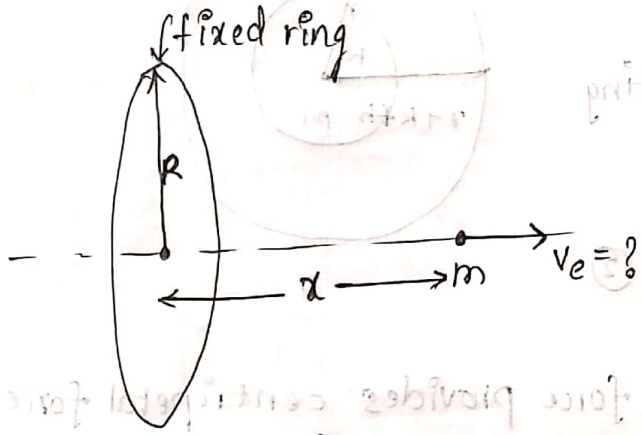
$$T.E = 0$$

$$U + K = 0$$

$$-\frac{GMm}{R} + \frac{1}{2}mv_e^2 = 0$$

$$\boxed{v_e = \sqrt{\frac{2GM}{R}}}$$

ex:



$$T.E = 0$$

$$U + K = 0$$

$$-\frac{GMm}{\sqrt{R^2+x^2}} + \frac{1}{2}mv_e^2 = 0$$

$$\frac{v_e^2}{2} = \frac{GM}{\sqrt{R^2+x^2}}$$

$$\boxed{v_e = \sqrt{\frac{2GM}{\sqrt{R^2+x^2}}}}$$

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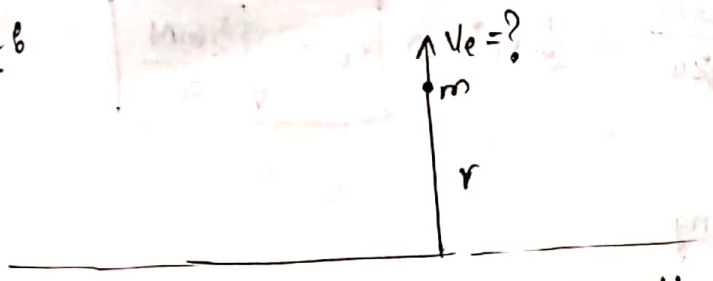
Ex:



Infinite thread

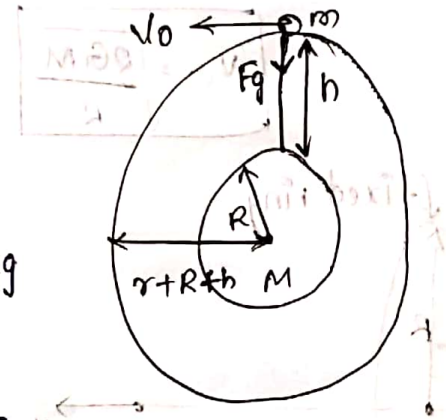
Absolute potential is undefined.

⇒ Escape velocity is also undefined.

31/12/24 Tuesday

Orbital Velocity:

It is the horizontal velocity required for a body to move in a particular orbit around the planet.



$$F_g = \frac{GMm}{(R+h)^2} \quad -①$$

Centripetal force acting on the body,

$$F_c = \frac{mv_0^2}{R+h} \quad -②$$

here gravitational force provides centripetal force.

$$\frac{GMm}{(R+h)^2} = \frac{mv_0^2}{R+h}$$

$$v_0 = \sqrt{\frac{GM}{R+h}}$$

$$\boxed{v_0 = \sqrt{\frac{gR^2}{R+h}}}$$

Clearly, v_0 depends on - ① Mass of planet
② Radius of planet

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⑤ height of orbit.

but it doesnot depends on mass of revolving body.

* For surface satilite,

$$h << R \Rightarrow h + R \approx R$$

$$V_0 = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

* for earth,

$$g = 9.8\ m/s^2,\ R = 6.4 \times 10^5\ m = 6400\ km$$

$$\boxed{V_0 = 7.92\ km/s}$$

points:

① Relation blw V_0 & V_e:

$$V_0 = \sqrt{\frac{GM}{R+h}} \quad \& \quad V_e = \sqrt{\frac{2GM}{R+h}}$$

$$\boxed{V_e = \sqrt{2}\, V_0}^{*} \qquad \boxed{V_e = 11.2\ km/s}$$

② Centripetal acceleration of Satillite:

$$a_c = \frac{V_0^2}{r} = \frac{1}{r} \times \frac{GM}{r} = \frac{GM}{r^2} = \frac{GM}{(R+h)^2}$$

$$a_c = \frac{gR^2}{(R+h)^2} = g_h$$

i.e, centripetal acceleration is equal to acceleration due to gravity at that altitude.

③ Angular velocity of Satillite:

$$\omega = \frac{V_0}{r} = \frac{1}{r}\sqrt{\frac{GM}{r}}$$

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$$\boxed{\omega = \sqrt{\frac{GM}{r^3}}}$$

④ Time period of revolution:

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^3}{GM}}$$

$$\boxed{T^2 = \frac{4\pi^2 r^3}{GM}}$$

⑤ linear momentum of satellite:

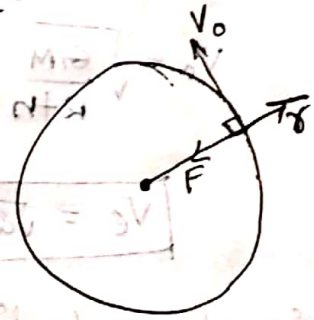
$$\bar{P} = m\bar{V}_0$$

$$P = m\sqrt{\frac{GM}{r}}$$

$$\boxed{P = \sqrt{\frac{GMm^2}{r}}}$$

⑥ Angular momentum of satellite:

about centre of planet,



$$L = mV_0 r$$

$$L = m\sqrt{\frac{GM}{r}}\, r$$

$$\boxed{L = \sqrt{GMm^2 r}}$$

⑦ Torque acting on the satillite:

$$\bar{\tau} = \bar{r} \times \bar{F} = 0 \qquad [\bar{r} \parallel -\bar{F}]$$

$$\frac{d\bar{L}}{dt} = 0$$

$$\boxed{L = \text{const}}$$

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8) Energies of a Satellite in an orbit :

* Potential energy,

$$U = -\frac{GMm}{r}$$

* Kinetic energy,

$$K = \frac{1}{2} m v_o^2 = \frac{1}{2} m \left(\frac{GM}{r}\right) = \frac{GMm}{2r}$$

$$\boxed{K = \frac{GMm}{2r}} \quad ; \text{always +ve.}$$

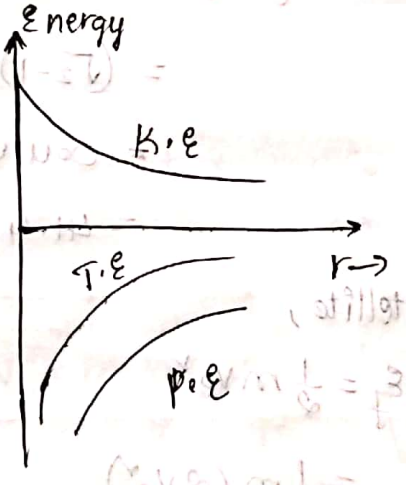
* Total Energy,

$$U + K = E_T$$

$$E_T = -\frac{GMm}{r} + \frac{GMm}{2r}$$

$$\boxed{E_T = -\frac{GMm}{2r}}$$

clearly, $K : E = -T : E$, $P : E = 2T : E$



* Ratio of energies in an orbit,

$$U : K : E_T = -2E : -E : E$$

$$\boxed{U : K : E_T = 2 : -1 : 1}$$

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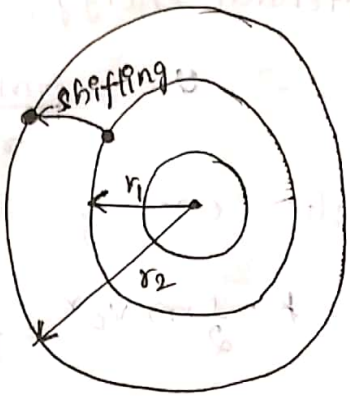
(9) Work done in shifting the satellite from one orbit to another is,

$$W = E_f - E_i$$

$$W = -\frac{GMm}{2r_2} - \left(-\frac{GMm}{2r_1}\right)$$

$$= -\frac{GMm}{2}\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$



$$\boxed{W = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

$r_1 = R + h_1$, $r_2 = R + h_2$.

* Binding energy of satillite in that orbit.

$$\boxed{B.E = -(T.E)}$$

* In order to escape the satillite from that orbit,

a) Additional velocity given to satillite,

$$\Delta V = V_e - V_0 = \sqrt{2}V_0 - V_0$$

$$= (\sqrt{2} - 1)V_0$$

$$= 0.414\, V_0$$

$$= 41.4\%\, V_0$$

K.E of satellite,

$$K.E_f = \frac{1}{2}mV_e^2$$

$$= \frac{1}{2}m(2V_0^2)$$

$$= 2 \times \frac{1}{2}mV_0^2$$

$$\boxed{K.E_f = 2K.E_i}$$

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Sec, K.E becomes doubled (increases by 100%)

* when satelite shifts from lower orbit to higher orbit.

① v_0 (↓) ✓

② a_c (↓) ✓

③ ω (↓) ✓

④ T (↑) ✓

⑤ P (↓) ✓

⑥ L (↑) ✓

⑦ γ (remains same) ✓

⑧ P.E (↑) ✓

⑨ K.E (↓) ✓

⑩ T.E (↑) ✓

Weightlessness of a body in artificial satellite :

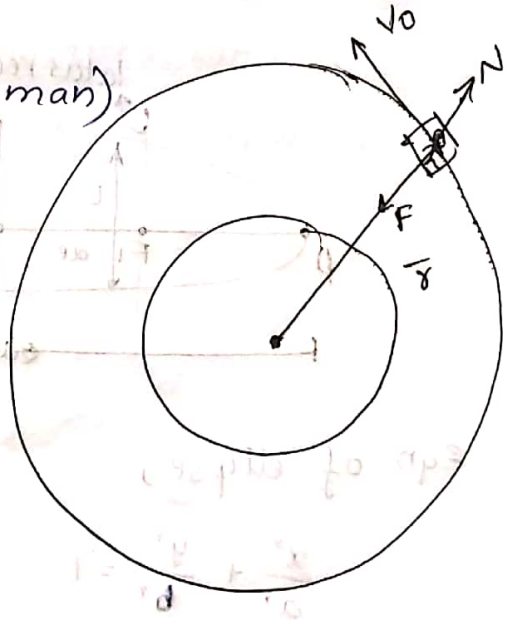
(Net force on the man)

$$F - N = ma_c$$

$$\frac{GMm}{r^2} - N = m\frac{v_0^2}{r}$$

$$\frac{GMm}{r^2} - N = \frac{m}{r} \times \frac{GM}{r}$$

$$\boxed{N = 0}$$



⇒ (apparent wt = 0) where as true wt ≠ 0.

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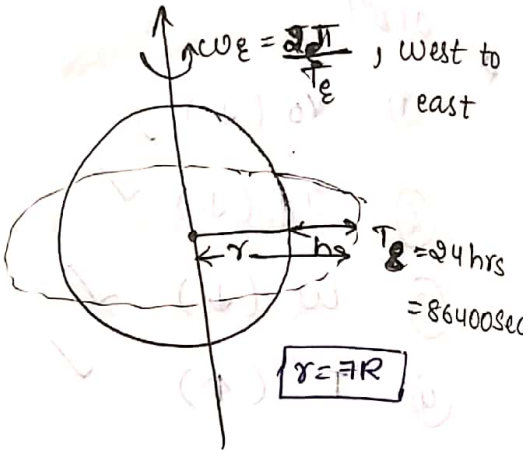
* Geo Stationary Satellite :

T_E = 24 hrs



$F_g = F_c$

$\frac{GMm}{r^2} = mr\omega^2 = mr\frac{4\pi^2}{T^2}$

after solving,

$r \approx 42000$ Km

$R + h = 42000$ Km

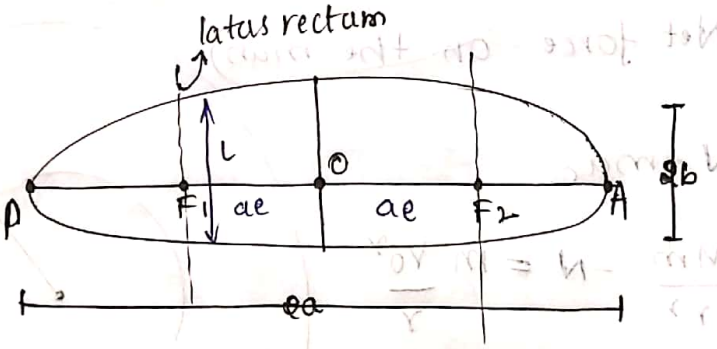
$h = 35600$ Km

Kepler's Law of planetary motion :

I - Law (Law of orbits) :

Statement :

"All planets revolve around the sun in an elliptical orbit with the sun at one of the foci."



* Eqn of ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

* eccentricity, $e = \frac{OF_1}{OP} = \frac{OF_2}{OA}$

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$OF_1 = OP \times e \Rightarrow \boxed{OF_1 = ae}$

$OF_2 = OA \times e \Rightarrow \boxed{OF_2 = ae}$

Eq $\boxed{b^2 = a^2(1-e^2)}$

$e = 0$, for circle.

$e < 1$, for ellipse.

$e = 1$, for parabola.

$e > 1$, for hyperbola.

* Length of latus rectum,

$$\boxed{L = \frac{2b^2}{a}}$$

* P = perigee (nearest pt from sun)

F_1P = Periheleon distance (closest distance) $= r_p = r_{min}$.

$$\boxed{r_p = r_{min} = a - ae = a(1-e)}$$

* A = Apogee (farthest pt from sun)

F_1A = Apeheleon distance (maximum distance) $= r_{max} = r_a$

$$r_a = a + ae$$

$$\boxed{r_{max} = r_a = a(1+e)}$$

$$\boxed{\frac{r_{max}}{r_{min}} = \frac{1+e}{1-e}} \quad ***$$

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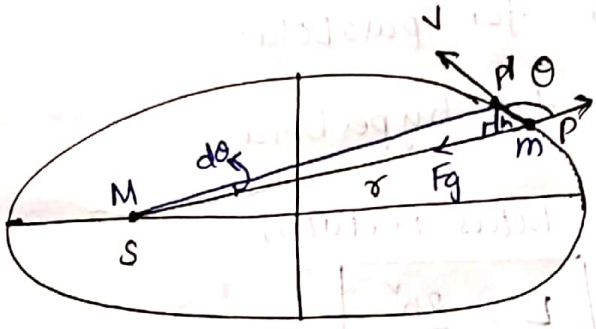
II-Law (law of areas):

Statement:

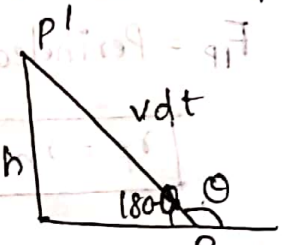
It is based on law of conservation of angular momentum.

"The line joining the sun and the planet called radius vector sweep out equal areas in equal intervals of time".



* In a very small time interval dt.

planet moves from P to P' with speed V



$$\sin(180-\theta) = \frac{h}{vdt}$$

$$h = (vdt)\sin\theta$$

* Area of Δle,

$$dA = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times r \times h$$

$$dA = \frac{1}{2} \times r \times v \cdot dt \cdot \sin\theta$$

$$\frac{dA}{dt} = \frac{vr\sin\theta}{2}$$

$$\frac{dA}{dt} = \frac{mvr\sin\theta}{2m}$$

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$$\frac{d\bar{A}}{dt} = \frac{m(\bar{r}\times\bar{v})}{2m}$$

$$\boxed{\frac{d\bar{A}}{dt} = \frac{\bar{L}}{2m}}$$

Torque acting on planet due to gravitational force,

$$\bar{\tau} = \bar{r}\times\bar{F} = \bar{0} \quad (\bar{r} \parallel -\bar{F})$$

$$\bar{\tau} = \frac{d\bar{L}}{dt} = \bar{0}$$

$$\boxed{\bar{L} = \text{const}}$$

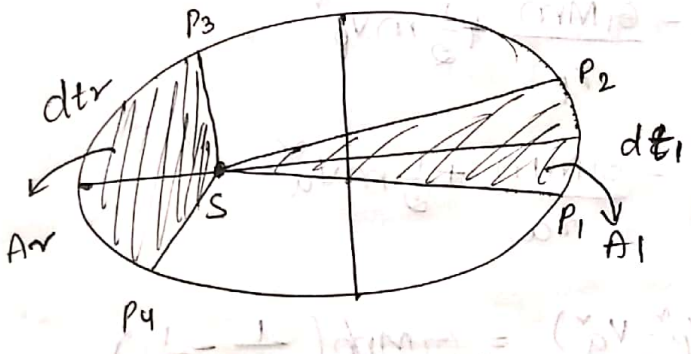
$$\frac{d\bar{A}}{dt} = \frac{\bar{L}}{2m} = \text{const}$$

└ areal velocity of radius vector.

$\therefore dA \propto dt$

i.e, $\boxed{\frac{A_1}{t_1} = \frac{A_2}{t_2} = - - - - -}$

Ex:



If $dt_1 > dt_2 \Rightarrow A_1 > A_2$

$dt_2 > dt_1 \Rightarrow A_2 > A_1$

$dt_1 = dt_2 \Rightarrow A_1 = A_2$

* From L.C.A.M,

$\bar{L} = m(\bar{r}\times\bar{v}) = \text{const.}$

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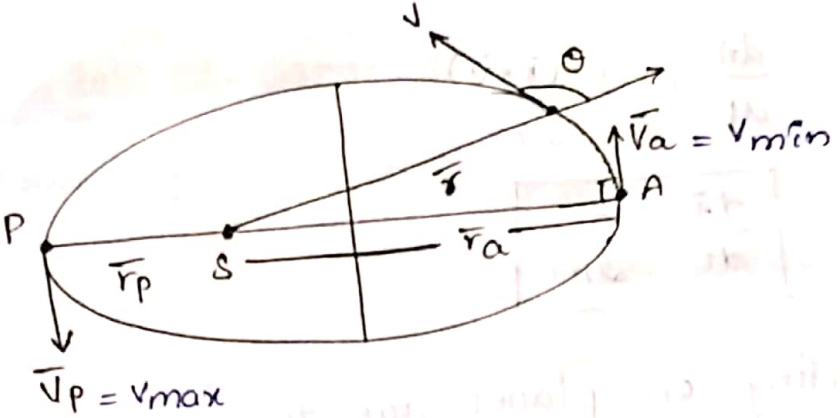

$$L = m_p V_p r_p = m V_a r_a = m v r \sin\theta$$

$$\boxed{V_p r_p = V_a r_a} \quad *$$

$$r_p < r_a$$

$$\Rightarrow V_p > V_a$$

* Since only gravitational force is acting on the planet;

T.E of planet = const.

i.e, $$T.E = \boxed{-\frac{GMm}{r} + \frac{1}{2}mv^2}$$

$$= -\frac{GMm}{r_p} + \frac{1}{2}mV_p^2$$

$$= -\frac{GMm}{r_a} + \frac{1}{2}mV_a^2$$

$$\frac{1}{2}m(V_p^2 - V_a^2) = GMm\left(\frac{1}{r_p} - \frac{1}{r_a}\right)$$

$$\frac{1}{2}\left(V_p^2 - \frac{V_p^2 r_p^2}{r_a^2}\right) = GM\left(\frac{r_a - r_p}{r_a r_p}\right)$$

$$\frac{V_p^2}{2}\left(\frac{r_a^2 - r_p^2}{r_a^2}\right) = GM\left(\frac{r_a - r_p}{r_a r_p}\right)$$

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$$v_p^2 = \frac{2GM(r_a)}{(r_a+r_p)\,r_p}$$

$$v_p = \sqrt{\frac{2GM\,a(1+e)}{a\times a(1-e)}}$$

$$\boxed{v_p = \sqrt{\frac{GM(1+e)}{a(1-e)}}} \quad **$$

Mely,

$$\boxed{v_a = \sqrt{\frac{GM(1-e)}{a(1+e)}}}$$

Total energy of planet in an orbit,

$$T\cdot E = -\frac{GMm}{r_a} + \frac{1}{2}mv_a^2$$

$$= -\frac{GMm}{a(1+e)} + \frac{1}{2}m\,\frac{GM(1-e)}{a(1+e)}$$

$$= \frac{GMm}{a(1+e)}\left[-1+\frac{1}{2}-\frac{e}{2}\right]$$

$$= \frac{GMm}{a(1+e)}\left[-\frac{(e+1)}{2}\right]$$

$$\boxed{T\cdot E = -\frac{GMm}{2a}} \quad ***$$

$$a = \frac{r_{max}+r_{min}}{2} \Rightarrow \boxed{r_{avg} = a}$$

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III-law (law of periods):

Statement:

The square of the period of the planet revolving in a particular orbit is directly proportional to cube of the length of semi-major axis of that orbit.

$$T^2 \propto a^3$$

i.e, $$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{a_1}{a_2}\right)^3$$

here, $$T^2 = \frac{4\pi^2}{GM} a^3$$

Ex: ① 2 satellites S_1 & S_2 revolve round a planet in coplanar circular orbits in the same sense. Their period of revolutions are 1 hour & 8 hour respectively. The radius of the orbit of $S_1 = 10^4$ Km. when S_2 is closest to S_1. Find

a) the speed of S_2 relative to S_1 = $\pi \times 10^4$ Km/hr

b) The Angular speed of S_2 as actually observed by an astronaut in S_1 = $\pi/3$ rad/hr.

② A particle is projected from pt A, that is at a distance 4R from the centre of the earth with speed v_1 in a direction making 30° with the line joining the centre of earth and pt A, as shown. Find the speed v_1 of particle if particles

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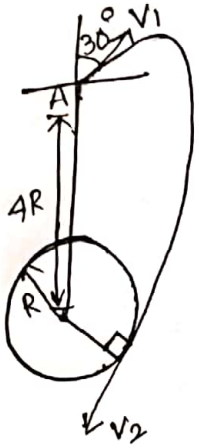
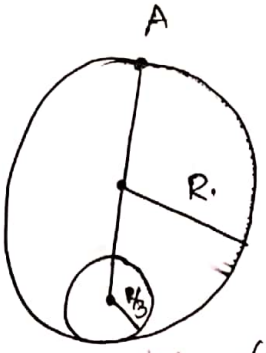
passes grazing the surface of the earth. Consider gravitational interaction only between these two.

$\left\{ \text{use } \frac{GM}{R} = 6.4 \times 10^7 \ m^2/s^2 \right\}$



(3)



Consider a uniform sphere of mass M and Radius R. A cavity of Radius R/3 is made in the sphere as shown.

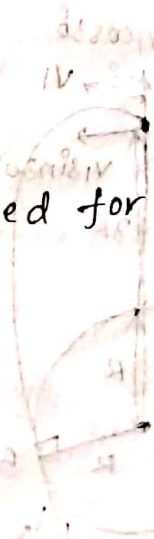
Options:-

(A) gravitational field inside the cavity is uniform. (do not change)

(B) ✗ gravitational field inside the cavity is non-uniform.

(C) ✗ escape velocity of a particle projected from point A is $\sqrt{\frac{88GM}{15R}}$.

(D) ✗ Escape velocity is defined for earth & particle system only.



Sol:

1) $T_1 = 1hr$, $T_2 = 8hr$

$r_1 = 10^4 km$

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3$$

$$\Rightarrow \left(\frac{1}{8}\right)^2 = \left(\frac{10^4}{r_2}\right)^3$$

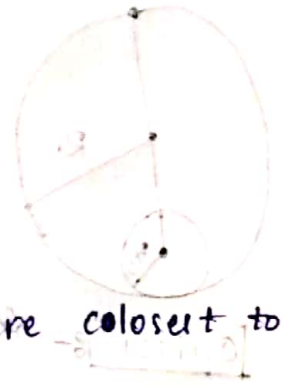
$$\boxed{r_2 = 4 \times 10^4 \text{ km}}$$

(a) $V_{rel} = V_{21} = V_2 - V_1$

$$= \frac{2\pi r_2}{T_2} - \frac{2\pi r_1}{T_1}$$

$$|\vec{V}_{rel}| = |\pi \times 10^4 - 2\pi \times 10^4|$$

$$\boxed{V_{rel} = \pi \times 10^4 \text{ km/hr}}$$



(b)

$$|\omega_{rel}| = \frac{|v_{rel}|}{|r_{rel}|}$$ ← (when they are coloseit to each other)

$$= \frac{\pi \times 10^4}{3 \times 10^4}$$

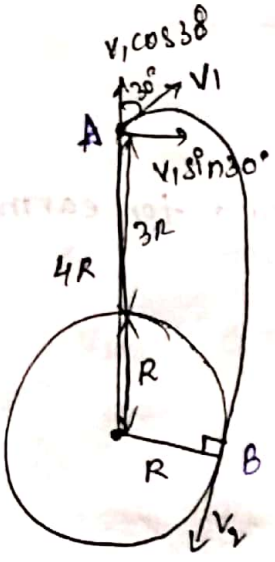
$$\boxed{\omega_{rel} = \pi/3}$$

when not closest then

$$\omega_{rel} = \omega_2 - \omega_1$$

$$= 2\pi\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

2)



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* L.C.A.M (about centre of earth):

$$m(v_1 \sin 30^\circ)(4R) = m v_2 R$$

$$v_1 \times \frac{1}{2} \times 4R = v_2 R$$

$$\boxed{v_2 = 2v_1}$$

* L.C.E:

$$T.E_A = T.E_B$$

$$-\frac{GMm}{4R} + \frac{1}{2} m v_1^2 = -\frac{GMm}{R} + \frac{1}{2} m v_2^2$$

$$-\frac{GM}{4R} + \frac{1}{2} v_1^2 = -\frac{GM}{R} + \frac{1}{2}(4 v_1^2)$$

$$\frac{GM}{R} - \frac{GM}{4R} = \left(2 - \frac{1}{2}\right) v_1^2$$

$$\frac{3GM}{4R} = \frac{3}{2} v_1^2$$

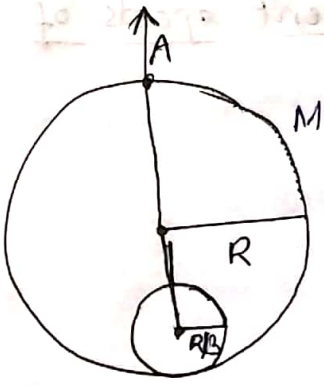
$$v_1^2 = \frac{GM}{2R}$$

$$v_1^2 = \frac{3.2}{6.4} \times 10^7$$

$$v_1^2 = 32 \times 10^6$$

$$\boxed{v_1 = 4\sqrt{2} \times 10^3 \text{ m/s}}$$

3)



M → total mass of sphere with out cavity.

Removed mass = $\frac{M}{27}$

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$V_{Total} = V_{ce} + V_{Rem}$ — potentials.

$$-\frac{GM}{R} = -\frac{GM}{\frac{5R}{2}} + V_{Rem}$$

$$-\frac{GM}{R} = -\frac{GM}{4.5R} + V_{Rem}$$

$$V_{Rem} = \frac{GM}{4.5R} - \frac{GM}{R}$$

$$\boxed{V_{Rem} = -\frac{4.4GM}{4.5R}}$$

$T.E = 0$

$U + K = 0$

$$-\frac{4.4GM}{4.5R} \times m + \frac{1}{2} m v_e^2 = 0$$

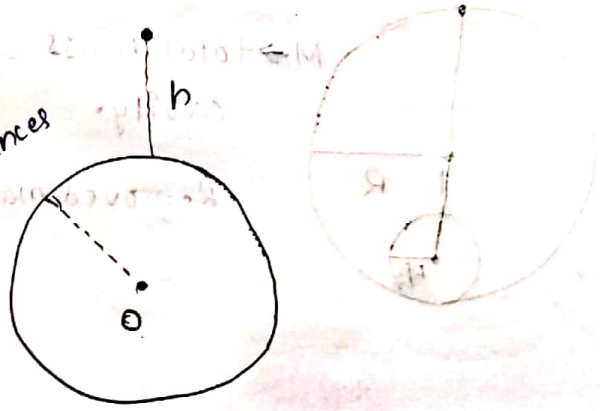
$$v_e^2 = \frac{8.8GM}{4.5R}$$

escape velocity

$$\boxed{v_e = \sqrt{\frac{8.8GM}{4.5R}}}$$

Trajecteries of projectile for different speeds of projection:

Assumed that earth is at rest & there is no other influences except earth's influence.



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Case (1):

* Body is dropped from height 'h' ; i.e, $\boxed{V = 0}$
* It travels along straight line.
* To find speed of body while hitting the earth surface,

$$-\frac{GMm}{R+h} + 0 = -\frac{GMm}{R} + \frac{1}{2}mv^2$$

$$\boxed{v = \sqrt{2GM\left(\frac{1}{R} - \frac{1}{R+h}\right)}}$$

* To find time taken by the body to reach the ground,

$$-\frac{GMm}{R+h} = -\frac{GMm}{R+x} + \frac{1}{2}mv^2$$

$$\boxed{v = \sqrt{2GM\left(\frac{1}{R+x} - \frac{1}{R+h}\right)}}$$

now, $v = -\frac{dx}{dt}$

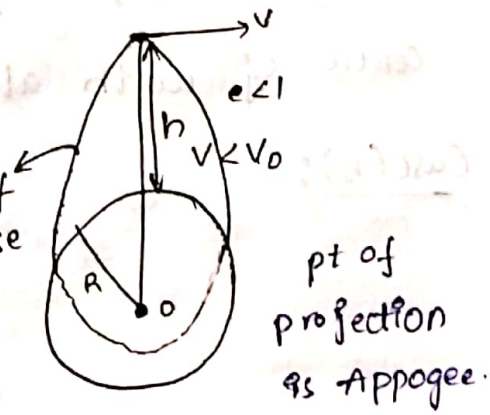
on integrating with limits $x = h$ at $t = 0$

$x = 0$ at $t = t$

we get t.

Case (2):

If $0 < v < v_0$, here body follows along the elliptical arc of ellipse path and strikes the earth surface.



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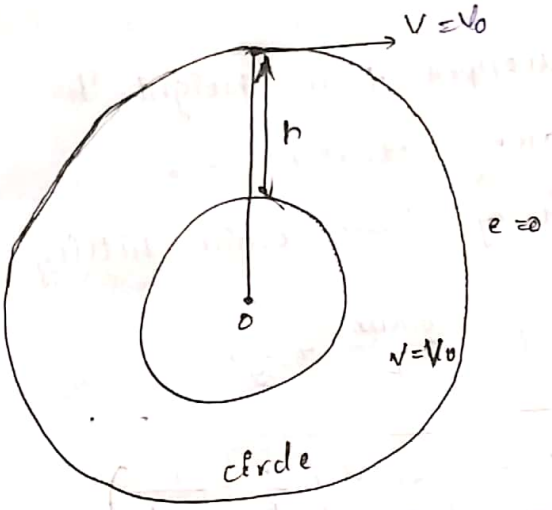
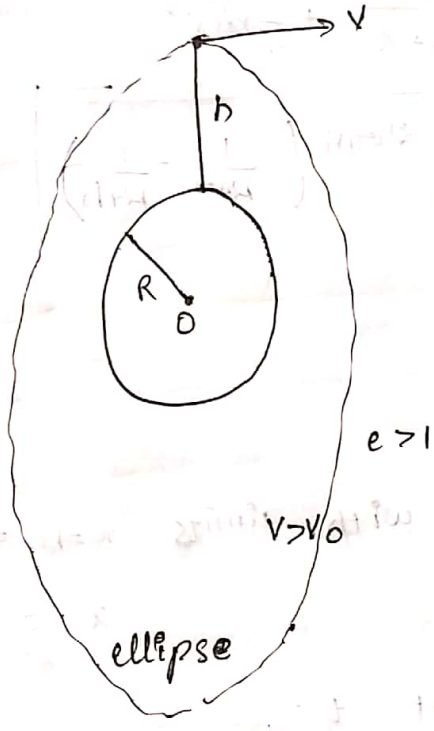
Case (3):



If $V = V_0$, Body follows circular path with centre at centre of earth. It is called "first cosmic velocity".

i.e, $V_I = V_0 = \sqrt{\frac{GM}{r}}$.

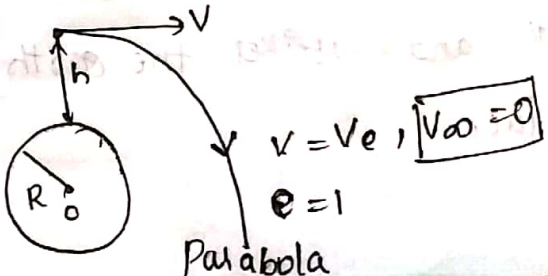
Case (4):



pt of projection is perigee.

If $V_0 < V \leq V_e$, body follows in elliptical path with centre of earth at one of the foci.

Case (5):



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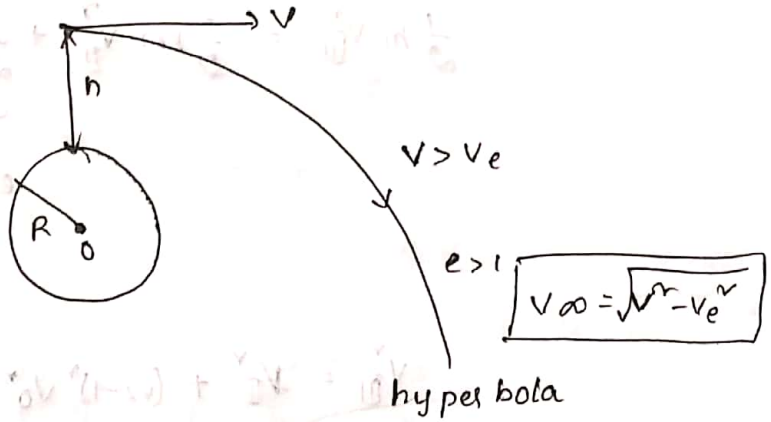
If $V = V_e$; body escapes from gravitational field of earth and follows in "parabola path". this is called "second cosmic velocity" (V_{II})

case (6) :



If $V > V_e$, body escapes from gravitational influence and follows "hyperbolic path".

3rd Cosmic Velocity :

$$\bar{V}_a = \bar{V}_{ab} + \bar{V}_b$$

* Velocity req to escape from gravitational influence of solar system is called 3rd cosmic velocity.
* It is the velocity with which when a body is projected from earth it may escape from our solar system.
* Velocity of the earth relative to the sun is measured its value as 29.76 Km/s.
* when a body is thrown from earth's surface at speed V_0 relative to earth, it will escape from earth gravitational attraction & will reach to infinity with no K.E in it w.r.to earth.
* If the body is projected from earth surface with such a speed so that it is able to overcome earth's attraction

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and at infinity it is left with extra speed of $(\sqrt{2}-1)v_0$ then it will also be able to escape from the gravitational attraction of sun.

∴ for III cosmic velocity,

$$\frac{1}{2}mv_{III}^2 = \frac{1}{2}mv_{II}^2 + \frac{1}{2}m[(\sqrt{2}-1)v_0]^2$$

(v_0: orbital velocity of earth w.r.t sun)

extra energy given from the surface

$$v_{III}^2 = v_{II}^2 + (\sqrt{2}-1)^2 v_0^2$$

$v_{II} = 11.2$ km/sec

$v_0 = 29.76$ km/sec

$$\Rightarrow v_{III}^2 = (11.2)^2 + (0.414)^2 (29.76)^2$$

$$\boxed{v_{III} = 16.75 \text{ km/sec}}$$

***

Ex: A satellite of mass m is orbiting the earth in a circular orbit of radius r. It starts loosing energy slowly at const rate c due to friction. If M_e & R_e denotes the mass & radius of the earth respectively, show that the satellite falls on the earth in a time given by $t = \frac{GmM_e}{2c}\left(\frac{1}{R_e} - \frac{1}{r}\right)$

Sol: Energy lost $= E_i - E_f$

$$ct = -\frac{GMm}{r} + \frac{1}{2}mv_i^2 - \left(-\frac{GMm}{R_e} + \frac{1}{2}mv_f^2\right)$$

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$$Ct = GMm\left(\frac{1}{Re} - \frac{1}{r}\right) + \frac{1}{2}m(V_i^2 - V_f^2)$$

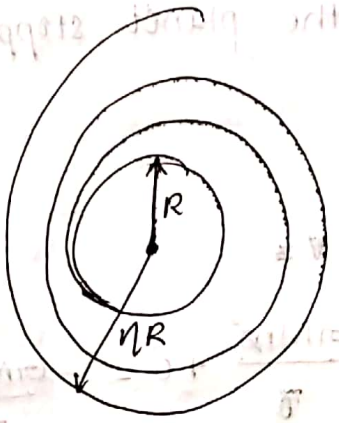
$$= GMm\left(\frac{1}{Re} - \frac{1}{r}\right) + \frac{1}{2}m\left(\frac{GM}{r^2} - \frac{GM}{Re^2}\right)$$

$$Ct = \frac{GMm}{2}\left(\frac{1}{Re} - \frac{1}{r}\right)$$

$$\boxed{t = \frac{GMm}{2C}\left(\frac{1}{Re} - \frac{1}{r}\right)}$$

Ex: Resistance force due to cosmic dust is $\boxed{F = \alpha v^2}$

Find the time taken by the satellite on reaching the surface of planet.



Sol: Given,

$$F = \alpha v^2$$

$$ma = \alpha v^2$$

$$a = \frac{\alpha v^2}{m}$$

$$v_i = \sqrt{\frac{GM}{\eta R}}$$

$$v_f = \sqrt{\frac{GM}{R}}$$

$$\frac{dv}{dt} = \frac{\alpha v^2}{m}$$

$$\int_{v_i}^{v_f} \frac{dv}{v^2} = \frac{\alpha}{m}\int_0^t dt$$

$$-\left[\frac{1}{v}\right]_{v_i}^{v_f} = \frac{\alpha}{m}(t)$$

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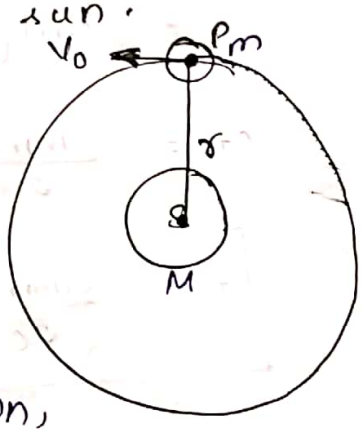
Ex: If a planet coas suddenly stopped on its orbit supposed to be Circular. Find the time taken by the planet to reach on to the sun.



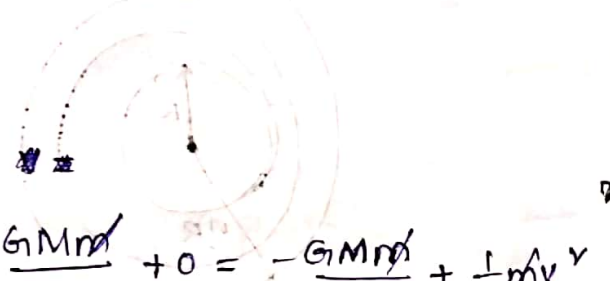
Sol: Before stopping,

$$V_0 = \sqrt{\frac{GM}{r}}$$

& Time period of revolution,

$$T = \frac{2\pi r}{V_0} = \frac{2\pi}{\sqrt{GM}}(r^{3/2})$$

After the planet stopped & dropped, it follows stline path.



$R+h \approx h$

$r = R+h \approx h$

$$-\frac{GMm}{r} + 0 = -\frac{GMm}{x} + \frac{1}{2}mv^2$$

$x \to 0 \to r$
$t \to t \to 0$

$$\frac{v^2}{2} = GM\left(\frac{1}{x} - \frac{1}{r}\right)$$

$$v = \sqrt{2GM\left(\frac{1}{x} - \frac{1}{r}\right)}$$

$x = r\sec\theta$

$dx = 2r\sec\theta\tan\theta$

$$\frac{dx}{dt} = \sqrt{2GM}\sqrt{\frac{1}{r}\left(\cos^2\theta - 1\right)}$$

$$\frac{dx}{dt} = \sqrt{\frac{2GM}{r}}(\sin\theta)$$

$$\int \frac{dx}{\sin\theta} =$$

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$x = r\sin^2\theta$

$$-\frac{dx}{dt} = \sqrt{\frac{2GM}{r}}\sqrt{\text{cosec}^2\theta - 1}$$

$$dx = 2r\sin\theta\cos\theta \cdot d\theta$$

$$-\frac{2r\sin\theta\cos\theta \cdot d\theta}{dt} = \sqrt{\frac{2GM}{r}}(\cot\theta)$$

$$-\frac{2r\sin\theta\cos\theta \cdot d\theta}{dt} = \sqrt{\frac{2GM}{r}} \times \frac{\cos\theta}{\sin\theta}$$

$$-2r\int_{\pi/2}^{0}\sin^2\theta \cdot d\theta = \sqrt{\frac{2GM}{r}}\cdot\int_{0}^{t} dt$$

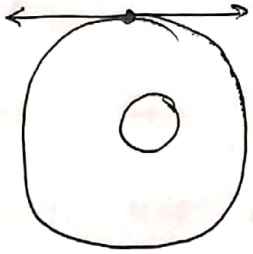
$$+\frac{2r}{2}\int_{0}^{\pi/2}(1-\cos 2\theta)\cdot d\theta = \sqrt{\frac{2GM}{r^3}}(+t)$$

$$\frac{r^{3/2}}{\sqrt{2GM}}\left[\theta - \frac{\sin 2\theta}{2}\right]_{0}^{\pi/2} = t$$

$$\frac{r^{3/2}}{\sqrt{2GM}}\left[\frac{\pi}{2}\right] = t$$

$$\frac{2\pi r^{3/2}}{2\times 2\sqrt{2}\sqrt{GM}} = t$$

$$\boxed{t = \frac{T}{4\sqrt{2}}}$$



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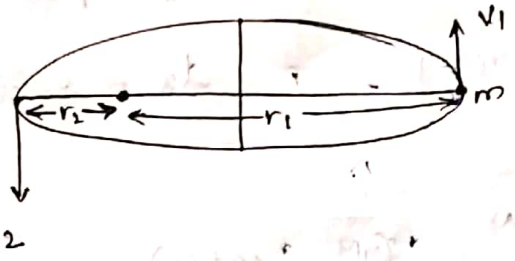

L·C·A·M,

$$m_1 v_1 r_1 = m_2 v_2 r_2$$

$$v_1 r_1 = v_2 r_2 \quad -①$$

L·C·E,

$$-\frac{GMm}{r_1} + \frac{1}{2}m v_1^2 = -\frac{GMm}{r_2} + \frac{1}{2}m v_2^2$$

$$\frac{1}{2}(v_1^2 - v_2^2) = GM\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\frac{1}{2}\left(v_1^2 - \frac{v_1^2 r_1^2}{r_2^2}\right) = GM\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\frac{1}{2}v_1^2\left(\frac{r_2^2 - r_1^2}{r_2^2}\right) = GM\left(\frac{r_2 - r_1}{r_1 r_2}\right)$$

$$\frac{v_1^2}{2}\left(\frac{(r_2 - r_1)(r_2 + r_1)}{r_2}\right) = GM\left(\frac{r_2 - r_1}{r_1}\right)$$

$$\boxed{v_1 = \sqrt{\frac{2GM(r_2)}{(r_1 + r_2)r_1}}}$$

Angular momentum,

$$L = m_1 v_1 r_1$$

$$= m r_1 \sqrt{\frac{2GMr_2}{(r_1 + r_2)r_1}}$$

$$\boxed{L = \sqrt{\frac{2GMm^2 r_2 r_1}{(r_1 + r_2)}}}$$

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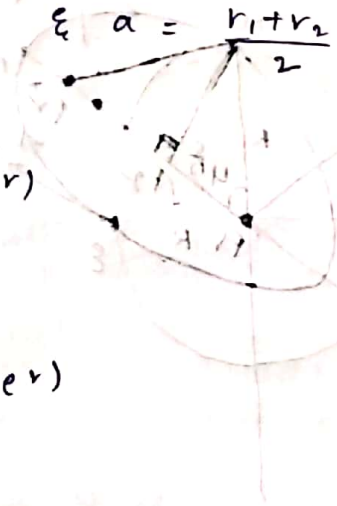
and $r_1 = a(1+e)$

$r_2 = a(1-e)$

$\Rightarrow r_1 r_2 = a^2(1-e^2)$



we have,

$b^2 = a^2(1-e^2)$

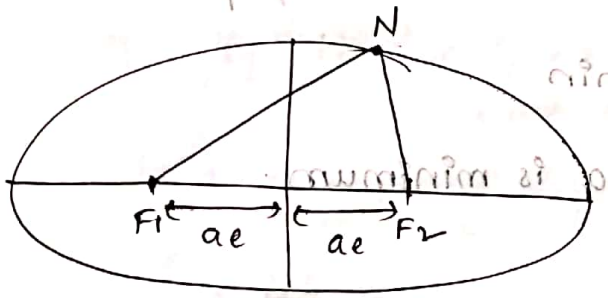
$r_1 r_2 = b^2$

length of semi-latus rectum,

$$r_\perp = \frac{b^2}{a} = \frac{r_1 r_2}{\frac{(r_1+r_2)}{2}}$$

$$\boxed{r_\perp = \frac{2r_1 r_2}{r_1+r_2}}$$

*



At all positions,

$$\boxed{F_1N + NF_2 = 2a}$$

& $\boxed{F_1F_2 = 2ae}$ *

*** Ex: Find the min velocity of projection of particle from the place at equator so that it is able to reach the north pole.

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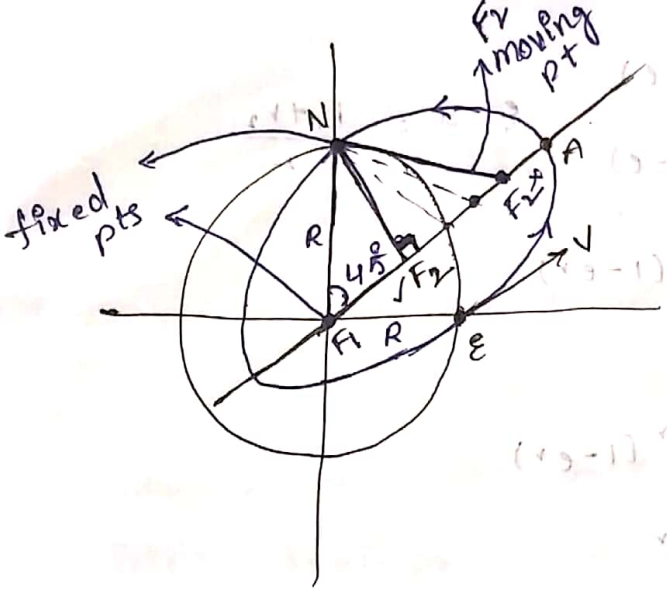
8.16



we have,

T.ε of particle in elliptical orbit,

$$\varepsilon = -\frac{GMm}{2a} \quad \text{—①}$$

For min velocity of projection,

K.ε min, where P.ε of particle remains sa

⇒ T.ε min

i.e, a is minimum.

we have,

$$F_1N + NF_2 = 2a$$

⇒ NF_2 is minimum.

i.e, NF_2 is ⊥ler to major axis.

∴ From Δle F_1NF_2, $NF_2 = \frac{R}{\sqrt{2}}$ & $F_1F_2 = \frac{R}{\sqrt{2}}$

now,

$$F_1N + NF_2 = 2a$$

$$R + \frac{R}{\sqrt{2}} = 2a$$

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$$a = \frac{R(\sqrt{2}+1)}{2\sqrt{2}}$$

now, $T.E = -\frac{GMm}{2a}$

T.E of particle in elliptical orbit,

$$-\frac{GMm}{R} + \frac{1}{2}mv_{min}^2 = -\frac{GMm}{2a}$$

$$\frac{1}{2}v_{min}^2 = GM\left(\frac{1}{R} - \frac{1}{2a}\right)$$

$$v_{min} = \sqrt{2GM\left(\frac{1}{R} - \frac{1}{2a}\right)}$$

(or) $$v_{min} = \sqrt{\frac{2GM_0}{R}(\sqrt{2}-1)}$$ — [sub 'a' in above eqn]

* eccentricity of the ellipse:

$$F_1F_2 = 2ae$$

$$\frac{R}{\sqrt{2}} = \frac{R(\sqrt{2}+1)}{\sqrt{2}}e$$

$$e = \frac{1}{\sqrt{2}+1}$$

$$e = \sqrt{2}-1$$

* max distance of particle from centre of earth,

$$d_{max} = F_1A = a(1+e)$$

$$= \frac{R}{2\sqrt{2}}(\sqrt{2}+1)(\sqrt{2}) = \frac{R}{2}(\sqrt{2}+1)$$

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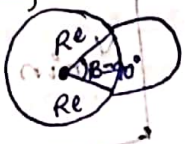
* min distance of particle from centre of earth = R.

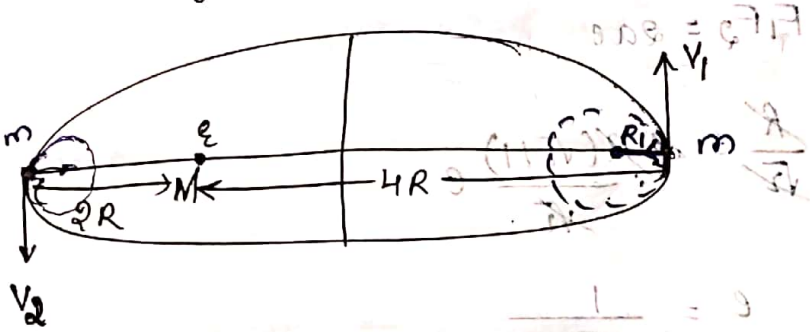
Ex① A particle is projected from the surface of the earth of radius R_e & mass m_e with speed $\sqrt{\frac{GM_e}{R_e}}$ at certain angle from local horizontal as shown in the figure such that the angle subtended by arc blw launching and landing site at earth's centre is $\beta = 90°$. Calculate the {maximum seperation of particle} from the centre of earth.



Ans: $\frac{(\sqrt{2}+1)R_e}{\sqrt{2}}$

② The max & min distances of the satellite from centre of earth are 4R & 2R. where R is radius of earth & M is mass of the earth. Find Radii of curvatures at the pts of max & min distances from the centre of the earth.

②



L.C.A.M, $m v_1 (4R) = m v_2 (2R)$

$$2v_1 = v_2$$

L.C.E,

$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{2R} + \frac{1}{2}mv_2^2$$

$$\frac{1}{2}v_1^2 = -\frac{GM}{4R} + \frac{1}{2}(4v_1^2)$$

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$$\frac{3V_1^2}{2} = \frac{GM}{2R}$$

$$\boxed{V_1 = \sqrt{\frac{GM}{6R}}}$$

$$\boxed{V_2 = \sqrt{\frac{2GM}{3R}}}$$

Now, $\frac{GMm}{(4R)^2} = \frac{mV_1^2}{R_1}$

$$\frac{GM}{16R} = \frac{1}{R_1}\left(\frac{GM}{6R}\right)$$

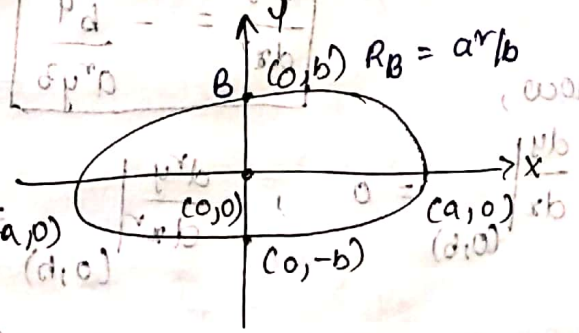
$$\boxed{R_1 = \frac{8R}{3}}$$

$$\frac{GMm}{(2R)^2} = \frac{mV_2^2}{R_2}$$

$$\frac{GM}{4R} = \frac{2GM}{3R} \times \frac{1}{R_2}$$

$$\boxed{R_2 = \frac{8R}{3}}$$

* general case :



eqn of ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

radius of curvature of any trajectory eqn,

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$$R_C = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\left(\frac{d^2y}{dx^2}\right)\right|}$$

now, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} + \frac{2y}{b^2}\cdot\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \left(0 - \frac{2x}{a^2}\right)\left(\frac{b^2}{2y}\right)$$

$$\boxed{\frac{dy}{dx} = \left(\frac{-b^2}{a^2}\right)\left(\frac{x}{y}\right)}$$

&

$$\frac{d^2y}{dx^2} = \left(\frac{-b^2}{a^2}\right)\left(\frac{y - x\frac{dy}{dx}}{y^2}\right)$$

$$= \left(\frac{-b^2}{a^2}\right)\left(\frac{y - x\times\left(\frac{-b^2}{a^2}\right)\left(\frac{x}{y}\right)}{y^2}\right)$$

$$= \left(\frac{-b^2}{a^2}\right)\left(\frac{a^2y^2 + b^2x^2}{a^2y^3}\right)$$

$$= \left(\frac{-b^2}{a^2}\right)\left(\frac{a^2b^2}{a^2y^3}\right)$$

$$\boxed{\frac{d^2y}{dx^2} = -\frac{b^4}{a^2y^3}}$$

now,

$$\left.\frac{dy}{dx}\right|_{(0,b)} = 0 \quad , \quad \left.\frac{d^2y}{dx^2}\right|_{(0,b)} = \frac{-b^4}{a^2b^3} = -\frac{b}{a^2}$$

$$R_B = \frac{[1+0]^{3/2}}{|-b^2/a^2|} = +\frac{a^2}{b}$$

$$\boxed{R_B = +a^2/b}$$

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$$2R = a(1-e)$$

$$4R = a(1+e)$$

$$a^2(1-e^2) = 8R^2$$

$$b^2 = 8R^2$$

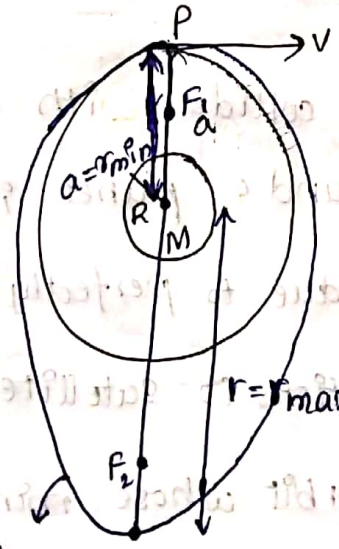
$$\boxed{b = 2\sqrt{2}R}$$

$$2a = 6R \Rightarrow \boxed{a = 3R}$$

$$\Rightarrow \begin{cases} R_B = \dfrac{a^2}{b} = \dfrac{9R^2}{2\sqrt{2}R} = \dfrac{9R}{2\sqrt{2}} \\ R_A = \dfrac{b^2}{a} = \dfrac{8R^2}{3R} = \dfrac{8R}{3} \end{cases}$$

Ex: A satellite is revolving round the earth in a circular orbit of radius a with velocity v_0. A particle is projected from the satellite in forward direction with $v_{rel} = \left[\sqrt{\frac{5}{4}} - 1\right] v_0$. Calculate during subsequent motion of particle, its min & max distances from earths centre.

Sol:



here,

$$v_0 = \sqrt{\frac{GM}{a}} = v_s$$

$$\bar{v}_{P/S} = \bar{v}_P - \bar{v}_S$$

$$\bar{v}_P = \bar{v}_{P/S} + \bar{v}_S$$

$$= \left(\sqrt{\frac{5}{4}} - 1\right) v_0 + v_0$$

$$\bar{v}_P = \sqrt{\frac{5}{4}}\, v_0$$

$$\boxed{v_P > v_0}$$

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$$v_p = \sqrt{\frac{5GM}{4a}}$$

* L.C.A.M,

$$mv_1 a = mv_2 r$$

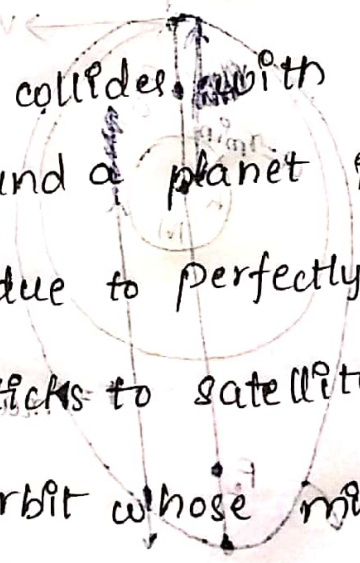
$$v_1 a = v_2 r \quad -①$$

* L.C.E,

$$-\frac{GMm}{a} + \frac{1}{2}mv_p^2 = -\frac{GMm}{r} + \frac{1}{2}mv_2^2$$

$$-\frac{GMm}{a} + \frac{1}{2}m\left(\frac{5}{4}v_0^2\right) = -\frac{GMm}{r} + \frac{1}{2}m\frac{v_p^2 a^2}{r^2}$$

Ex: A meteorite of mass (m) collides with a satellite of mass 10m orbiting around a planet in a circular path of radius R. due to perfectly inelastic collision the meteorite sticks to satellite & the satellite now follows an orbit whose min distance from the planet is R/2. Determine the velocity v of the meteorite before collision. Mass of the planet is M.



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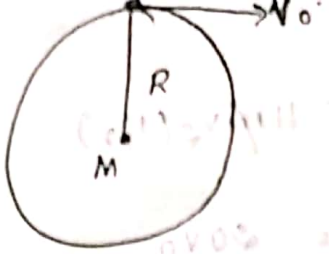
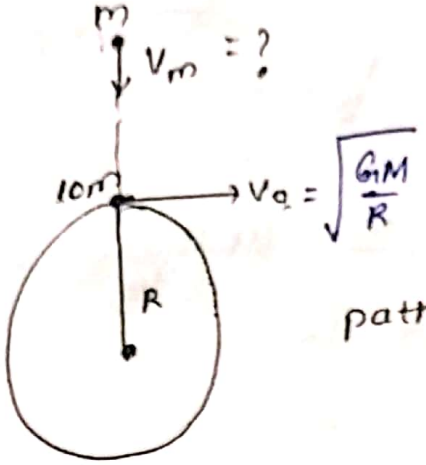
Sol:

1)

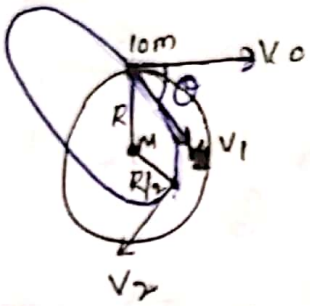


2)



path before collision

3)



path after collision.

* L.C.L.M,

$$10mV_0 = 11mV_1\cos\theta \quad -①$$

& $$mV_m = 11mV_2\sin\theta \quad -②$$

* L.C.A.M,

$$11mV_1\cos\theta(R) = 11mV_2(R/2) \quad -③$$

* L.C.E,

$$-\frac{GM(11m)}{R} + \frac{1}{2}(11m)(V_1^2) = -\frac{GM(11m)}{R/2} + \frac{1}{2}(11m)V_2^2 \quad -④$$

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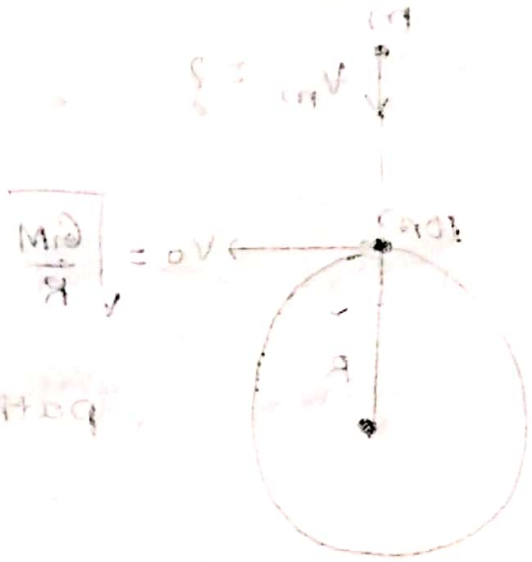
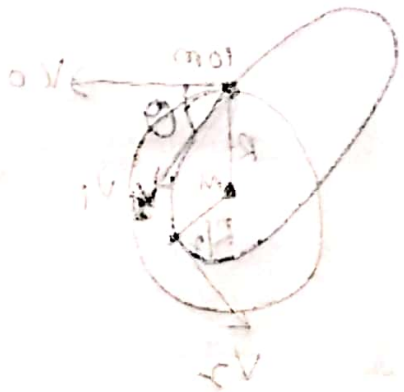
from ③ ,

$$10m/v_0 = 11m/v_2 (1/2)$$

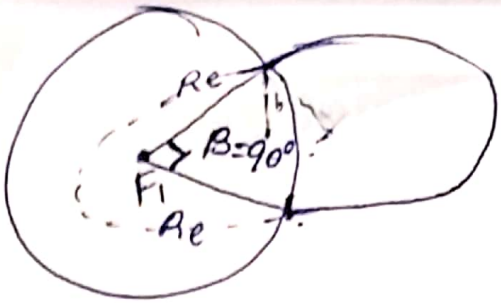
$$v_2 = \frac{20v_0}{11}$$





① Ans 8

$$T.E = -\frac{GMm}{2a}$$



$$T.E = U + K$$

$$-\frac{GMm}{2a} = -\frac{GMm}{R} + \frac{1}{2}mv^2$$

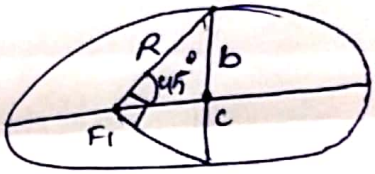
$$v = \sqrt{\frac{GM}{R}}$$

$$-\frac{GMm}{2a} = -\frac{GMm}{R} + \frac{GMm}{2R}$$

$$-\frac{GMm}{2a} = -\frac{GMm}{2R}$$

$$\boxed{a = R_e}$$

* eccentricity of the ellipse:



$$(F_1F_2 = 2ae) \quad \text{(or)}$$

$$\sin 45^\circ = \frac{b}{R}$$

$$b^2 = a^2(1-e^2)$$

$$\boxed{b = \frac{R_e}{\sqrt{2}}}$$

$$\frac{R^2}{2} = R^2(1-e^2)$$

$$1 - e^2 = \frac{1}{2}$$

$$\boxed{e = 1/\sqrt{2}}$$

* max distance, $d_{max} = a(1+e)$

$$= R\left(1 + \frac{1}{\sqrt{2}}\right)$$

$$\boxed{d_{max} = \frac{(\sqrt{2}+1)R_e}{\sqrt{2}}}$$

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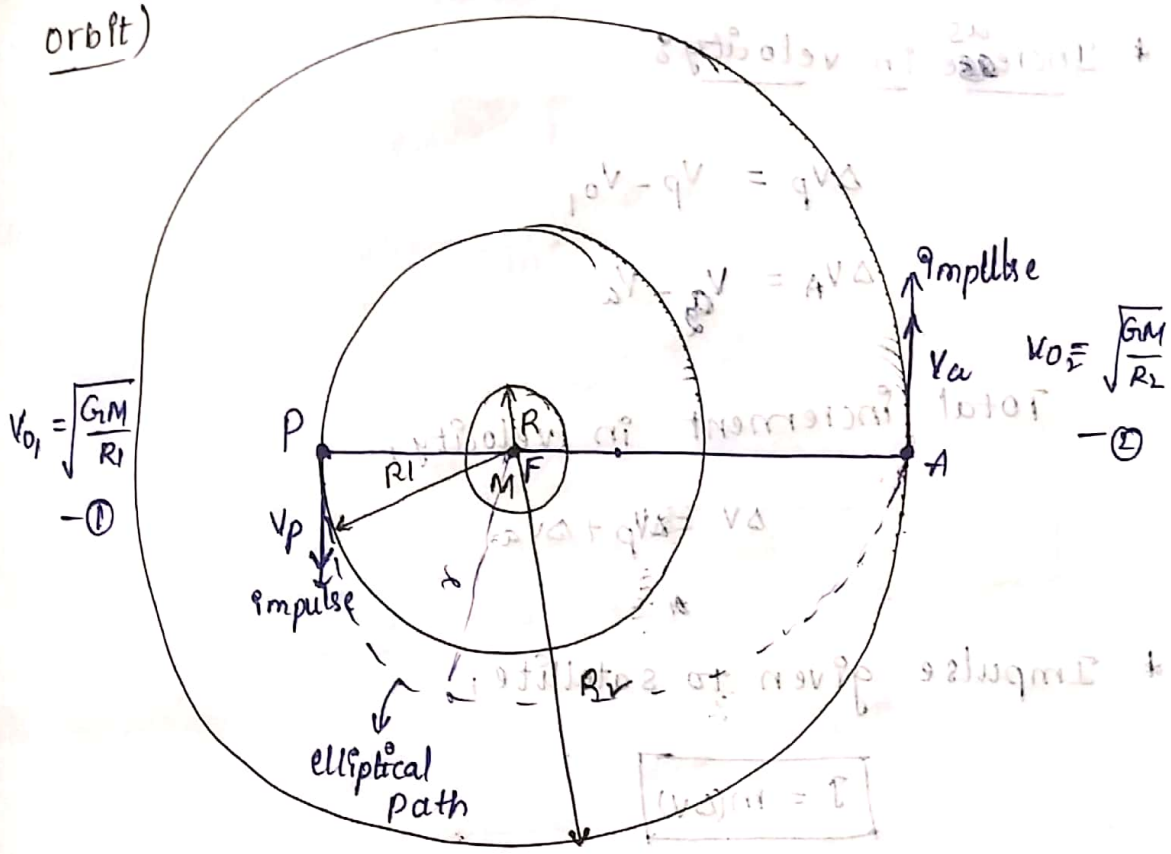
Concept of transfer of orbit (Hohmann transfer of orbit)



here,

$$R_1 + R_2 = 2a \quad -③$$

$$V_p = \sqrt{\frac{GM}{a}\left(\frac{1+e}{1-e}\right)}, \quad V_a = \sqrt{\frac{GM}{a}\left(\frac{1-e}{1+e}\right)}$$

$$b^2 = a^2(1-e^2)$$

* Energy of satellite in elliptical orbit,

$$\frac{-GMm}{r} + \frac{1}{2}mv^2 = \frac{-GMm}{2a}$$

$$\frac{v^2}{2} = GM\left(\frac{1}{r} - \frac{1}{2a}\right)$$

$$\boxed{v = \sqrt{2GM\left(\frac{1}{r} - \frac{1}{2a}\right)}}$$

* at P, $R_1 = r$, $v = V_p$

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* at A, $R_2 = r$, $V = V_a$

* Increase in velocity as

$$\Delta V_p = V_p - V_{o_1}$$

$$\Delta V_A = V_{o_2} - V_a$$

Total increment in velocity,

$$\Delta V = \Delta V_p + \Delta V_a$$

*

* Impulse given to satellite,

$$\boxed{J = m(\Delta V)}$$

* work done (energy required) in transfering the satellite,

$$W = \mathcal{E}_f - \mathcal{E}_i \; ; \quad \mathcal{E}_i = -\frac{GMm}{2R_1}$$

$$W = -\frac{GMm}{2}\left(\frac{1}{R_2} - \frac{1}{R_1}\right) \qquad \mathcal{E}_f = -\frac{GMm}{2R_2}$$

$$\boxed{W = \frac{GMm}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

Ex: $R_1 = 2R_e$, $R_2 = 4R_e$, Mass of space vehicle, $m = 300kg$

$g = 10 m/s^2$, $R_e = 6400Km$. Find :-

ΔV_p, ΔV_A, $\mathcal{E}_{req}$.

Ans: 876 m/s | 734 m/s | 2.4×10^9 J

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Sol:

$$\varepsilon_{req} = \frac{GMm}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

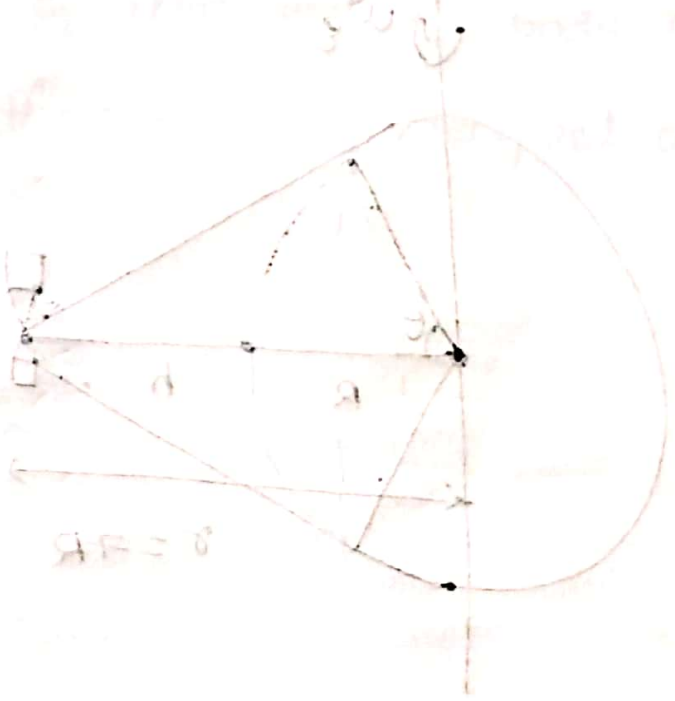
$$= \frac{Re\,gm}{2}\left(\frac{1}{4Re}\right)$$

$$= \frac{gmRe}{8}$$

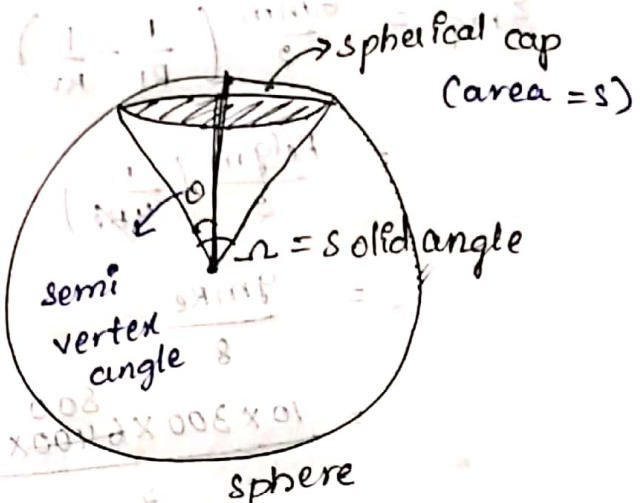
$$= \frac{10 \times 300 \times 6400 \times 10^3}{8}$$

$$= 24 \times 10^8$$

$$\boxed{\varepsilon_{req} = 2.4 \times 10^9\ J}$$

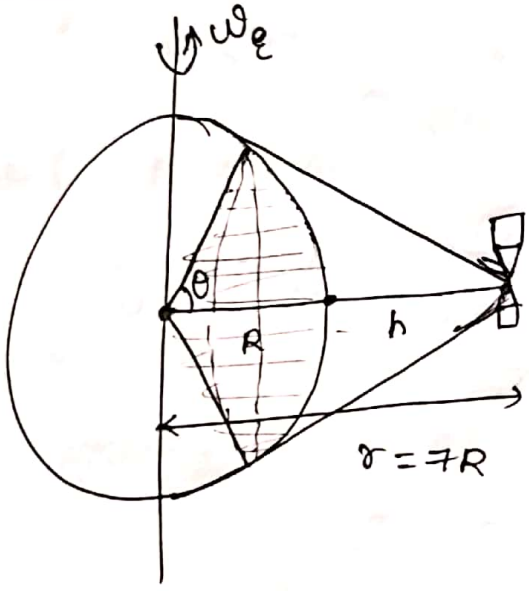


Broadcasting Region of a satellite



$$\Omega = 2\pi(1-\cos\theta)$$

$$\Omega = \frac{S}{R^2}$$



$$\Omega = 2\pi(1-\cos\theta)$$

$$\Omega = 2\pi\left(1-\frac{R}{R+h}\right)$$

$$\Omega = \frac{2\pi h}{R+h}$$

$$S = \Omega R^2$$

$$S = \frac{2\pi h R^2}{R+h}$$

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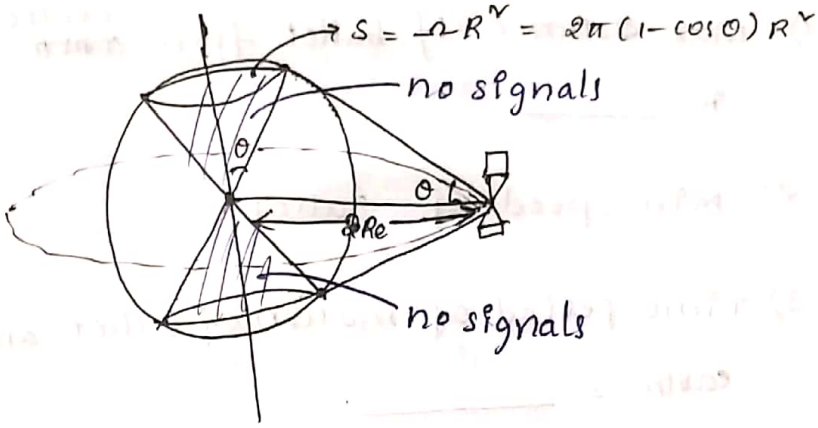

Area covered by satellite emitting signals,

$$\boxed{A = 4\pi R^2 - 2S}$$

Black hole: 3|1|25 | Friday

It is body from which nothing is escaped even light.

*From Einstein's theory of relativety, no body can travel with a speed greates than (or) equal to speed of light.

* for black body,

$$\boxed{V_e \geq c}$$

$$\sqrt{\frac{2GM}{R}} \geq c$$

$$\frac{2GM}{R} \geq c^2$$

$$\boxed{\frac{2GM}{c^2} \geq R}$$

M — mass of black hole.

R — radius of black (~~body~~) hole

Ex: An earth satellite is revolving in a circular orbit of radius 'a' with a velocity v_0. Mass of earth is M. A gun in the satellite is directly aimed towards earth. A bullet is fired from the gun with muzzle velocity $\frac{v_0}{2}$. Assume that mass of the satellite is very-very large w.r.to the mass of the bullet,

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Find 1) Max distance of bullet from centre of earth is _____

2) Min speed of bullet is _____.

3) Time period of motion of bullet around earth = _____.

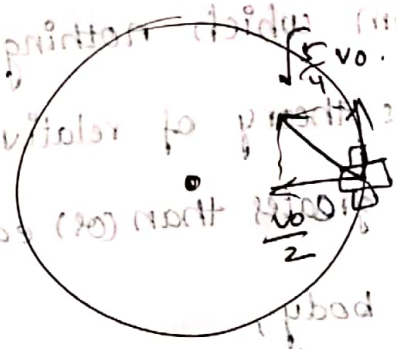
Sol:-

$$\vec{V_{bs}} = \vec{V_b} - \vec{V_s}$$

$$\vec{V_b} = \vec{V_{bs}} + \vec{V_s} \quad ; \quad |\vec{V_b}| = \sqrt{V_0^2 + V_0^2}$$

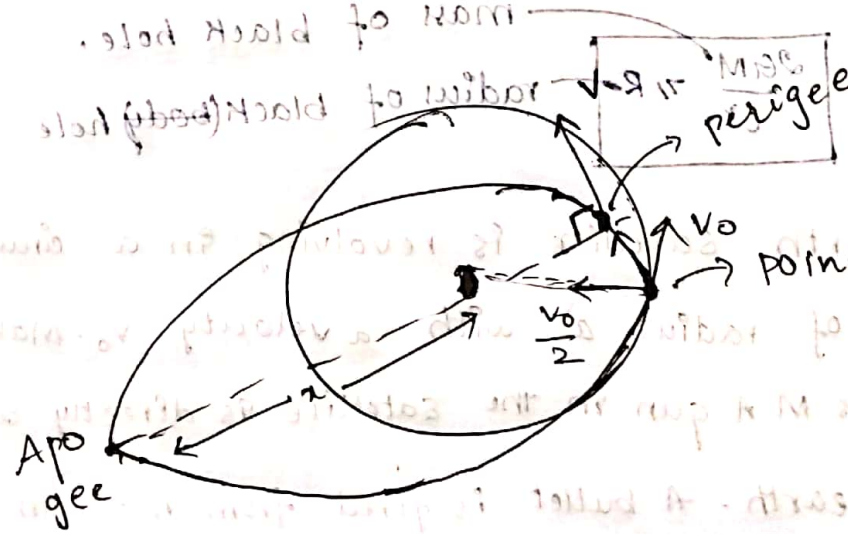
$$V_b = \sqrt{\frac{5}{4}}\, V_0$$

(Case seen from earth)



at point 'P'

The velocity vector of bullet is not perpendecular to the radius vector. So, it is not 'perigee'. it is the general point on the elliptical path.



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from law of conservation of Angular momentum, blw point 'p' and Apogee,

$$mv_0 r = mv(x) \rightarrow (1)$$

$mv_0 r$ ['comp' which is 1ler to r]

from law of Conservation of Energy,

$$-\frac{GMm}{a} + \frac{1}{2} m\left(\sqrt{\frac{5}{4}} v_0\right)^2 = -\frac{GMm}{x} + \frac{1}{2} mv^2 \rightarrow (2)$$

By solving (1) & (2),

we get

$$x = 2a, \frac{2a}{3}.$$

max. distance $\rightarrow$ $2a$.

min distance $\rightarrow$ $\frac{2a}{3}$.

Gen. point distance $\rightarrow$ a.

* for max. speed of bullet,

Apply L.C.E blw point 'p' and perigee.

for min speed of bullet,

Apply L.C.E blw point 'p' and Apogee.

$$T = \frac{2\pi}{\sqrt{Gm}} [a']^{3/2},$$

$$a' = \frac{r_{max} + r_{min}}{2}$$

Scanned with CamScanner